Question No : 1 of 34

The graph of an odd function is symmetrical about ----------

Answer (Please select your correct option)

- x-axis
- y-axis
- origin
- None of these

even function y axis
odd function origion

Start Time 9:41 AM

87:00
Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

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MTH631 Real Analysis II

Question No : 10 of 34

The rate of change of $f(x,y)$ in the direction of a unit vector is called the ---------- of $f(x,y)$.

Answer (Please select your correct option)

- differential
- directional derivative
- mixed derivative
- None of these

Start Time: 9:41 AM

74:00 Time Left

10

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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Question No : 11 of 24

A function of several variables may have first order partial derivatives at a point X_0 but ———— continuous at X_0.

Answer (Please select your correct option)

- fail to be
- must be
- always not
- None of these

page 102

Start Time: 9:41 AM

74:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Question No : 12 of 34

A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0 page 105
- constant function
- None of these

Start Time 9:41 AM

74:00 Time Left

12

Question Summary : (Attempted Question)

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$
- None of these

Start Time: 9:41 AM

74:00 Time Left

13

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r}$ = ...

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

Start Time: 9:41 AM

74:00 Time Left

14

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the

Answer (Please select your correct option)

- critical point
- point of inflection
- point of relative extrema
- None of these

Start Time: 9:41 AM

74:00

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Question Summary : (Attempted Questions)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 5 of 34 Mark

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that _____

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

Start Time: 9:41 AM

78:00 Time Left

Question Summary : (Attempted Question)

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Question No : 3 of 50

_____ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval
- Any finite sequence of R
- None of these

Start Time: 9:41 AM

81:00
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3

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$ ✓ p 87
- $-\frac{1}{2}$
- 0
- 1

Start Time 9:41 AM

80:00 Time Left

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 2 of 34

If H is an open covering of a compact subset S then, S can be covered by ----------

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H
- atleast one set from H
- None of these

p 79

Start Time 9:41 AM

83:00 Time Left

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 15 of 34

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called _____

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

p 126

Start Time: 9:41 AM

71:00 Time Left

Question Summary : (Attempted Question)

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Question No : 12 of 34

A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

Start Time 9:41 AM

74:00 Time Left

12

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 15 of 34

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called _____

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

Start Time: 9:41 AM

71:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$
- None of these

Start Time: 9:41 AM

74:00 Time Left

13

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r}$ = ...

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

Start Time: 9:41 AM

74:00 Time Left

14

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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MTH631 Real Analysis II

Question No : 18 of 34 Marks: 1

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ----------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$ ✓ p 137
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

Start Time 9:41 AM

65:00 Time Left

18

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 19 of 34

An absolutely convergent integral is always ----------

Answer (Please select your correct option)

- divergent
- convergent
- conditionally convergent
- None of these

Start Time: 9:41 AM

62:00 Time Left

19

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is ----------

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent
- None of these

p 137

Start Time: 9:41 AM

62:00 Time Left

20

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 34 of 34 Marks: 10 (Budgeted Time Min)

Suppose $U_0 \in T$ is a limit point of T, g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then, show that $f \circ g$ is continuous at U_0.

Answer (Please click here to Add Answer)

Start Time: 9:41 AM

62:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

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MTH631 Real Analysis II

Question No : 33 of 34 Marks: 5 (Budgeted Time

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^3 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0$

Answer (Please click here to Add Answer)

Start Time: 9:41 AM

62:00

Time Left

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 32 of 34 **Marks:**

Define a locally integrable function. Also show that the function $f(x) = \log x$ is locally integrable on $[0, 1)$.

Answer (Please click here to Add Answer)

100%

Start Time: 9:41 AM

61:00
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32

Question Summary : (Attempted Question)

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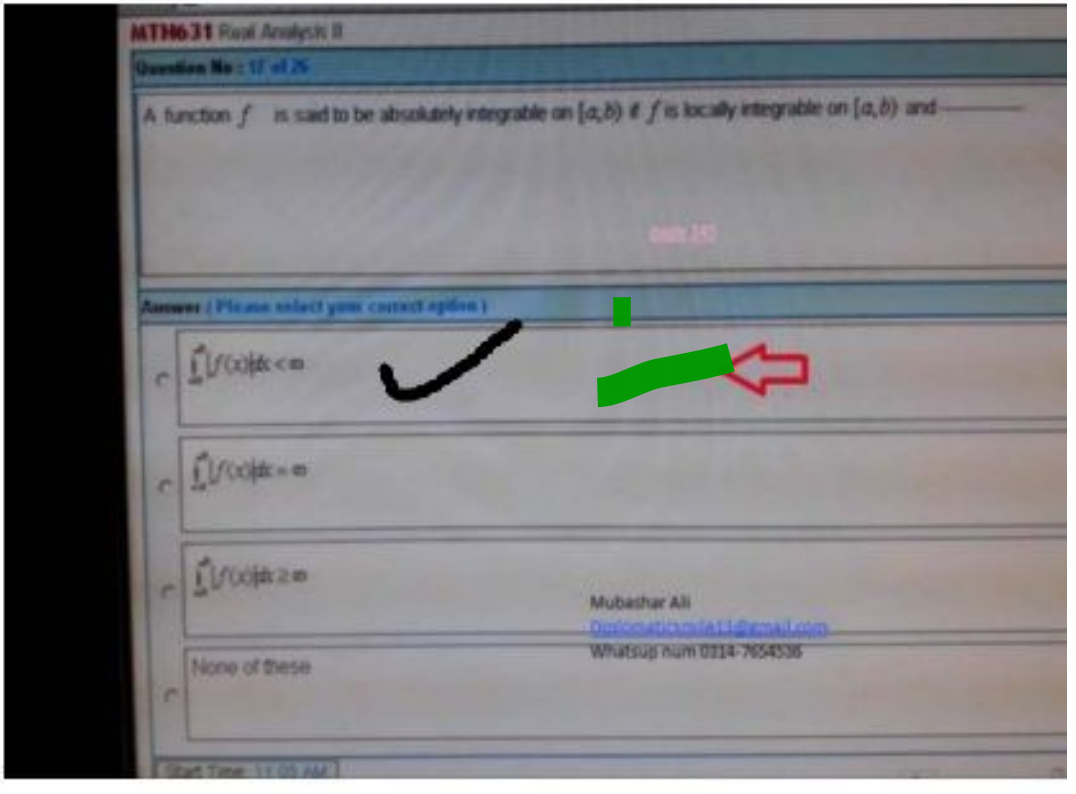
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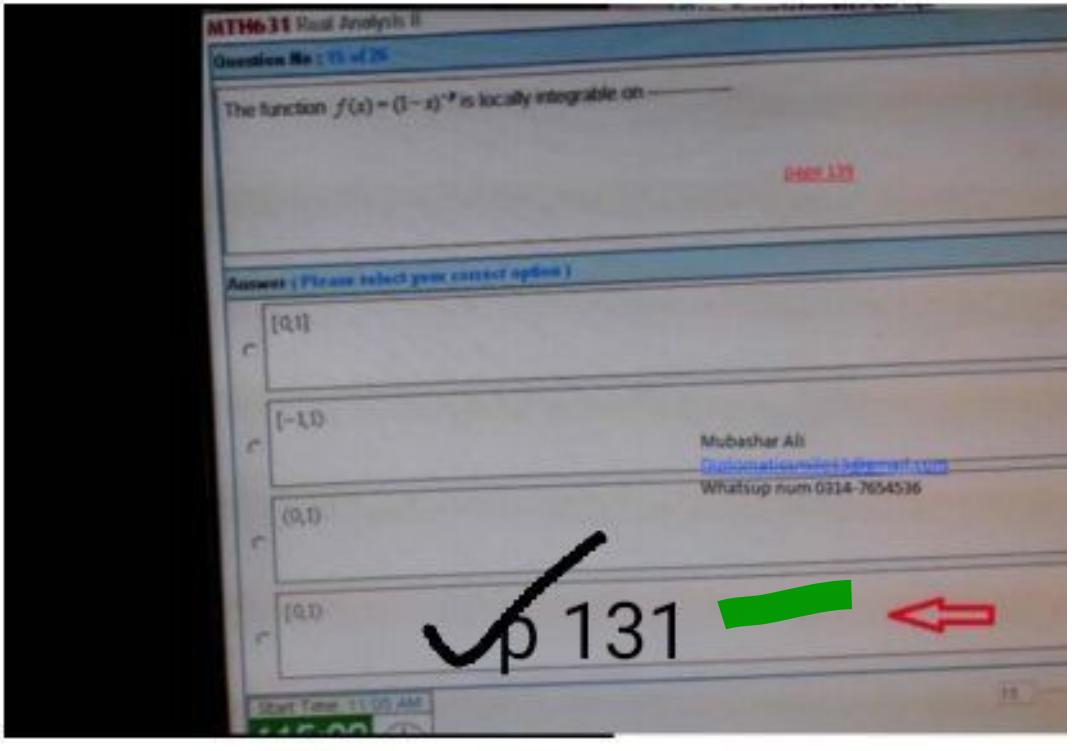
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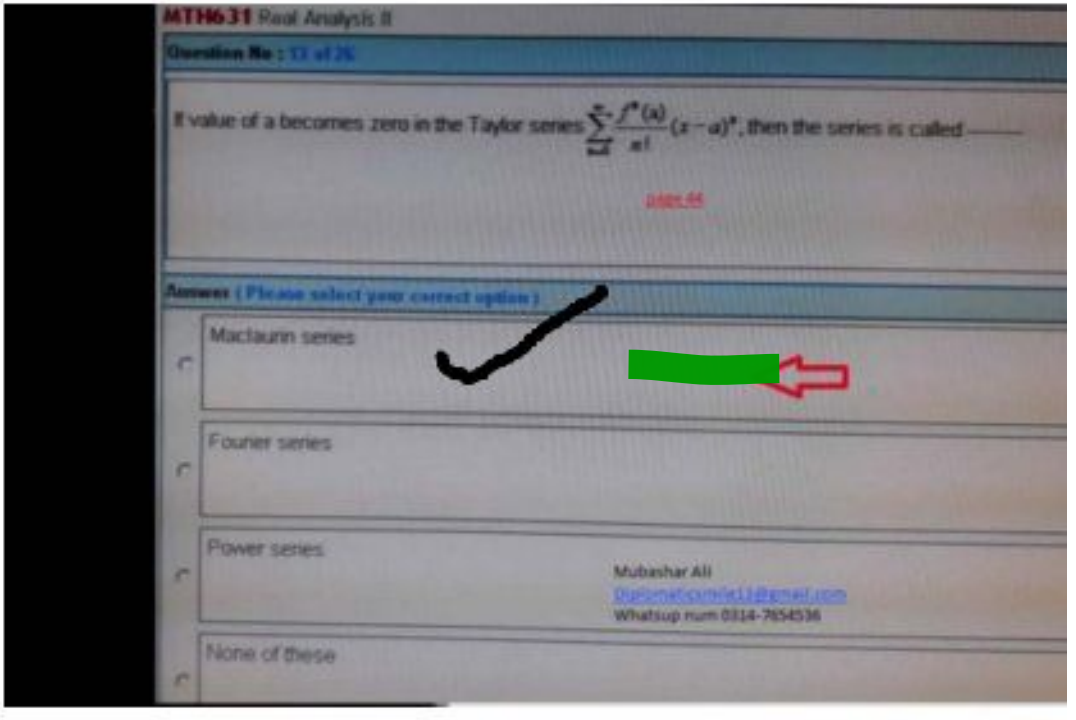
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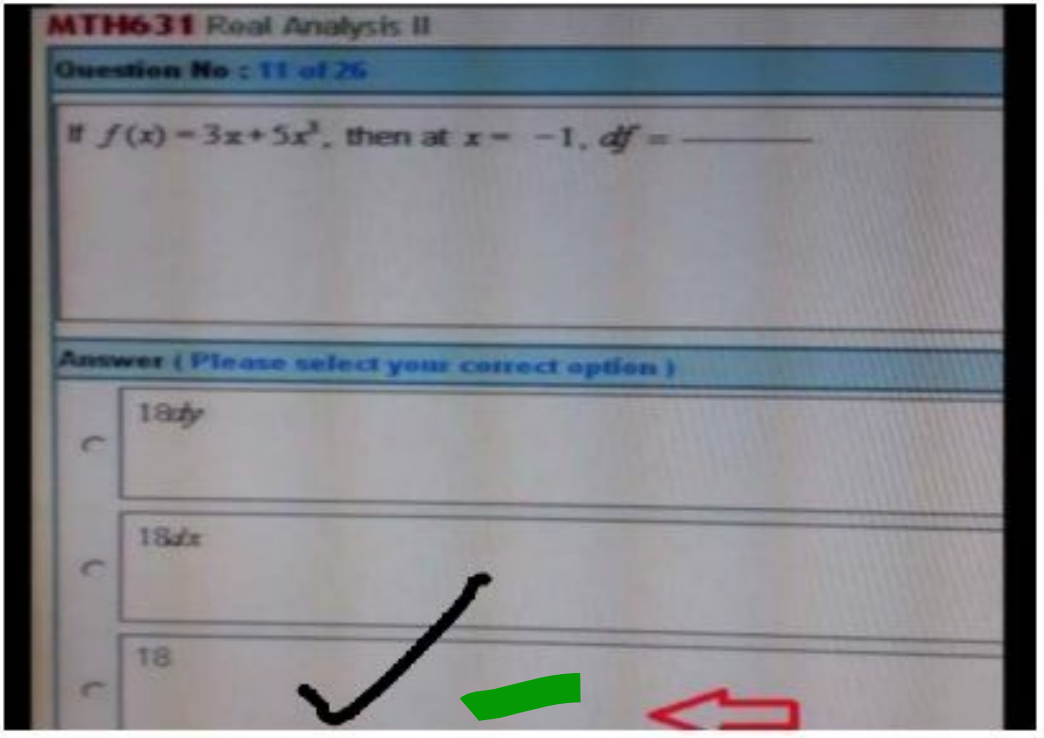
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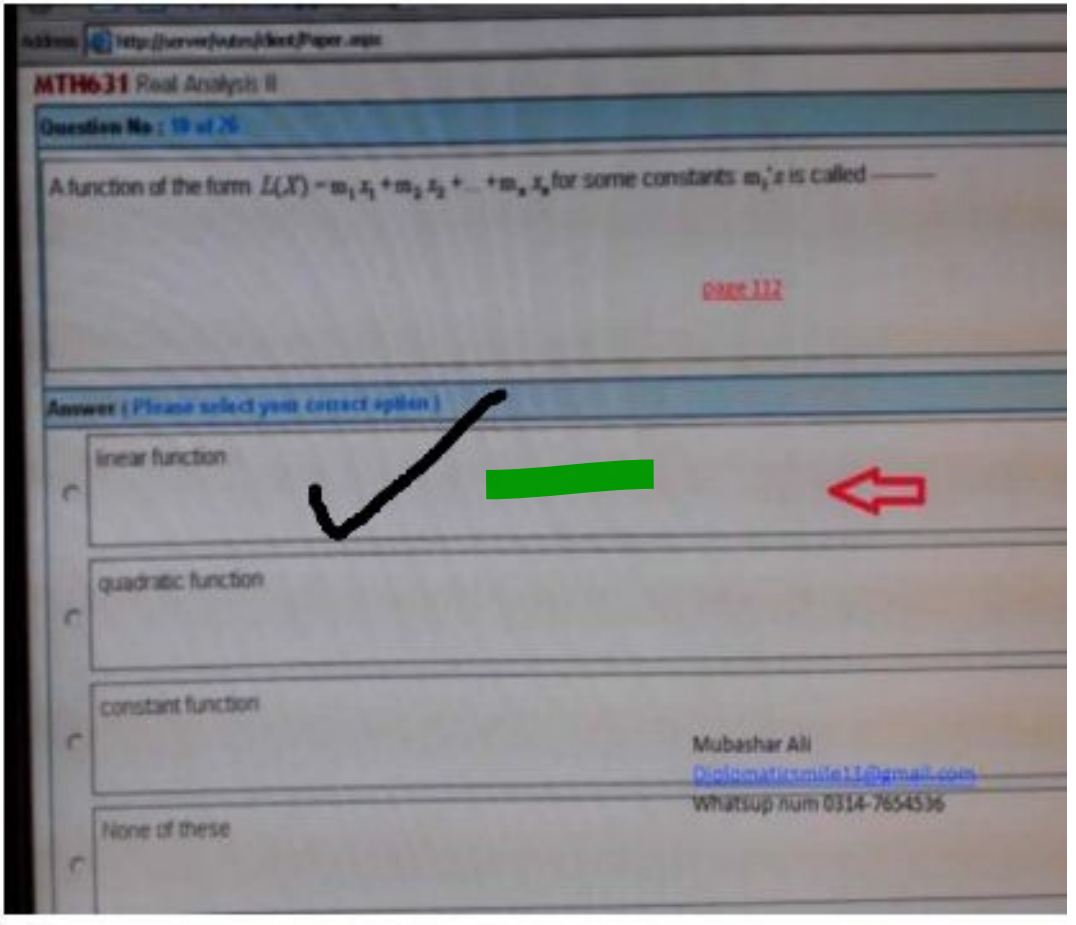
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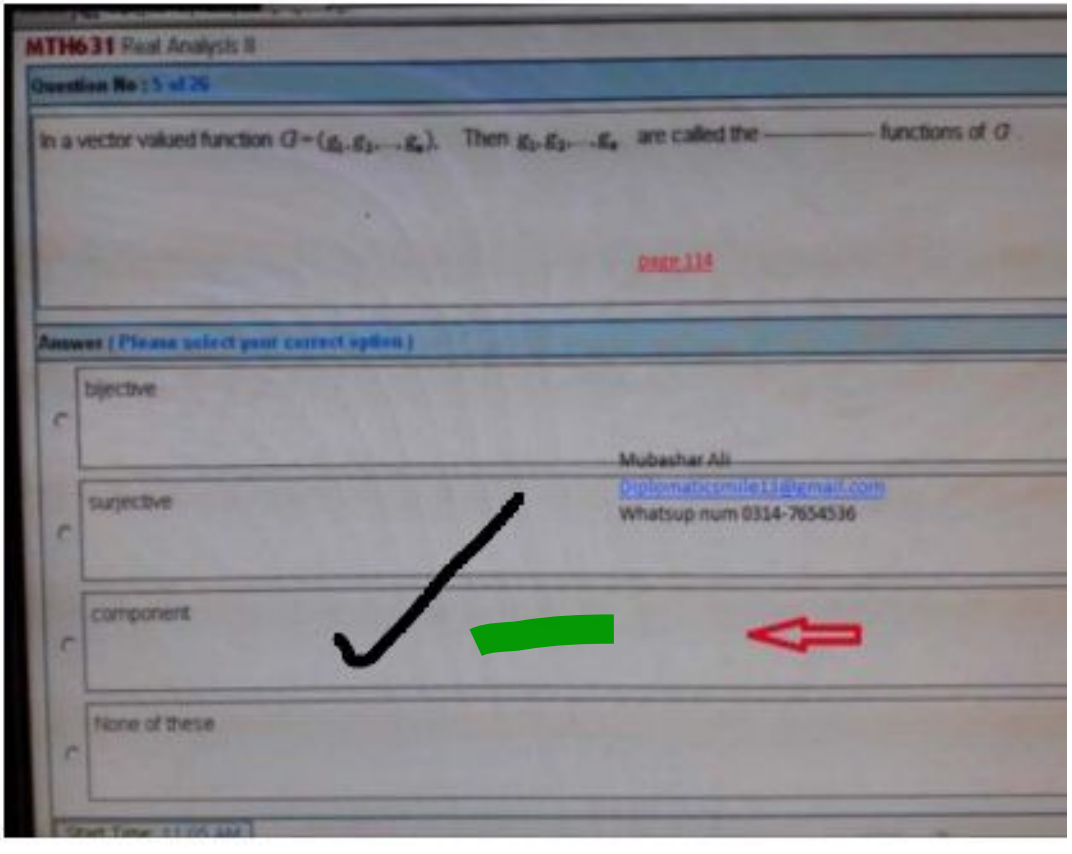
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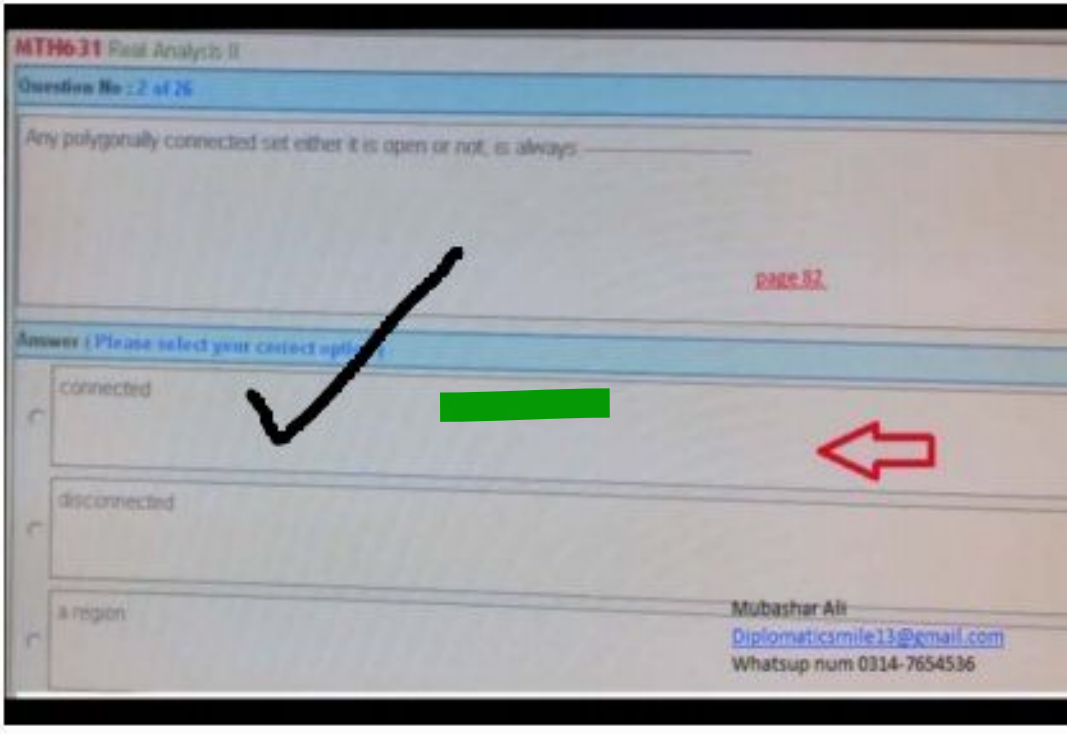
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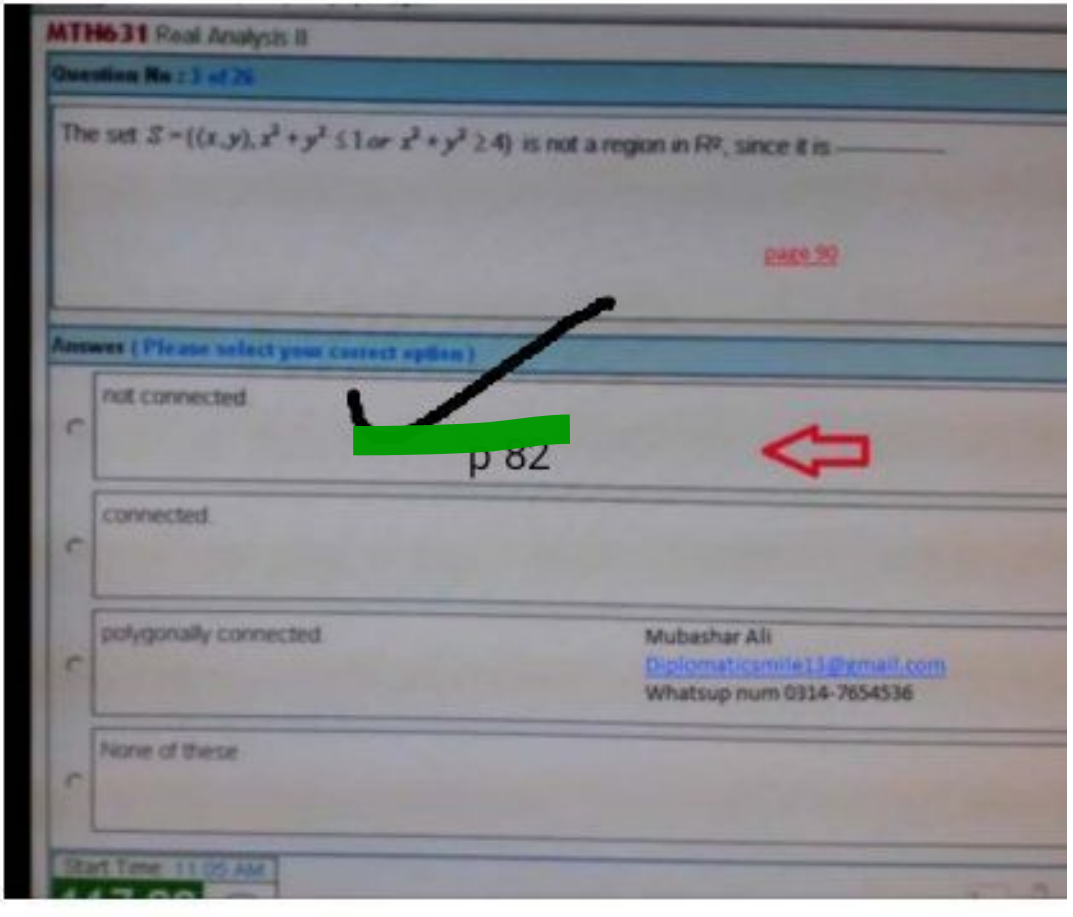
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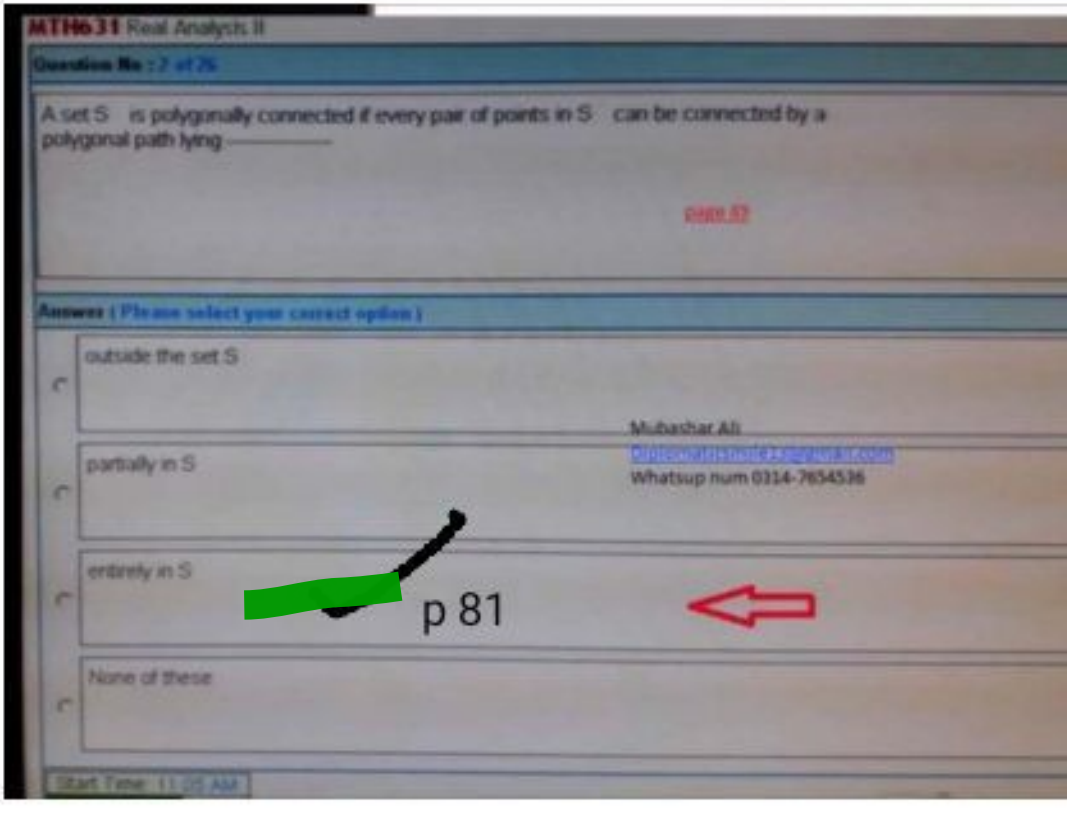
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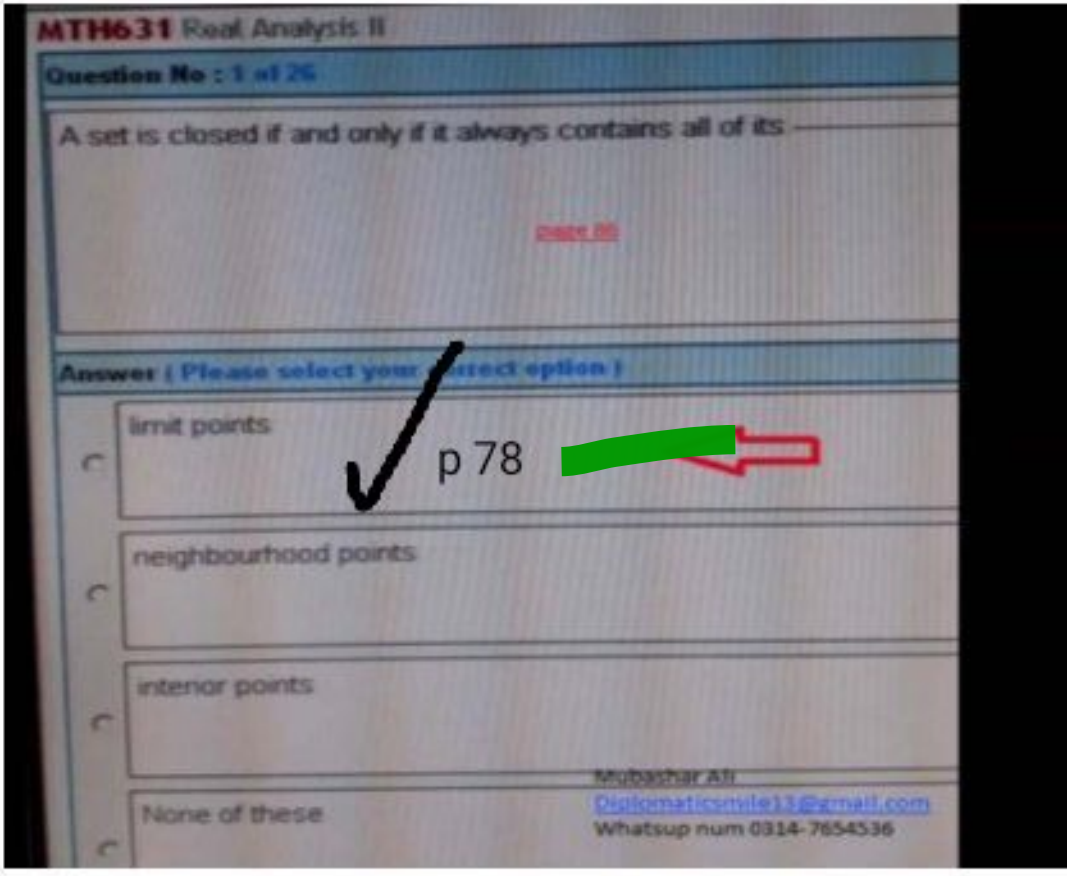
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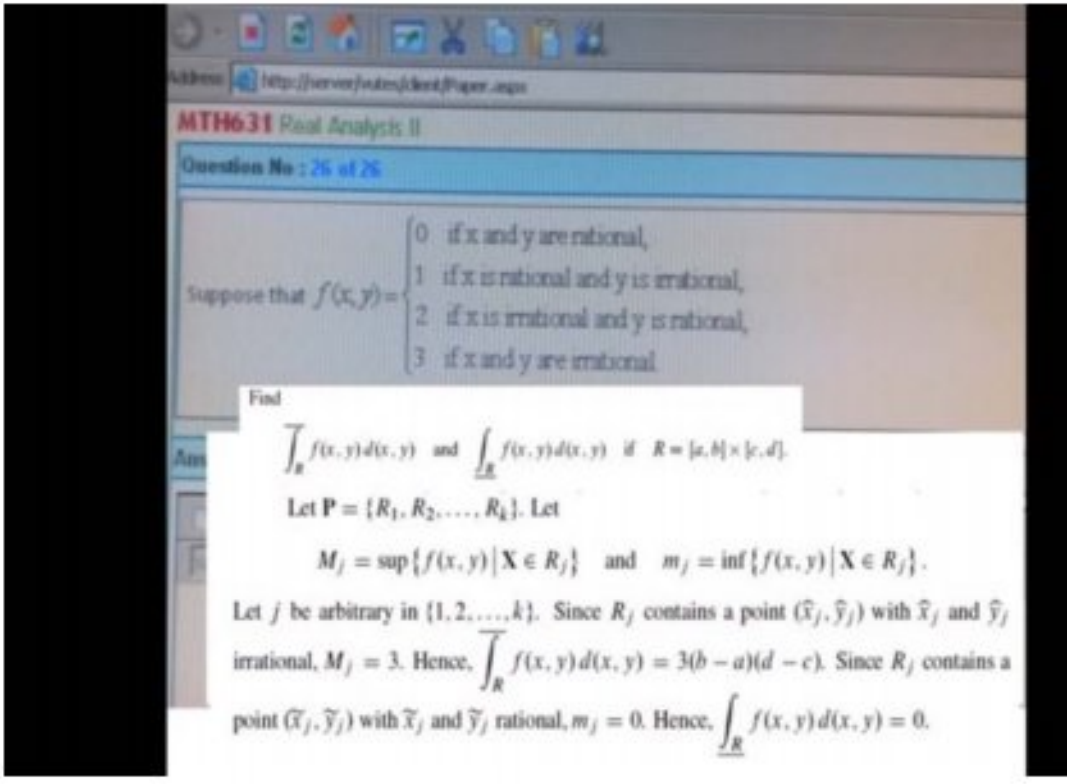
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MTH631 Real Analysis II

Question No : 26 of 26

Suppose that $f(x,y)=\begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational,} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational,} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational,} \\ 3 & \text{if } x \text{ and } y \text{ are irrational.} \end{cases}$

Find

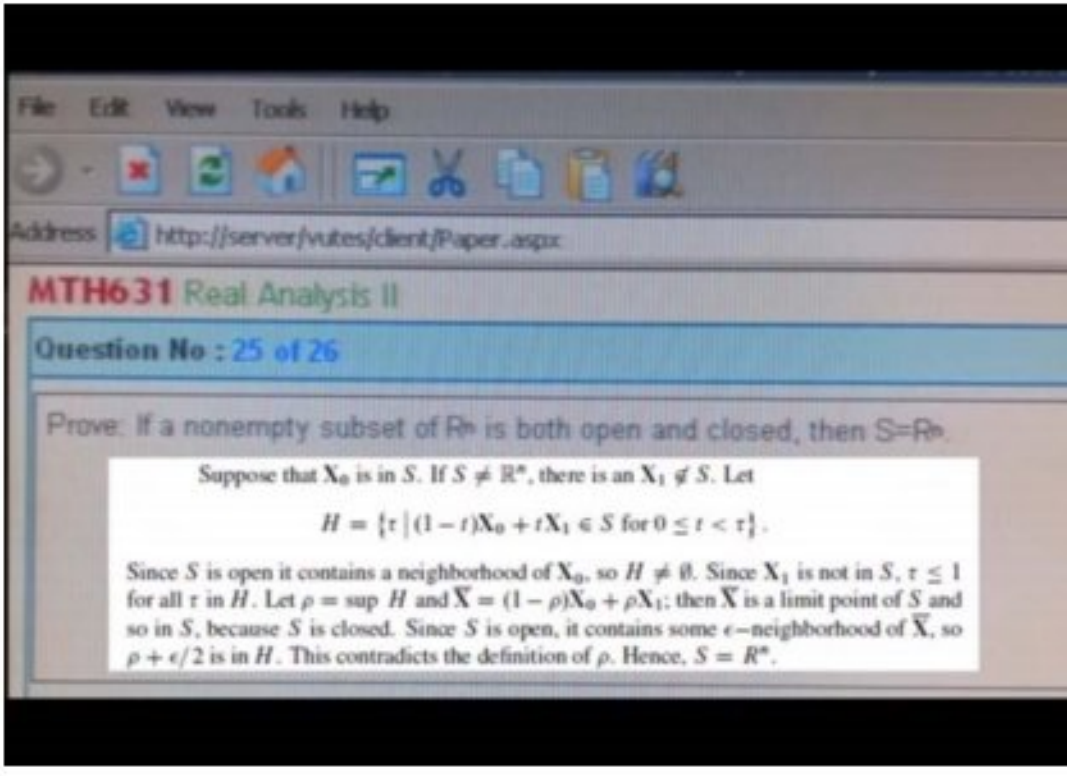
$\overline{\int_R} f(x,y)\,d(x,y)$ and $\underline{\int_R} f(x,y)\,d(x,y)$ if $R=[a,b]\times[c,d]$.

Let $\mathbf{P}=\{R_1,R_2,\ldots,R_k\}$. Let

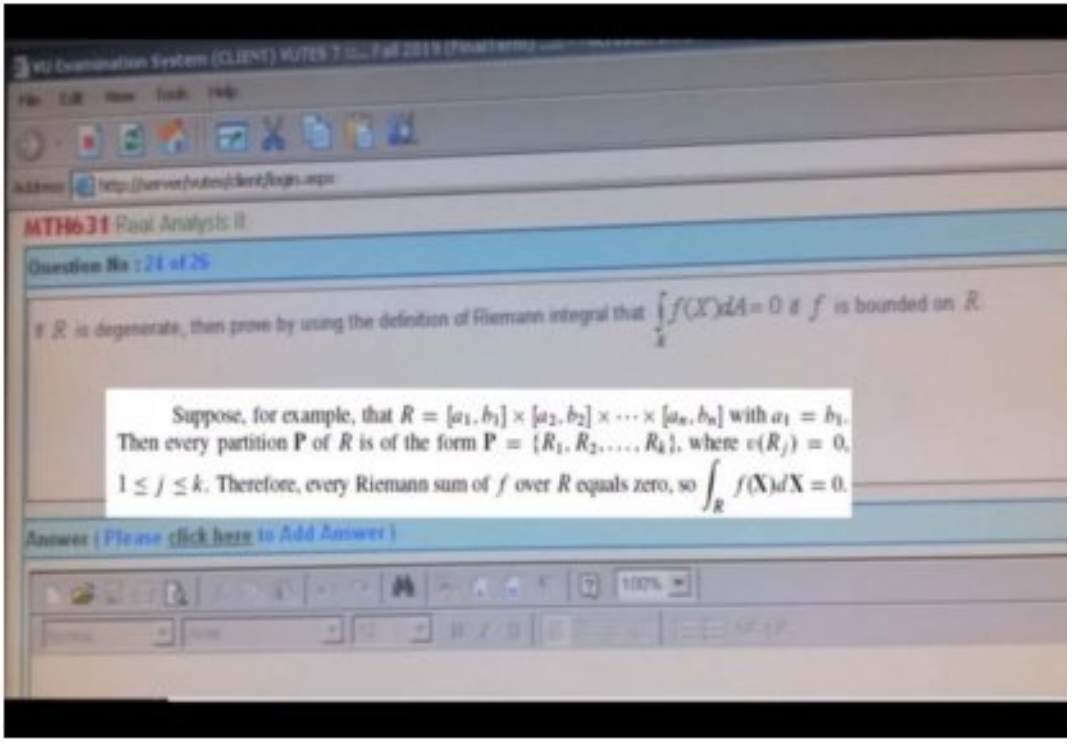
$$M_j=\sup\{f(x,y)\,|\,\mathbf{X}\in R_j\} \quad \text{and} \quad m_j=\inf\{f(x,y)\,|\,\mathbf{X}\in R_j\}.$$

Let j be arbitrary in $\{1,2,\ldots,k\}$. Since R_j contains a point $(\hat{x}_j,\hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational, $M_j=3$. Hence, $\overline{\int_R} f(x,y)\,d(x,y)=3(b-a)(d-c)$. Since R_j contains a point $(\tilde{x}_j,\tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational, $m_j=0$. Hence, $\underline{\int_R} f(x,y)\,d(x,y)=0$.

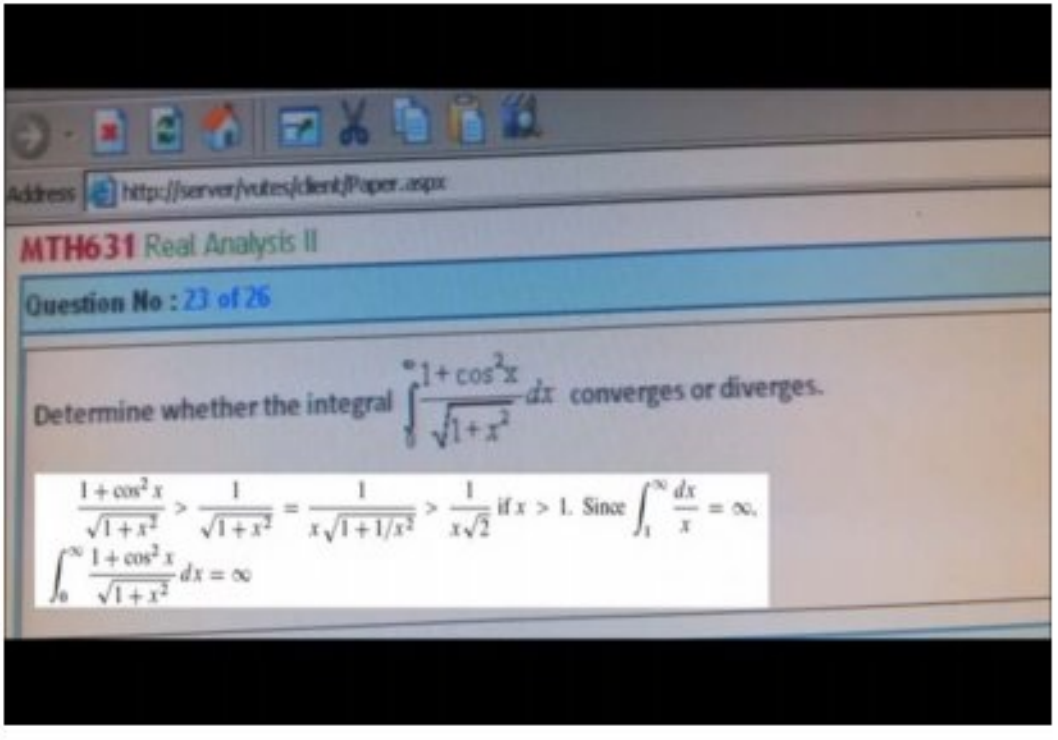
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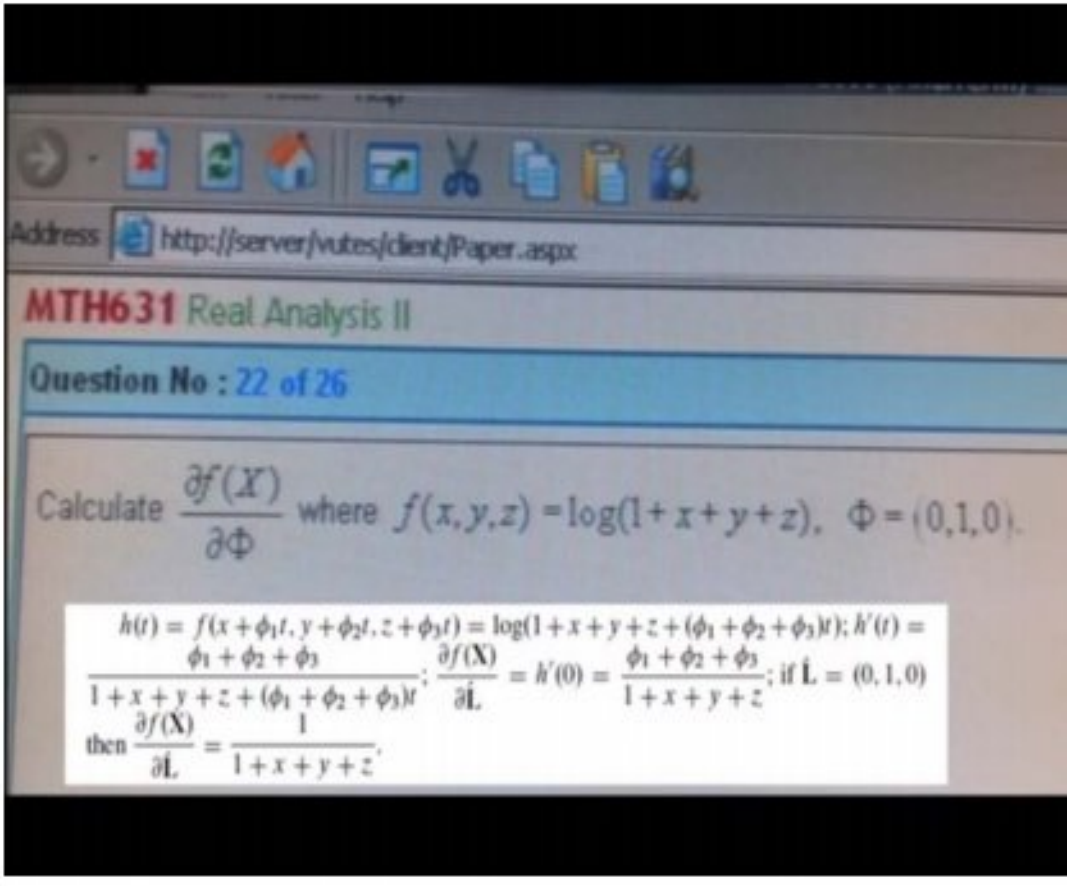
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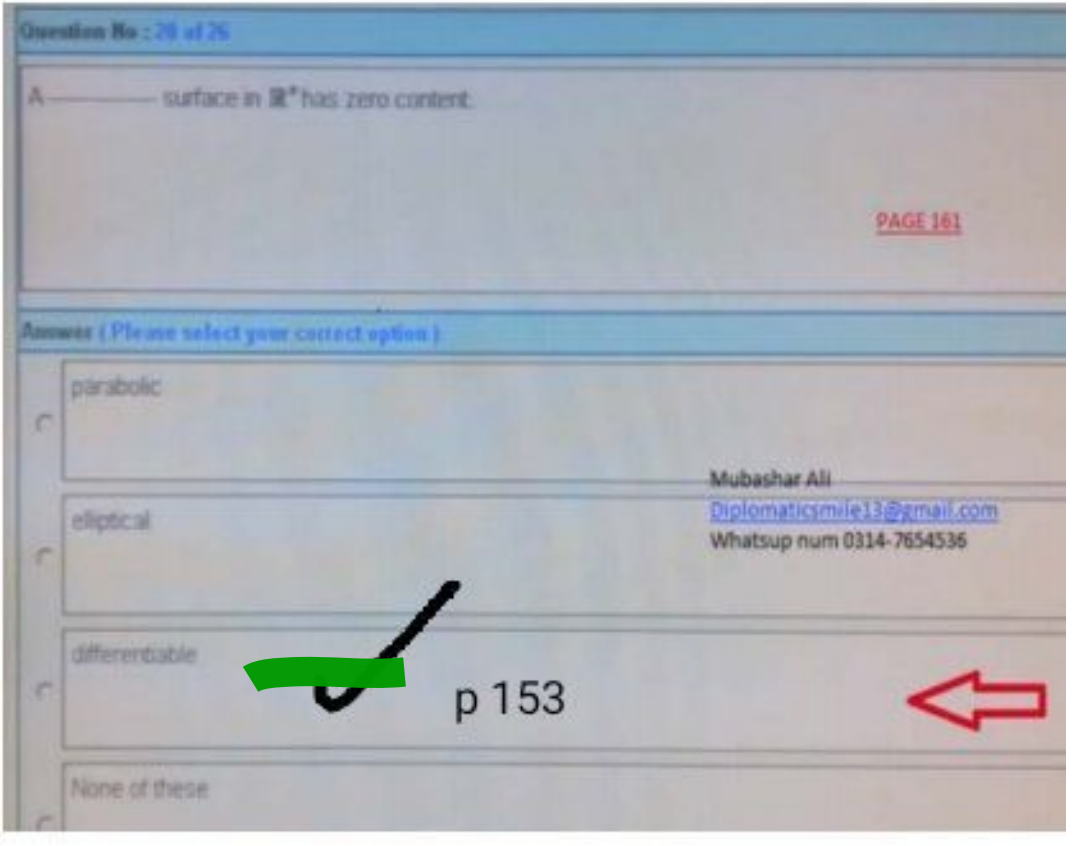
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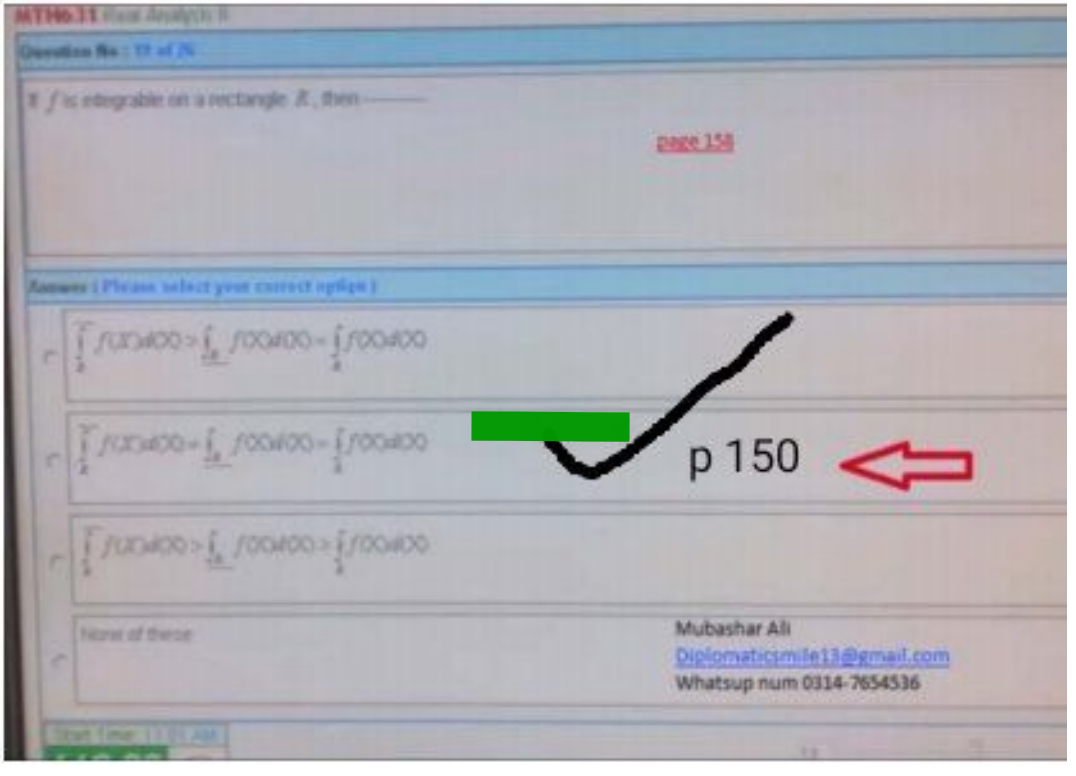
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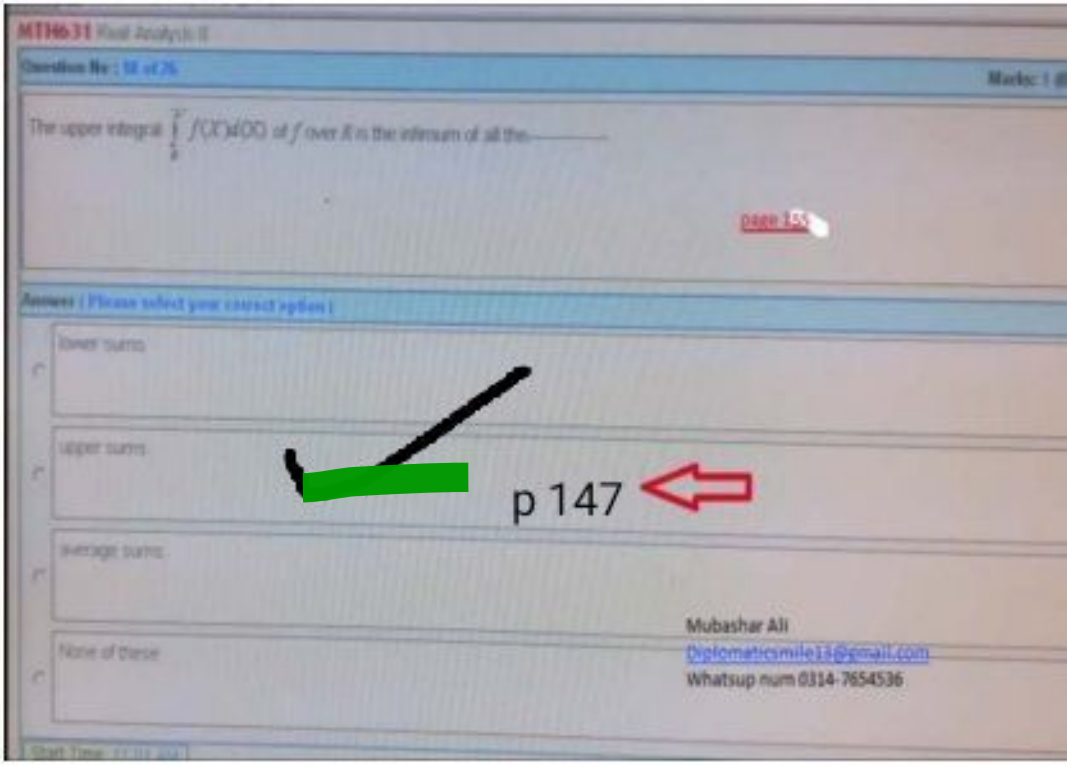
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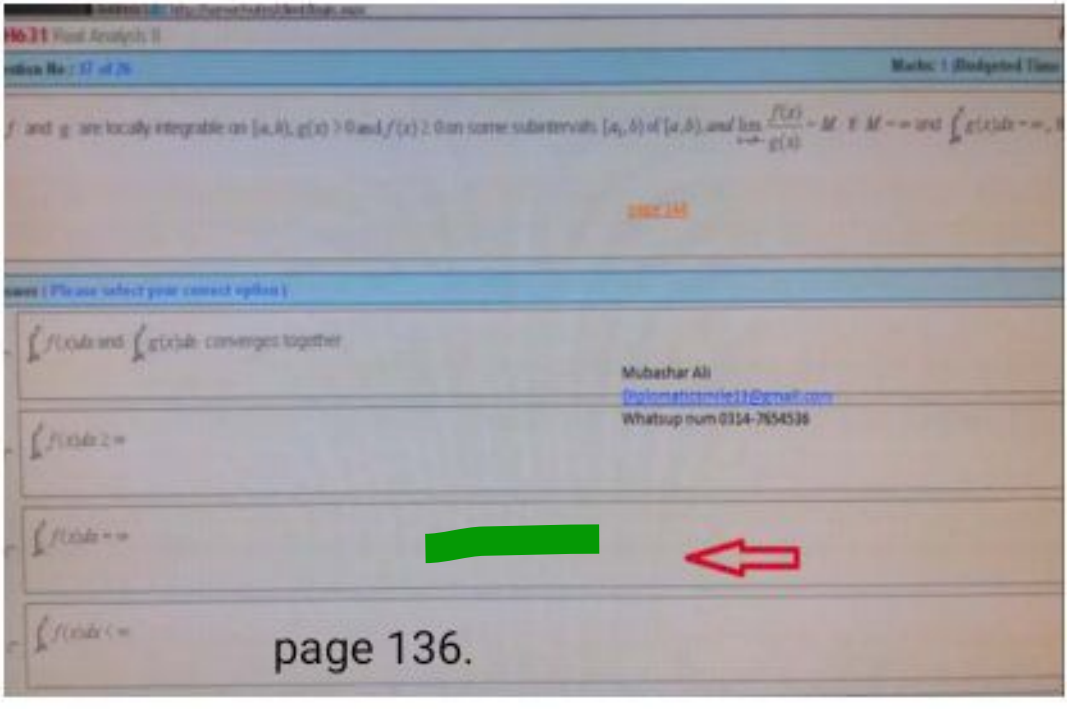
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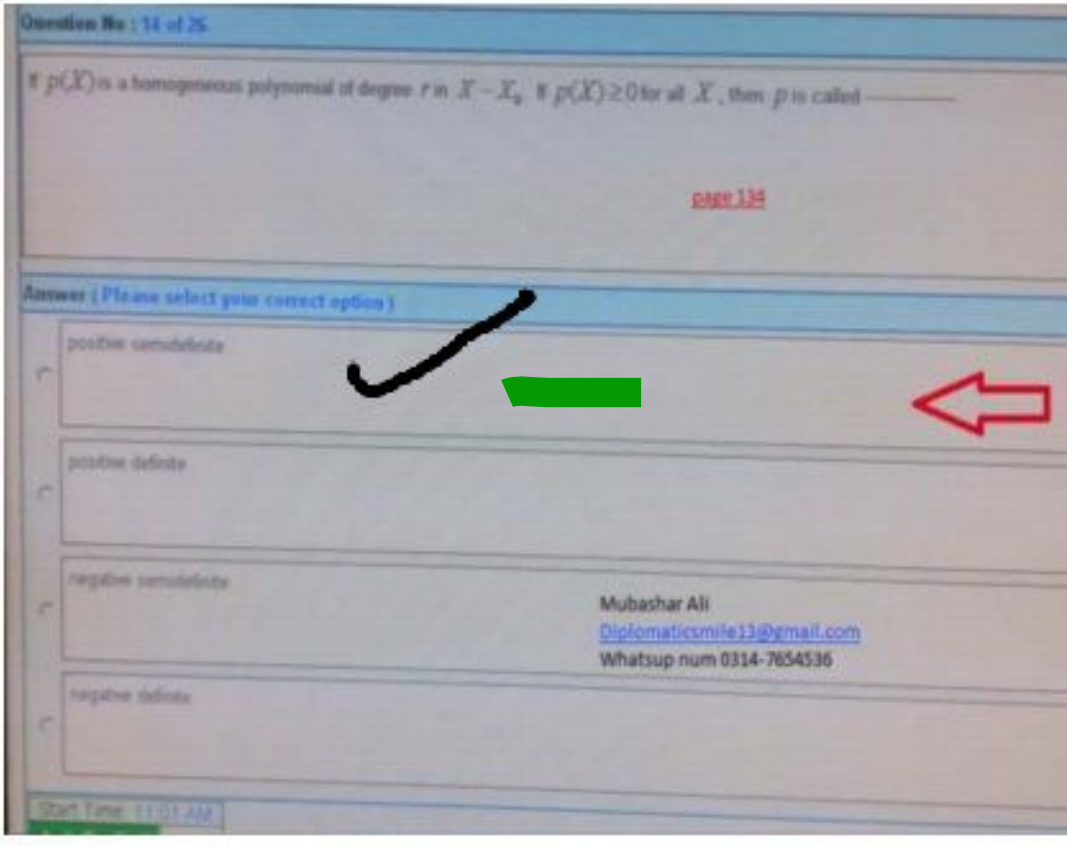
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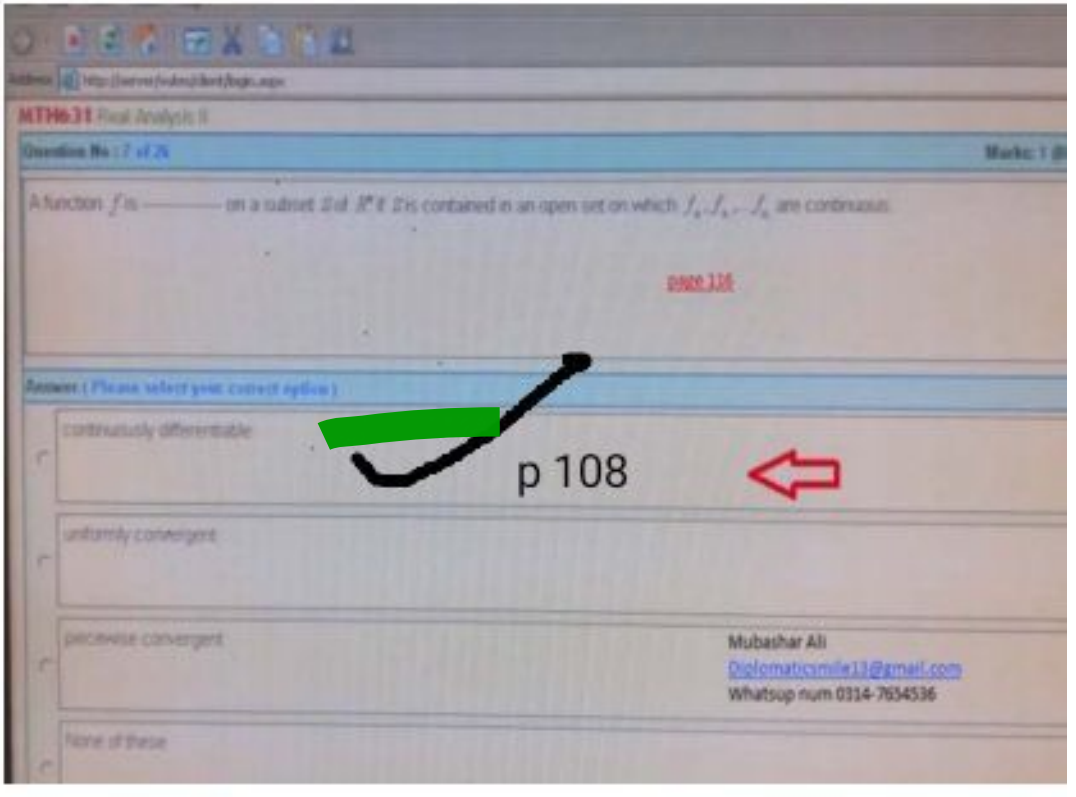
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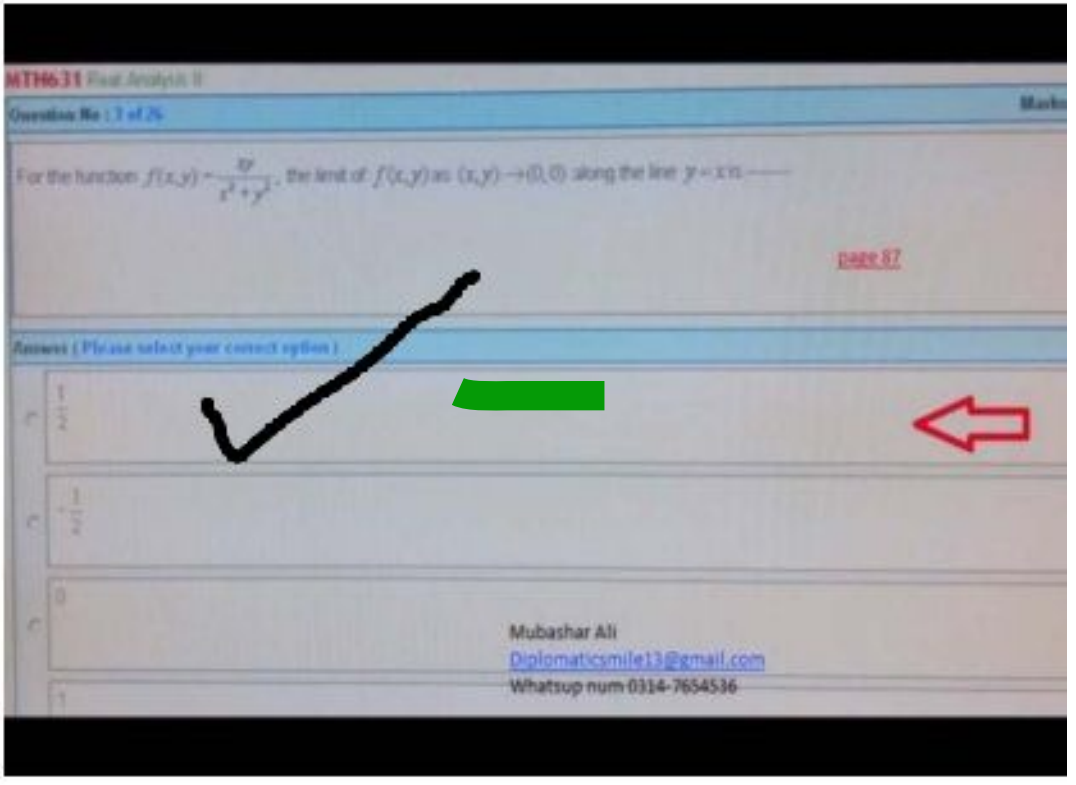
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MTH631 Real Analysis II

Question No : 1 of 26

The singleton set is always —

page 81

Answer (Please select your correct option)

- disconnected
- polygonally disconnected
- connected
- None of these

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MTH631 Real Analysis II

Question No : 2 of 26

Any polygonally connected set either it is open or not, is always _____

page 82

Answer (Please select your correct option)

- connected
- disconnected
- a region
- None of these

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MTH631 Real Analysis II

Question No : 3 of 26 **Marks:** 1 (Budgeted Ti

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ——

page 87

Answer (Please select your correct option)

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

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MTH631 Real Analysis II

Question No : 4 of 26 **Marks:** 1 (Budgeted

If f and g are continuous functions at the point X_0 in R^*, then $\frac{f}{g}$ is continuous at X_0 provided that _____

page 100

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

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MTH631 Real Analysis II

Question No : 5 of 26 **Marks: 1 (Budgeted Ti**

Suppose $U_0 \in T$ is a limit point of T. g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then ——— is continuous at U_0.

page 102

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Answer (Please select yo[illegible]orrect option)

- $f \circ g$
- $\frac{f}{g}$
- $\frac{g}{f}$
- None of these

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MTH631 Real Analysis II

Question No : 6 of 26

Marks: 1 (Budgeted T

A set A ⊆ R of real numbers is ———— if there exists a real number M ⊆ R, such that $x \leq M$ for every $x \in A$.

Answer (Please select your correct option)

- bounded below
- uniformly continuous
- bounded above
- None of these

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MTH631 Real Analysis II

Question No : 7 of 26 **Marks:** 1 (Bud

A function f is ———— on a subset S of R^n if S is contained in an open set on which $f_1, f_2, \ldots f_n$ are continuous

page 116

Answer (Please select your correct option)

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

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MTH631 Real Analysis II

Question No : 8 of 26 **Marks:** 1 (Budgeted

A critical point at which a function attains its minimum value among all points where it is defined is called _____

Answer: (Please select your correct option)

- global maximum
- global minimum
- supremum
- None of these

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MTH631 Real Analysis II

Question No : 18 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants m_i's is called _____

page 112

Answer (Please select your correct option)

- linear function
- quadratic function
- constant function
- None of these

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MTH631 Real Analysis II

Question No : 18 of 25

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants m_i's is called ———

page 112

Answer (Please select your correct option)

- linear function
- quadratic function
- constant function
- None of these

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2.17.1 Linear Function

A *linear function* is a function of the form

$$L(\mathbf{X}) = m_1x_1 + m_2x_2 + \cdots + m_nx_n,$$

where m_1, m_2, ..., m_n are constants. From definition of differentiable at $\mathbf{X}_0$ if and only if there is a linear function L s can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

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MTH631 Real Analysis II

Question No : 9 of 25

For the function $f(x,y) = 3x^3y^3 + xy$ in two variables, $f_{xy}(0,0) =$ .....

Answer (Please select your correct option)

- 1
- -1 correct
- 0
- None of these

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it is an *rth-order partial derivative of* f. The function

$$f(x,y) = 3x^2y^3 + xy$$

has partial derivatives ever[illegible]. It[illegible] first-order partial derivatives are

$$f_x(x,y) = 6xy^3 + y, \quad f_y(x,y) = 9x^2y^2 + x.$$

Its second-order partial derivatives are

$$\begin{aligned} f_{xx}(x,y) &= 6y^3, & f_{yy}(x,y) &= 18x^2y, \\ f_{xy}(x,y) &= 18xy^2 + 1, & f_{yx}(x,y) &= 18xy^2 + 1. \end{aligned}$$

There are eight third-order partial derivatives. Some examples are

$$f_{xxy}(x,y) = 18y^2, \quad f_{xyx}(x,y) = 18y^2, \quad f_{yxx}(x,y) = 18y^2.$$

Compute $f_{xx}(0,0)$, $f_{yy}(0,0)$, $f_{xy}(0,0)$, and $f_{yx}(0,0)$ if

$$f(x,y) = \begin{cases} \frac{(x^2y+xy^2)\sin(x-y)}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

If $(x,y) \neq (0,0)$, the ordinary rules for differentiation, applied sepa and y, yield

$$\begin{aligned} f_x(x,y) &= \frac{(2xy+y^2)\sin(x-y)+(x^2y+xy^2)\cos(x-y)}{x^2+y^2} \\ &\quad - \frac{2x(x^2y+xy^2)\sin(x-y)}{(x^2+y^2)^2}, \quad (x,y) \neq (0,0), \end{aligned}$$

and

$$\begin{aligned} f_y(x,y) &= \frac{(x^2+2xy)\sin(x-y)-(x^2y+xy^2)\cos(x-y)}{x^2+y^2} \\ &\quad - \frac{2y(x^2y+xy^2)\sin(x-y)}{(x^2+y^2)^2}, \quad (x,y) \neq (0,0). \end{aligned}$$

These formulas do not apply if $(x,y) = (0,0)$, so we find $f_x(0,0)$ and $f_$ their definitions as difference quotients:

$$\begin{aligned} f_x(0,0) &= \lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0, \\ f_y(0,0) &= \lim_{y\to 0} \frac{f(0,y) - f(0,0)}{y} = \lim_{y\to 0} \frac{0-0}{y} = 0. \end{aligned}$$

Setting $y = 0$ in (2.32) and (2.33) yields

$$f_x(x,0) = 0, \quad f_y(x,0) = \sin x, \quad x \neq 0,$$

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$$f_y(0,0) = \lim_{y\to 0}\frac{f(0,y)-f(0,0)}{y} = \lim_{y\to 0}\frac{0-0}{y} = 0.$$

Setting $y = 0$ in (2.32) and (2.33) yields

$$f_x(x,0) = 0, \quad f_y(x,0) = \sin x, \quad x \neq 0,$$

2.16. Directional Derivative

so

$$f_{xx}(0,0) = \lim_{x\to 0}\frac{f_x(x,0)-f_x(0,0)}{x} = \lim_{x\to 0}\frac{0-0}{x} = 0.$$

$$f_{yx}(0,0) = \lim_{x\to 0}\frac{f_y(x,0)-f_y(0,0)}{x} = \lim_{x\to 0}\frac{\sin x - 0}{x} = 1.$$

Setting $x = 0$ in (2.32) and (2.33) yields

$$f_x(0,y) = -\sin y, \quad f_y(0,y) = 0, \quad y \neq 0,$$

so

$$f_{xy}(0,0) = \lim_{y\to 0}\frac{f_x(0,y)-f_x(0,0)}{y} = \lim_{y\to 0}\frac{-\sin y - 0}{y} = -1,$$

$$f_{yy}(0,0) = \lim_{y\to 0}\frac{f_y(0,y)-f_y(0,0)}{y} = \lim_{y\to 0}\frac{0-0}{y} = 0.$$

2.16.2 Equality of Mixed Partial Derivatives

Theorem: Suppose that f, f_x, f_y, and f_{xy} exist on a neighborhood N o and f_{xy} is continuous at (x_0, y_0).

Then $f_{yx}(x_0, y_0)$ exists, and

$$f_{yx}(x_0,y_0) = f_{xy}(x_0,y_0).$$

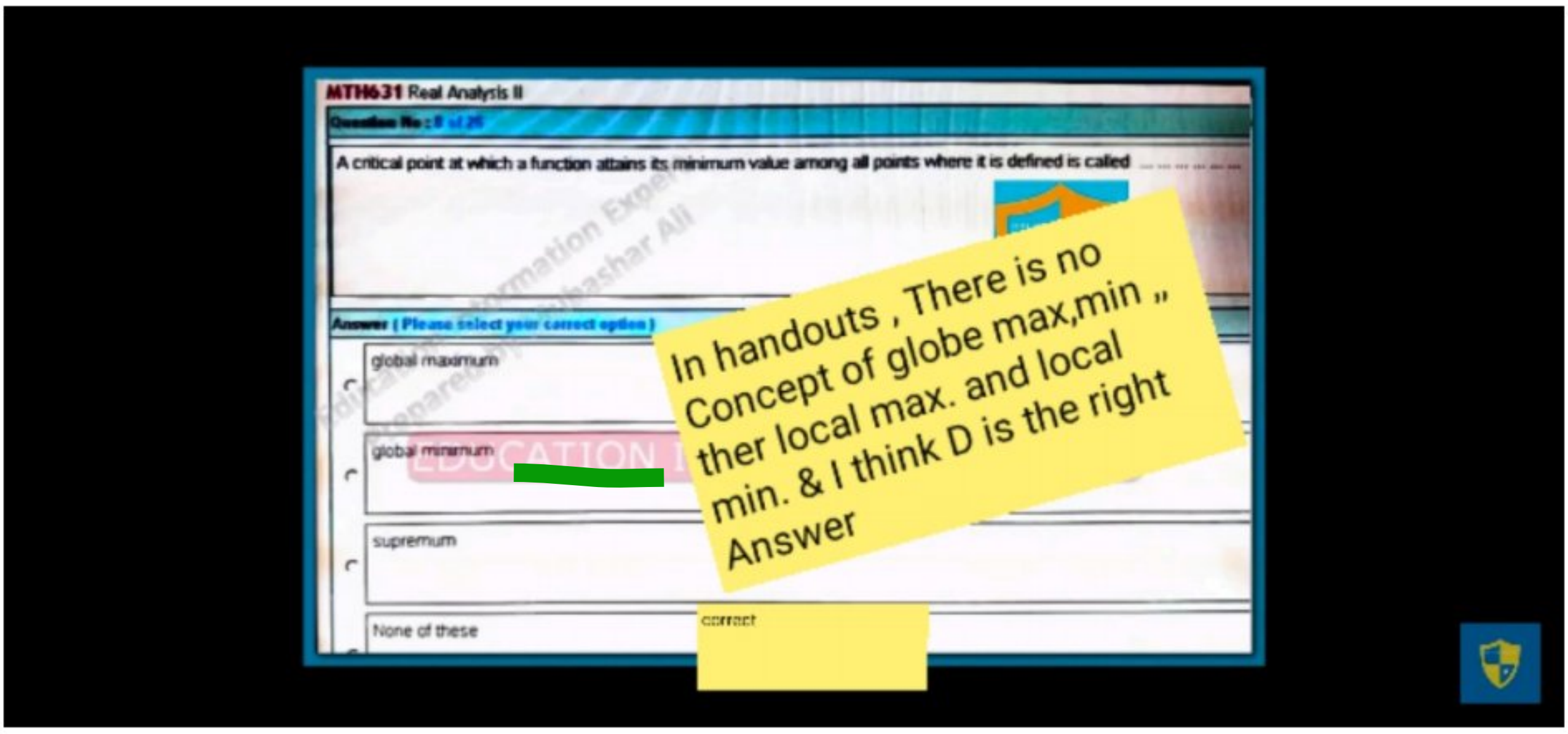
Proof: Suppose that $\varepsilon > 0$. Choose $\delta > 0$ so that the open

$$S_\delta = \{(x,y) : |x-x_0| < \delta, |y-y_0| < \delta\}$$

is in N.

if $|(\hat{x},\hat{y}) - f_{xy}(x_0,y_0)| < \varepsilon$ if $(\hat{x},\hat{y}) \in S_\delta$

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2.18 Maxima and Minima

We say that $\mathbf{X}_0$ is a *local extreme point* of f if there is a $\delta > 0$ such that

$$f(\mathbf{X}) - f(\mathbf{X}_0)$$

does not change sign in $S_\delta(\mathbf{X}_0) \cap D_f$.

More specifically, $\mathbf{X}_0$ is a *local maximum point* if

$$f(\mathbf{X}) \leq f(\mathbf{X}_0)$$

or a *local minimum point* if

$$f(\mathbf{X}) \geq f(\mathbf{X}_0)$$

for all $\mathbf{X}$ in $S_\delta(\mathbf{X}_0) \cap D_f$.

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MTH631 Real Analysis II

Question No : 7 of 26

A $f_n . f_n . \ldots f_n$ exist on a neighborhood of X_0 and are continuous at X_0, then f is ——— at X_0

page 115

Answer (Please select your correct option)

- uniformly converget
- piecewise convergent
- differentiable
- None of these

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2.17.3 A sufficient Condition for Differentiability

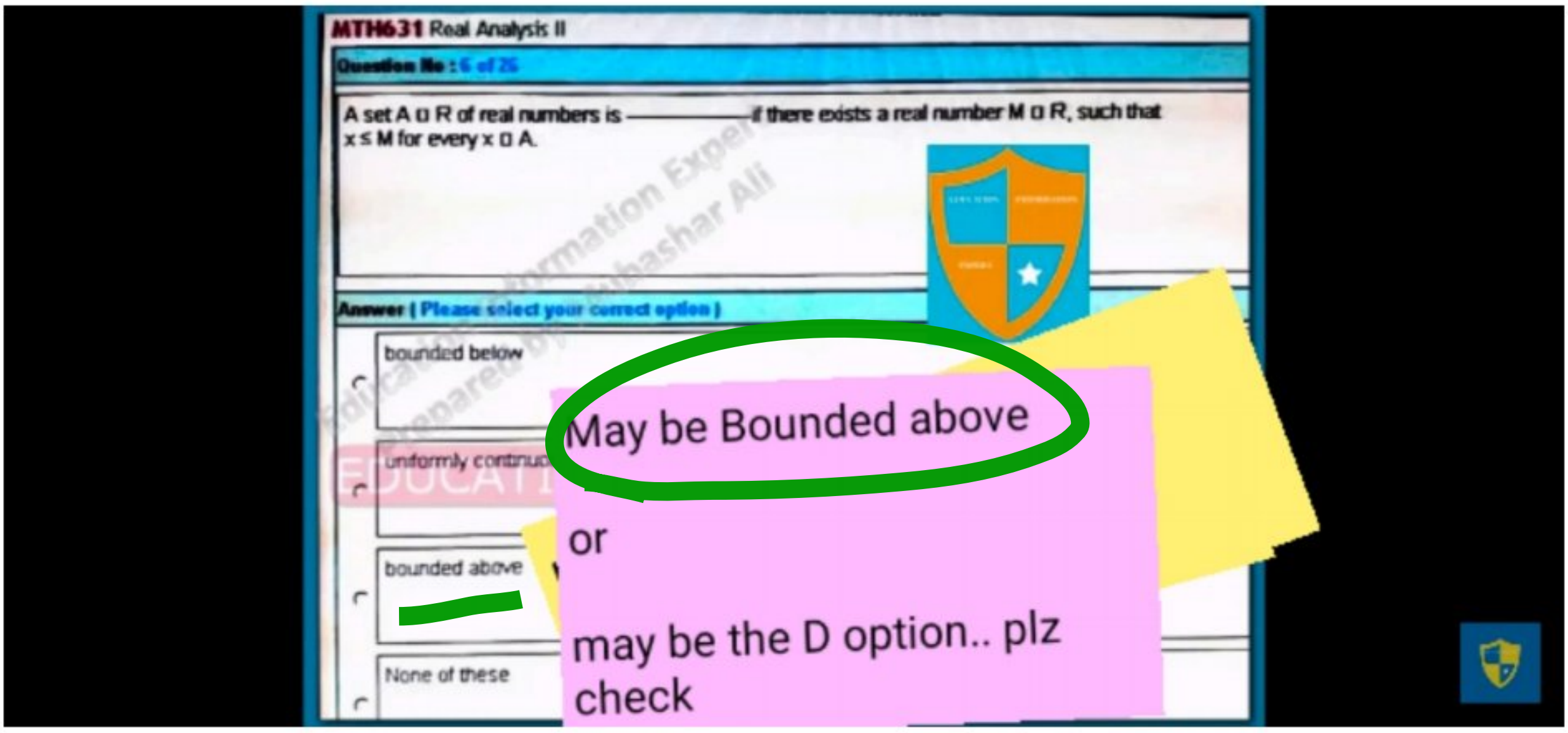
Theorem: If $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$.

Proof: Let $\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$ and suppose that $\varepsilon > 0$. Our assumptions imply that there is a $\delta > 0$ such that $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are defined in the n-ball

$$S_\delta(\mathbf{X}_0) = \{\mathbf{X} : |\mathbf{X} - \mathbf{X}_0| < \delta\}$$

and

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1.12 Pointwise and Uniform Bounded Functions

A sequence of functions $\{F_n\}$ on the set S is said to be pointwise bounded on S if the sequence of functions is bounded for every $x \in S$, that is, if there exists a finite valued function $\phi(x)$ defined on S such that

$$|F_n(x)| < \phi(x), \quad x \in S, n = 1, 2, 3, \ldots.$$

We say that $\{F_n\}$ is uniformly bounded on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, \ldots.$$

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conclude that $f(\mathbf{C}) = u$ for some $\mathbf{C}$ in S.

Theorem: A function f is uniformly continuous on a subset S of its domain in $\mathbb{R}^n$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{X}')| < \varepsilon$$

whenever

$$|\mathbf{X} - \mathbf{X}'| < \delta$$

and $\mathbf{X}, \mathbf{X}' \in S$.

Remark: We emphasize that δ must depend only on ε and S, and not on the particular points $\mathbf{X}$ and $\mathbf{X}'$.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is uniformly continuous on S.

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$|x - p_1| < \delta$.

Since $\{F_n\}$ is pointwise bounded, there exists $M_i < \infty$ such that

$$|F_n(p_i)| < M_i, n \in \mathbb{N}.$$

If we take

$$M = \max\{M_1, ..., M_r\},$$

then $|F_n(x)| < M + \varepsilon$ for every $x \in S$. This proves the first part of the theorem.

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We say that $\{F_n\}$ is uniformly bounded on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, \ldots.$$

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MTH631 Real Analysis II

Question No : 5 of 26

In a vector valued function $G=(g_1,g_2,\ldots,g_n)$. Then $g_1,g_2,\ldots,g_n$ are called the ———— functions of G.

page 114

Answer (Please select your correct option)

- bijective
- surjective
- component
- None of these

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2.19 Differentiable Vector Valued Function

A vector-valued function $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ is *differentiable* at

$$\mathbf{U}_0 = (u_{10}, u_{20}, \ldots, u_{m0})$$

if its component functions $g_1, g_2, \ldots, g_n$ are differentiable at $\mathbf{U}_0$.

Lemma: Suppose that $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ is differentiable at

$$\mathbf{U}_0 = (u_{10}, u_{20}, \ldots, u_{m0}),$$

and define

$$M = \left(\sum_{i=1}^{n} \sum_{j=1}^{m} \left(\frac{\partial g_i(\mathbf{U}_0}{\partial u_j} \right)^2 \right)^{1/2} .$$

Then, if $\epsilon > 0$, there is a $\delta > 0$ such that

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MTH631 Real Analysis II

Question No : 4 of 25

The quotient of two continuous functions is a ———— function wherever the denominator is non-zero.

Answer (Please select your correct option)

- differentiable
- continuous
- composite
- None of these

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MTH631 Real Analysis II

Question No : 3 of 25

The set $S = \{(x,y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in R^2, since it is _____

page 90

Answer (Please select your correct option)

- not connected
- connected
- polygonally connected
- None of these

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Example: Intervals are the only regions in $\mathbb{R}$. The n-ball $B_r(\mathbf{X}_0)$ is a region in $\mathbb{R}^n$, as is its closure $\overline{S}_r(\mathbf{X}_0)$. The set $S = \{(x, y) : x^2 + y^2 \leq 1 \quad \text{or} \quad x^2 + y^2 \geq 4\}$ is not a region in $\mathbb{R}^2$, since it is not connected.

The set S_1 obtained by adding the line segment

$$L_1: \quad \mathbf{X} = t(0, 2) + (1 - t)(0, 1), \quad 0 < t < 1,$$

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MTH631 Real Analysis II

Question No : 2 of 25

A set S is polygonally connected if every pair of points in S can be connected by a polygonal path lying ————

page 89

Answer (Please select your correct option)

- outside the set S
- partially in S
- entirely in S
- None of these

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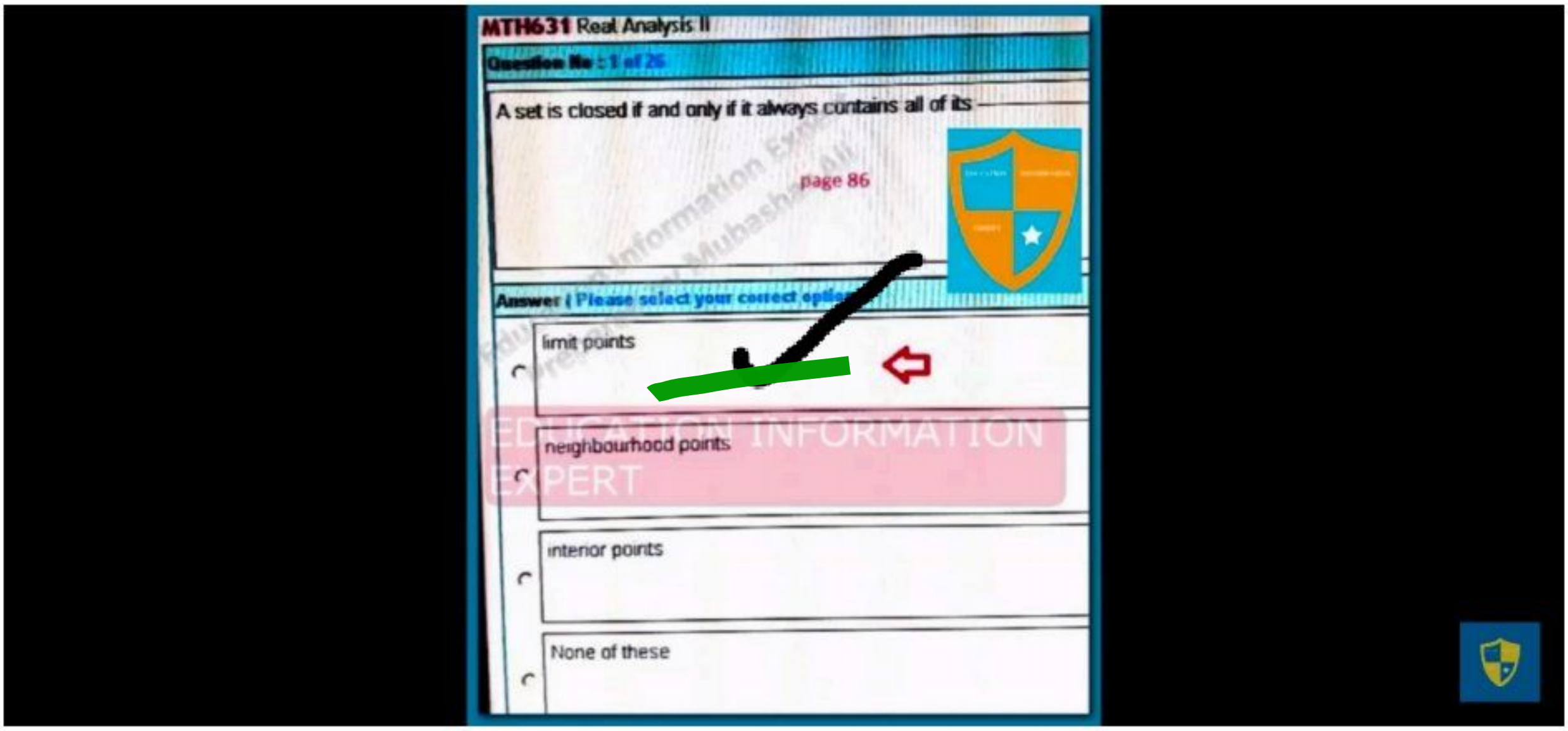
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2.8 Polygonally Connected Set

A set S is *polygonally connected* if every pair of points in S can be connected by a polygonal path lying entirely in S.

Theorem: An open set S in $\mathbb{R}^n$ is connected if and only if it is polygonally connected.

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limit $\mathbf{X}$. Since $\mathbf{X}$ is a limit point of every S_k and every S_k is closed, $\mathbf{X}$ is in e
S_k (**A set is closed if and only if it contains all its limit points**). There
$\overline{\mathbf{X}} \in I$, so $I \neq \emptyset$. Moreover, $\overline{\mathbf{X}}$ is the only point in I, since if $\mathbf{Y} \in I$, then

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MTH631 Real Analysis II

Question No : 19 of 26

The lower integral $\underline{\int_R} f(X)d(X)$ of f over R is the supremum of all the

page 135

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

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$$s(\mathbf{P}) = \sum_{j=1}^{k} m_j V(R_j).$$

The *lower integral of* f *over* R, denoted by

$$\underline{\int_R} f(\mathbf{X})\, d\mathbf{X},$$

is the supremum of all lower sums.

Theorem: Let f be bounded on a rectangle R and le

Then

1. The upper sum $S(\mathbf{P})$ of f over $\mathbf{P}$ is the supremu sums of f over $\mathbf{P}$.

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MTH631 Real Analysis II

Question No : 28 of 26

If f and g are integrable on S and $f(x) \leq g(x)$ for x in S, then ———

page163

Answer (Please select your correct option)

- $\int_S f(x)dx \geq \int_S g(x)dx$
- $\int_S f(x)dx \leq \int_S g(x)dx$
- $\int_S f(x)dx > \int_S g(x)dx$
- None of these

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Theorem: If f and g are integrable on S and $f(\mathbf{X}) \leq g(\mathbf{X})$ for $\mathbf{X}$ in S, then

$$\int_S f(\mathbf{X})\, d\mathbf{X} \leq \int_S g(\mathbf{X})\, d\mathbf{X}.$$

Theorem: If f is integrable on S, then so is $|f|$, and

$$\left|\int_S f(\mathbf{X})\, d\mathbf{X}\right| \leq \int_S |f(\mathbf{X})|\, d\mathbf{X}.$$

Theorem: If f and g are integrable on S, then so is the product fg.

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MTH631 Real Analysis II

Question No : 10 of 25

The upper integral $\overline{\int_R} f(X)d(X)$ of f over R is the infimum of all the———

page 135

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

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3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^{k} M_j V(R_j).$$

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MTH631 Real Analysis II

Question No : 17 of 26

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ———

page 145

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

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3.2 Absolute integrability

We say that f is *absolutely integrable on* $[a,b)$ if f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|\,dx < \infty$. In this case we also say that $\int_a^b f(x)\,dx$ *converges absolutely* or *is absolutely convergent*.

Remark: If f is nonnegative and integrable on $[a,b)$, then f is absolutely integrable on $[a,b)$, since $|f| = f$.

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Example: The empty set and singleton sets are connected, because they cannot represented as the union of two disjoint nonempty sets.

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2.6 Heine-Borel Theorem

We are going to state and prove the Heine-Borel theorem for $\mathbb{R}^n$.

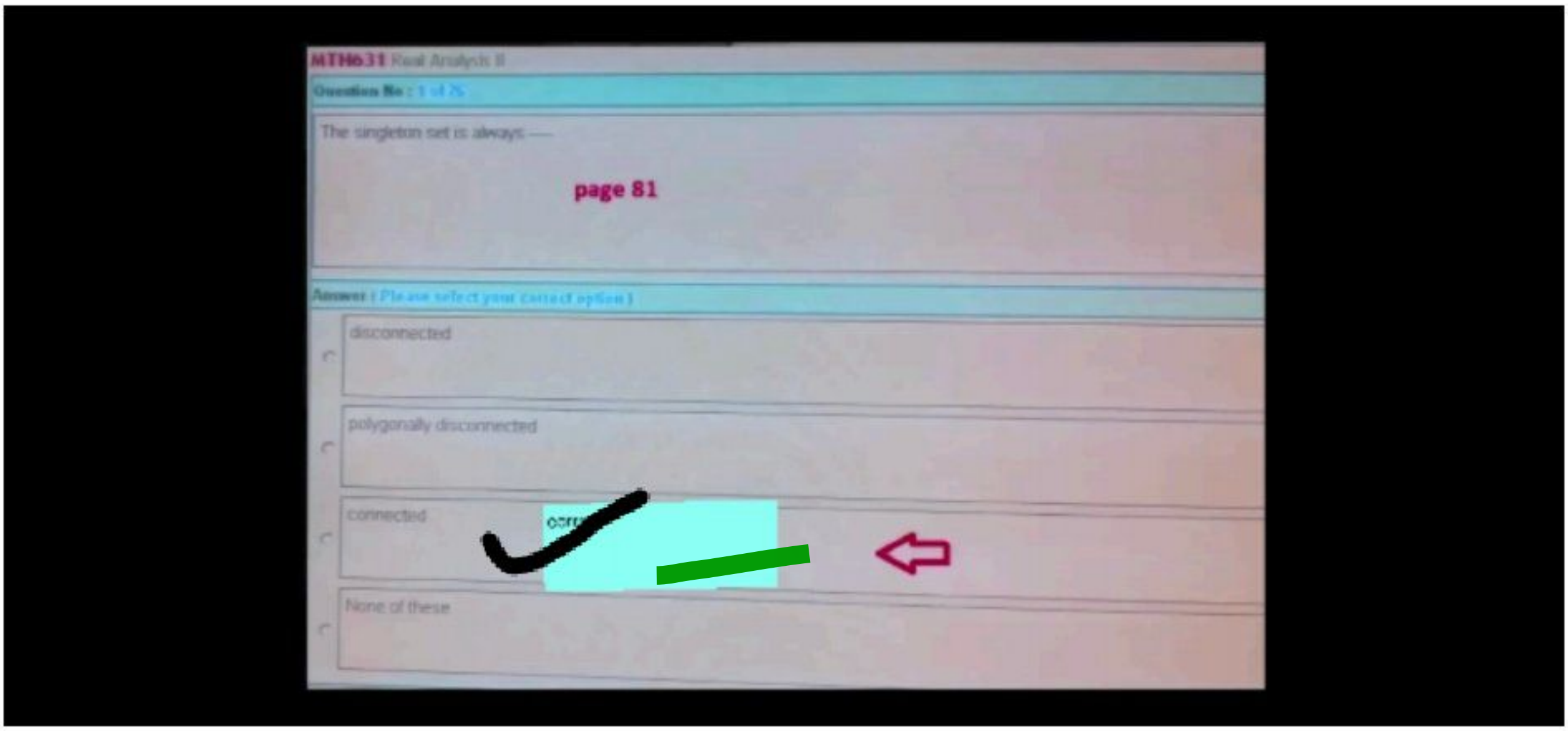
This theorem concerns *compact* sets. As in $\mathbb{R}$, a compact set in $\mathbb{R}^n$ is a closed and bounded set.

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of *supremum* and *infimum* of a function on a set S.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is bounded on S.

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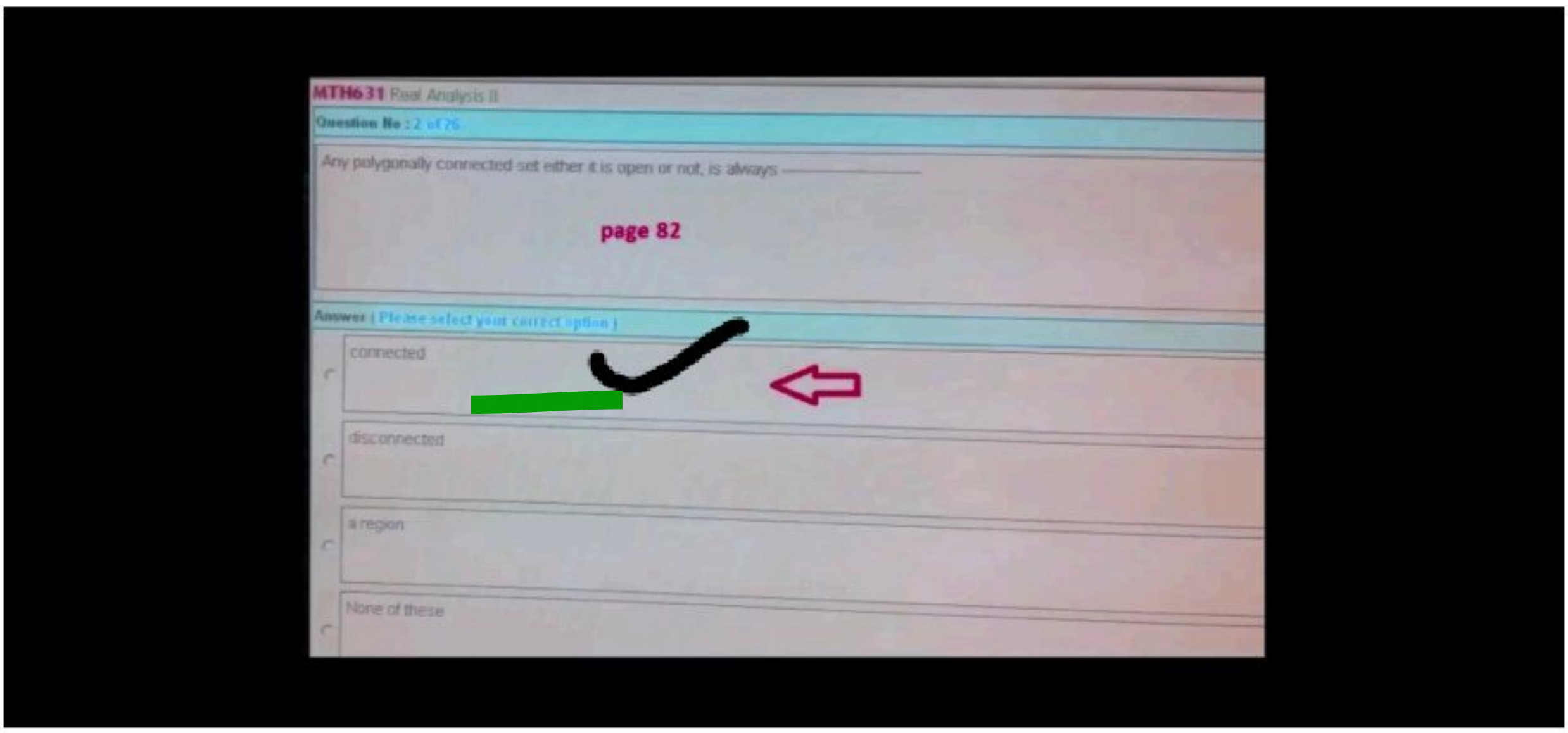


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f S can be expressed in this way, then S is *disconnected*.

ample: The empty set and singleton sets are connected, because
esented as the union of two disjoint nonempty sets.

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$B = \emptyset$, since if $B \neq \emptyset$, S would be
e, $A = S$, which completes the proof of necessity.

: Any polygonally connected set, open or not, is connected. T
A set (not open) may be connected but not polygonally connec

in $\mathbb{R}^n$. A *region* S in $\mathbb{R}^n$ is the union of

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Example: The function

$$f(x,y) = \frac{xy}{x^2+y^2}$$

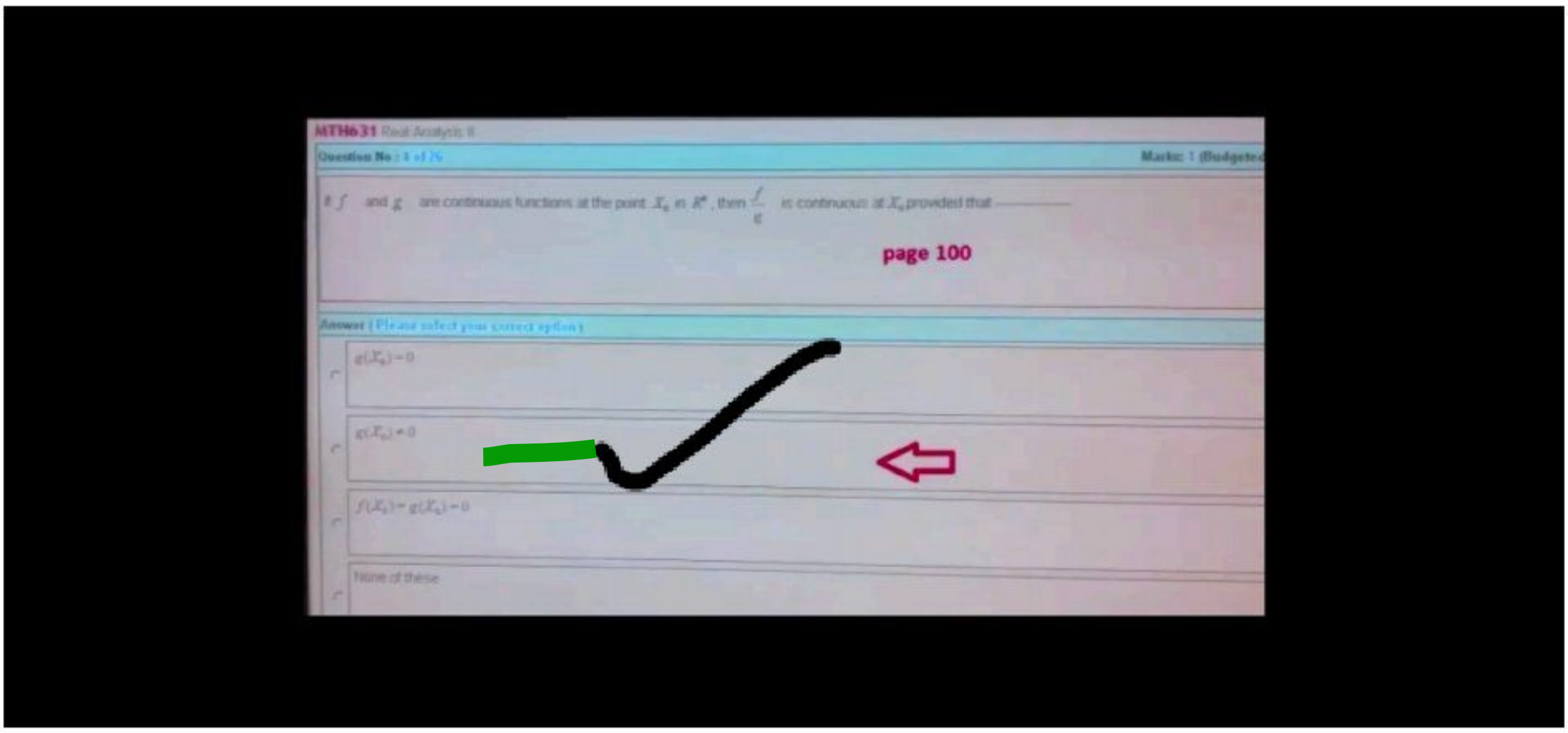
is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

If we try to answer this question by letting (x,y) approach $(0,0)$ along the line $y=x$, we see the functional values

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$$

and conclude that the limit is $1/2$.

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valid for all p and q. The quotient

$$f(x) = \frac{h(x)}{g(x)} \tag{1.51}$$

of two power series

$$\begin{aligned} h(x) &= \sum_{n=0}^{\infty} c_n (x - x_0)^n, \quad |x - x_0| < R_1, \\ g(x) &= \sum_{n=0}^{\infty} b_n (x - x_0)^n, \quad |x - x_0| < R_2, \end{aligned}$$

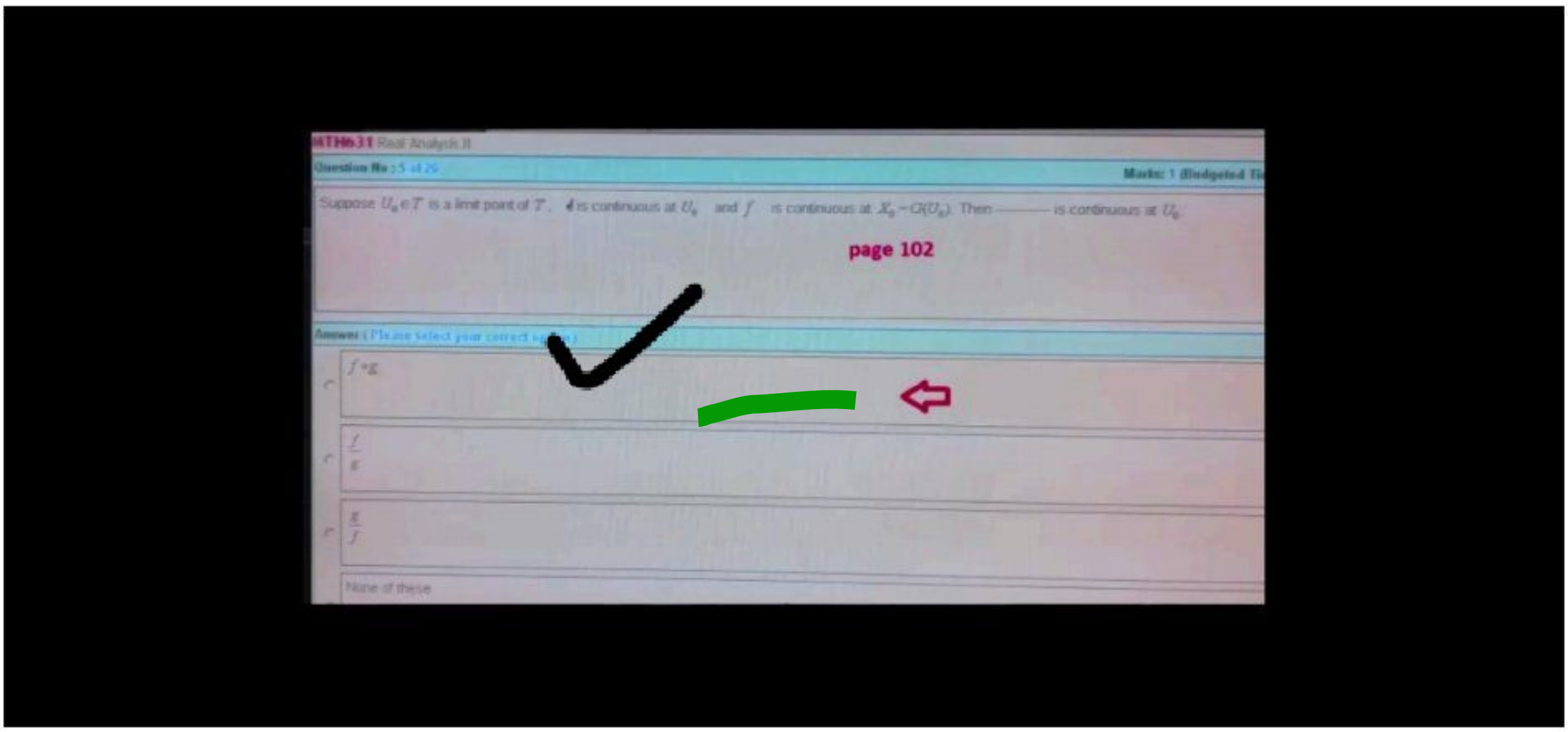
can be represented as a power series

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \tag{1.52}$$

with a positive radius of convergence, provided that

$$b_0 = g(x_0) \neq 0.$$

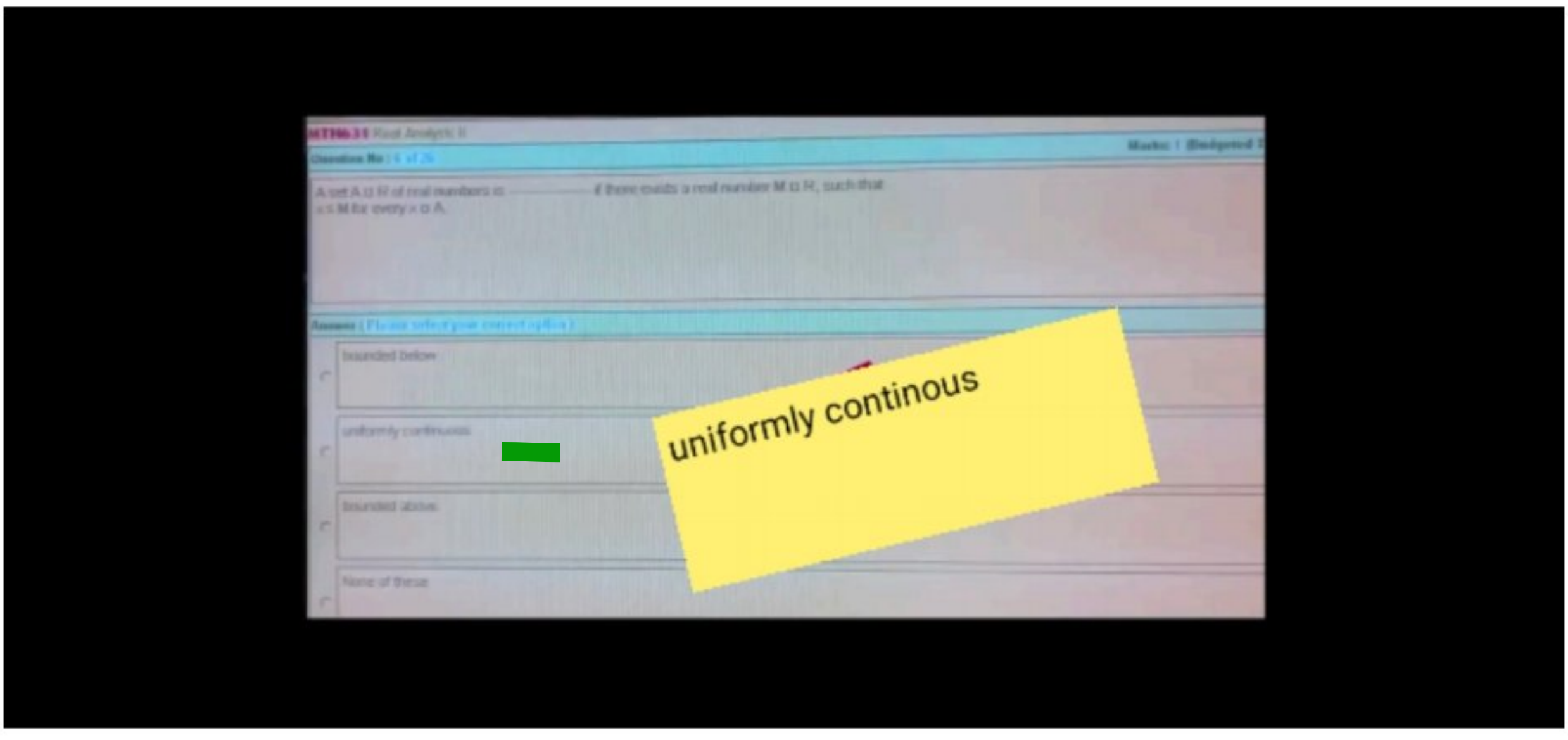
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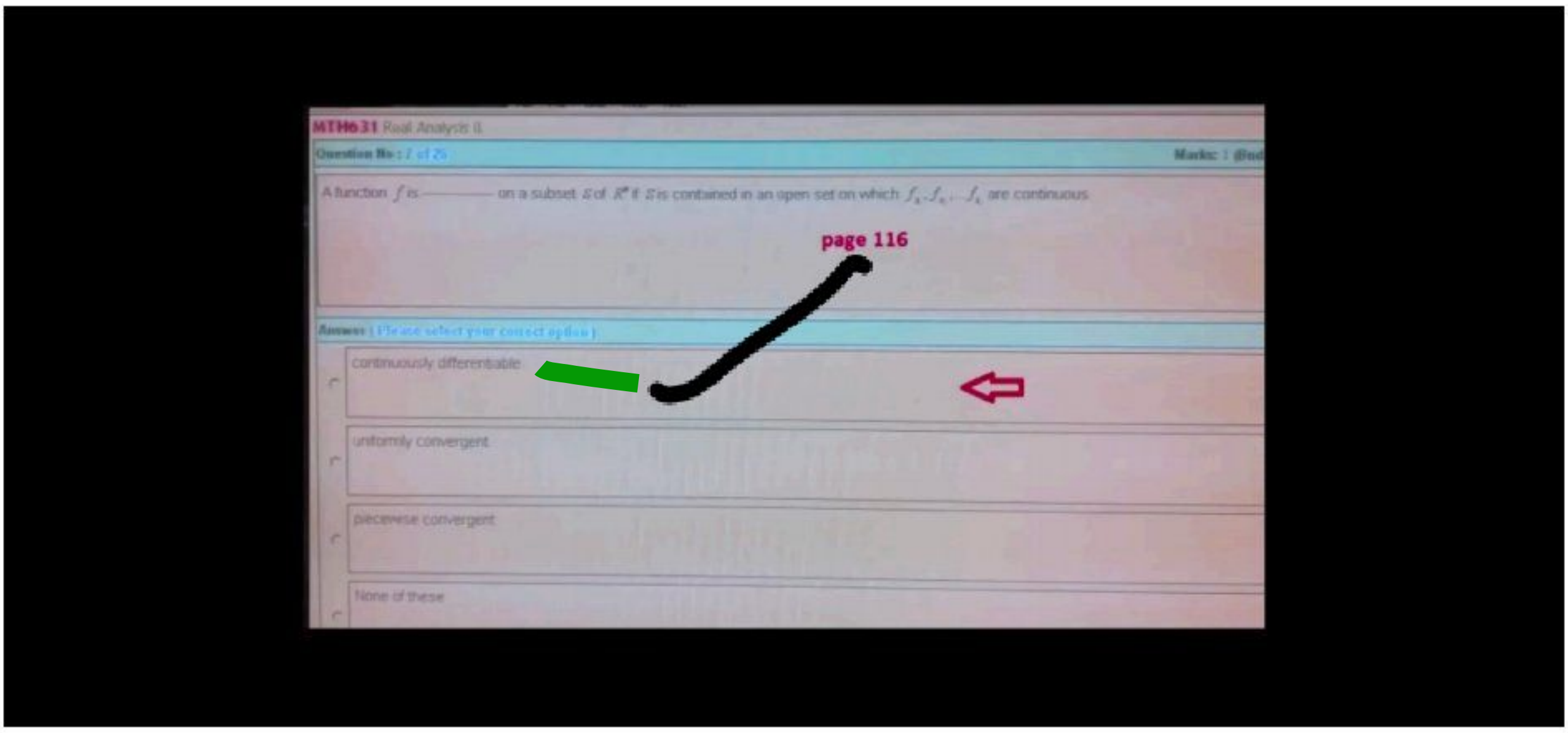
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Theorem: Suppose that $\mathbf{U}_0$ is in T and is a limit point of T, $\mathbf{G}$ is continuous at $\mathbf{U}_0$, and f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$. Then $h = f \circ \mathbf{G}$ is continuous at $\mathbf{U}_0$.

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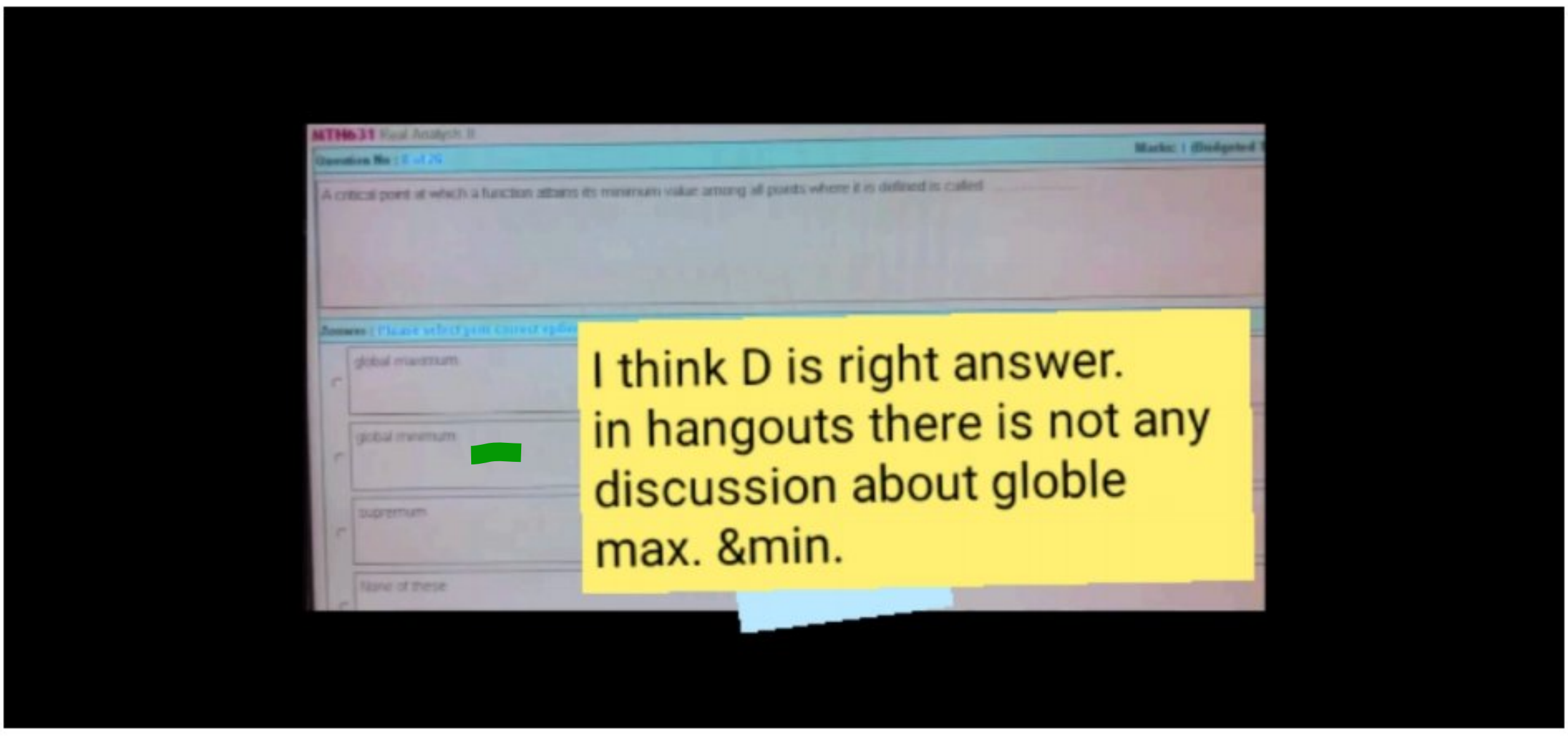
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2.17.4 Continuously Differentiable Function

We say that f is *continuously differentiable* on a subset S of $\mathbb{R}^n$ if S is contained in an open set on which $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are continuous.

The above theorem implies that such a function is differentiable at each $\mathbf{X}_0$ in S.

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Proof: See lecture.

2.16 Directional Derivative

Let $\mathbf{\Phi}$ be a unit vector and $\mathbf{X}$ a point in $\mathbb{R}^n$.

The directional derivative of f at $\mathbf{X}$ in the direction of $\mathbf{\Phi}$ is defined by

$$\frac{\partial f(\mathbf{X})}{\partial \mathbf{\Phi}} = \lim_{t \to 0} \frac{f(\mathbf{X} + t\mathbf{\Phi}) - f(\mathbf{X})}{t}$$

if the limit exists.

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Question 1

Prove that $d(S) = d(\overline{S})$ for any set S in $\mathbb{R}^n$.

Proof:

Since $S \subset \overline{S}$, (A) $d(S) \leq d(\overline{S})$. If $\mathbf{X}$ and $\mathbf{Y}$ are in $\overline{S}$ and $\epsilon > 0$, there are points $\mathbf{X}_1, \mathbf{Y}_1$ in S such that $|\mathbf{X} - \mathbf{X}_1| < \epsilon$ and $|\mathbf{Y} - \mathbf{Y}_1| < \epsilon$. Then

$$|\mathbf{X} - \mathbf{Y}| \leq |\mathbf{X} - \mathbf{X}_1| + |\mathbf{X}_1 - \mathbf{Y}_1| + |\mathbf{Y}_1 - \mathbf{Y}| < d(S) + 2\epsilon.$$

Hence, $d(\overline{S}) < d(S) + \epsilon$. Let $\epsilon \to 0+$ to conclude that $d(\overline{S}) \leq d(S)$, which, with (A), implies that $d(S) = d(\overline{S})$.

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Question 2

If a nonempty subset S of $\mathbb{R}^n$ is both open and closed, then $S = \mathbb{R}^n$.

Prove:

Suppose that $\mathbf{X}_0$ is in S. If $S \neq \mathbb{R}^n$, there is an $\mathbf{X}_1 \notin S$. Let

$$H = \{\tau \mid (1-t)\mathbf{X}_0 + t\mathbf{X}_1 \in S \text{ for } 0 \leq t < \tau\}.$$

Since S is open it contains a neighborhood of $\mathbf{X}_0$, so $H \neq \emptyset$. Since $\mathbf{X}_1$ is not in S, $\tau \leq 1$ for all τ in H. Let $\rho = \sup\ H$ and $\overline{\mathbf{X}} = (1-\rho)\mathbf{X}_0 + \rho\mathbf{X}_1$; then $\overline{\mathbf{X}}$ is a limit point of S and so in S, because S is closed. Since S is open, it contains some $\epsilon-$neighborhood of $\overline{\mathbf{X}}$, so $\rho + \epsilon/2$ is in H. This contradicts the definition of ρ. Hence, $S = R^n$.

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Question 3

$$S = \{(x, y, x) \mid |x| \leq 2, |y| \leq 1, |z-2| \leq 2\}$$

Ans:

$d(S)$ is the supremum of $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

where $-2 \leq x_1, x_2 \leq 2, -1 \leq y_1, y_2 \leq 1, 0 \leq z_1, z_2 \leq 4$;

to maximize this function let, for example,

$x_1 = -2, x_2 = 2, y_1 = -1, y_2 = 1, z_1 = 0, z_2 = 4$; thus, $d(S) = \sqrt{4^2 + 2^2 + 4^2} = 6$.

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Question 4

$$f(x, y, z) = \log(1 + x + y + z), \quad \mathbf{\Phi} = (0, 1, 0)$$

Ans:

$$\begin{aligned} h(t) &= f(x + \phi_1 t, y + \phi_2 t, z + \phi_3 t) \\ &= \log(1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t); \\ h'(t) &= \frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t}; \end{aligned}$$

$$\frac{\partial f(\mathbf{X})}{\partial \acute{\mathbf{L}}} = h'(0) = \frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z}; \text{ if } \acute{\mathbf{L}} = (0, 1, 0)$$

then

$$\frac{\partial f(\mathbf{X})}{\partial \acute{\mathbf{L}}} = \frac{1}{1 + x + y + z}.$$

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Question 5

If $h(u, v) = f(u^2 + v^2)$, then $vh_u - uh_v = 0$.

ANS:

$$h_u = 2uf'$$

$$h_v = 2vf'$$

$$vh_u - uh_v = 2(vu - uv)f'$$

$$vh_u - uh_v = 0.$$

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Question 6

If $h(u, v) = f(\sin u + \cos v)$, then $h_u \sin v + h_v \cos u = 0$.

Ans:

$$h_u = f' \cos u$$
$$h_v = -f' \sin v$$
$$h_u \sin v + h_v \cos u = (\sin v \cos u - \cos u \sin v) f'$$
$$h_u \sin v + h_v \cos u = 0.$$

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Question 7

If $h(u, v) = f(u/v)$, then $uh_u + vh_v = 0$.

Ans:

$$h_u = \frac{1}{v} f'$$

$$h_v = -\frac{u}{v^2} f'$$

$$uh_u + vh_v = \left(\frac{u}{v} - \frac{u}{v}\right) f'$$

$$uh_u + vh_v = 0.$$

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Question 8

If $h(u, v) = f(g(u, v), -g(u, v))$, then $dh = (f_x - f_y)\, dg$.

Ans:

$$h_u = (f_x - f_y)g_u$$

$$h_v = (f_x - f_y)g_v$$

$$dh = h_u\, du + h_v\, dv$$

$$dh = (f_x - f_y)(g_u\, du + g_v\, dv)$$

$$dh = (f_x - f_y)\, dg.$$

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Question 9

Let $h(r, \theta, z) = f(x, y, z)$, where $x = r\cos\theta$ and $y = r\sin\theta$.

Find h_r, h_θ, and h_z in terms of f_x, f_y, and f_z.

Ans:

$$h_r = f_x x_r + f_y y_r + f_z z_r$$

$$h_r = f_x \cos\theta + f_y \sin\theta$$

$$h_\theta = f_x x_\theta + f_y y_\theta + f_z z_\theta$$

$$h_\theta = f_x(-r\sin\theta) + f_y(r\cos\theta)$$

$$h_\theta = r(-f_x \sin\theta + f_y \cos\theta)$$

$$h_z = f_x x_z + f_y y_z + f_z = f_z.$$

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Question 10

Let $h(r, \theta, \phi) = f(x, y, z)$, where $x = r \sin\phi \cos\theta$, $y = r \sin\phi \sin\theta$, and $z = r \cos\phi$. Find h_r, h_θ, and h_ϕ in terms of f_x, f_y, and f_z.

ANS:

$$h_r = f_x x_r + f_y y_r + f_z z_r$$

$$h_r = f_x \sin\phi \cos\theta + f_y \sin\phi \sin\theta + f_z \cos\phi$$

$$h_\theta = f_x x_\theta + f_y y_\theta + f_z z_\theta$$

$$h_\theta = r \sin\phi(-f_x \sin\theta + f_y \cos\theta)$$

$$h_\phi = f_x x_\phi + f_y y_\phi + f_z z_\phi$$

$$h_\phi = r(f_x \cos\phi \cos\theta + f_y \cos\phi \sin\theta - f_z \sin\phi.$$

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Question 11

Let $h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U}))$ and find $d_{\mathbf{U}_0}h$ by writing h explicitly as a function of $\mathbf{U}$.

$$\begin{aligned} f(x, y) &= 3x^2 + 4xy^2 + 3x, \\ g_1(u, v) &= ve^{u+v-1}, \\ g_2(u, v) &= e^{-u+v-1}, \end{aligned} \qquad (u_0, v_0) = (0, 1)$$

Ans:

$$\begin{aligned} h(\mathbf{U}) &= 3v^2e^{2u+2v-2} + 4ve^{u+v-1}e^{-2u+2v-2} + 3ve^{u+v-1} \\ &= 3v^2e^{2u+2v-2} + 4ve^{-u+3v-3} + 3ve^{u+v-1} \\ h_u(\mathbf{U}) &= 6v^2e^{2u+2v-2} - 4ve^{-u+3v-3} + 3ve^{u+v-1} \\ h_u(\mathbf{U}_0) &= 5 \\ h_v(\mathbf{U}) &= (6v + 6v^2)e^{2u+2v-2} + (4 + 12v)e^{-u+3v-3} + (3 + 3v)e^{u+v-1} \\ h_v(\mathbf{U}_0) &= 34 \\ d_{\mathbf{U}_0}h &= h_u(\mathbf{U}_0)\,du + h_v(\mathbf{U}_0)\,dv = 5\,du + 34\,dv \end{aligned}$$

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Question 12

Let $h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U}))$ and find $d_{\mathbf{U}_0}h$ by writing h explicitly as a function of $\mathbf{U}$.

$$
\begin{aligned}
f(x, y, z) &= e^{-(x+y+z)}, \\
g_1(u, v, w) &= \log u - \log v + \log w, \\
g_2(u, v, w) &= -2\log u - 3\log w, \\
g_3(u, v, w) &= \log u + \log v + 2\log w,
\end{aligned}
\qquad (u_0, v_0, w_0) = (1, 1, 1)
$$

Ans:

$$x(\mathbf{U}) + y(\mathbf{U}) + z(\mathbf{U}) = (\log u - \log v + \log w) + (-2\log u - 3\log w) + (\log u + \log v + 2\log w)$$

$$x(\mathbf{U}) + y(\mathbf{U}) + z(\mathbf{U}) = 0$$

hence

$$h(\mathbf{U}) \equiv 1$$

$$d_{\mathbf{U}_0}h = 0.$$

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Question 13

Let $h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U}))$ and find $d_{\mathbf{U}_0}h$ by writing h explicitly as a function of $\mathbf{U}$.

$$f(x, y) = (x + y)^2,$$
$$g_1(u, v) = u \cos v, \qquad (u_0, v_0) = (3, \pi/2)$$
$$g_2(u, v) = u \sin v,$$

Ans:

$$h(\mathbf{U}) = u^2(\cos v + \sin v)^2$$
$$= u^2(1 + 2 \sin v \cos v)$$
$$= u^2(1 + \sin 2v)$$
$$h_u(\mathbf{U}) = 2u(1 + \sin 2v)$$
$$h_u(\mathbf{U}_0) = 6$$
$$h_v(\mathbf{U}) = 2u^2 \cos 2v;$$
$$h_v(\mathbf{U}_0) = -18$$
$$d_{\mathbf{U}_0}h = h_u(\mathbf{U}_0)\,du + h_v(\mathbf{U}_0)\,dv$$
$$d_{\mathbf{U}_0}h = 6\,du - 18\,dv.$$

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Question 14

Let $h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U}))$ and find $d_{\mathbf{U}_0} h$ by writing h explicitly as a function of $\mathbf{U}$.

$$
\begin{aligned}
f(x, y, z) &= x^2 + y^2 + z^2, \\
g_1(u, v, w) &= u \cos v \sin w, \\
g_2(u, v, w) &= u \cos v \cos w, \\
g_3(u, v, w) &= u \sin v;
\end{aligned}
\qquad (u_0, v_0, w_0) = (4, \pi/3, \pi/6)
$$

Ans:

$$h(\mathbf{U}) = u^2 \cos^2 v \sin^2 w + u^2 \cos^2 v \cos^2 w + u^2 \sin^2 v$$

$$h(\mathbf{U}) = u^2$$

$$h_u(\mathbf{U}) = 2u$$

$$h_v(\mathbf{U}) = 0$$

$$h_w(\mathbf{U}) = 0$$

$$h_u(\mathbf{U}_0) = 8;$$

$$h_v(\mathbf{U}_0) = 0;$$

$$h_w(\mathbf{U}_0) = 0$$

$$dh = 8\,du.$$

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Question 15

Show that $(0, 0)$ is a critical point of each of the following functions, and that they have positive semidefinite second differentials at $(0, 0)$.

$$p(x, y) = x^2 - 2xy + y^2 + x^4 + y^4;$$

Ans:

$$p(x, y) = x^2 - 2xy + y^2 + x^4 + y^4$$

$$p_x(x, y) = 2x - 2y + 4x^3$$

$$p_y(x, y) = -2x + 2y + 4y^3.$$

Since $p_x(0, 0) = p_y(0, 0) = 0$, $(0, 0)$ is a critical point of p.

$$p_{xx}(x, y) = 2 + 12x^2$$

$$p_{xy}(x, y) = -2$$

$$p_{yy}(x, y) = 2 + 12y^2$$

$$(d^{(2)}_{(0,0)} p)(x, y) = 2x^2 - 4xy + 2y^2$$

$(d^{(2)}_{(0,0)} p)(x, y) = 2(x - y)^2$ is positive semidefinite.

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Question 16

Show that $(0, 0)$ is a critical point of each of the following functions, and that they have positive semidefinite second differentials at $(0, 0)$.

$$q(x, y) = x^2 - 2xy + y^2 - x^4 - y^4.$$

Ans:

$$q(x, y) = x^2 - 2xy + y^2 - x^4 - y^4$$

$$q_x(x, y) = 2x - 2y - 4x^3$$

$$q_y(x, y) = -2x + 2y - 4y^3$$

Since $q_x(0,0) = q_y(0,0) = 0$, $(0, 0)$ is a critical point of q.

$$q_{xx}(x, y) = 2 - 12x^2$$

$$q_{xy}(x, y) = -2$$

$$q_{yy}(x, y) = 2 - 12y^2$$

$$(d^{(2)}_{(0,0)}q)(x, y) = 2x^2 - 4xy + 2y^2$$

$(d^{(2)}_{(0,0)}q)(x, y) = 2(x - y)^2$ is positive semidefinite.

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Question 17

If R is degenerate, then implies that $\int_R f(\mathbf{X})\,d\mathbf{X} = 0$ if f is bounded on R.

Proof:

Suppose, for example, that $R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ with $a_1 = b_1$.

Then every partition $\mathbf{P}$ of R is of the form $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$, where $v(R_j) = 0$, $1 \le j \le k$. Therefore, every Riemann sum of f over R equals zero, so $\int_R f(\mathbf{X})d\mathbf{X} = 0$.

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Question 18

Suppose that

$$f(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational,} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational,} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational,} \\ 3 & \text{if } x \text{ and } y \text{ are irrational.} \end{cases}$$

Find $\overline{\int_R} f(x,y)\,d(x,y)$ and $\underline{\int_R} f(x,y)\,d(x,y)$ if $R = [a,b] \times [c,d]$.

Ans:

Let $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$. Let

$$M_j = \sup\{f(x,y) \,|\, \mathbf{X} \in R_j\} \quad \text{and} \quad m_j = \inf\{f(x,y) \,|\, \mathbf{X} \in R_j\}.$$

Let j be arbitrary in $\{1,2,\ldots,k\}$. Since R_j contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational, $M_j = 3$. Hence, $\overline{\int_R} f(x,y)\,d(x,y) = 3(b-a)(d-c)$. Since R_j contains a point $(\tilde{x}_j, \tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational, $m_j = 0$. Hence, $\underline{\int_R} f(x,y)\,d(x,y) = 0$.

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Question 19

If S_1 and S_2 have zero content, then $S_1 \cup S_2$ has zero content.

Prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \ldots, T_r$ and $T_1', T_2', \ldots, T_s'$ such that $S_1 \subset \cup_{i=1}^{r} T_i$, $S_2 \subset \cup_{j=1}^{s} T_j'$, $\sum_{i=1}^{r} V(T_r) < \epsilon/2$, and $\sum_{j=1}^{s} V(T_j') < \epsilon/2$. Then $\{T_1, \ldots, T_m, T_1', \ldots, T_r'\}$ covers $S_1 \cup S_2$ with total content $< \epsilon$.

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Question 20

If S_1 has zero content and $S_2 \subset S_1$, then S_2 has zero content.

prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \ldots, T_r$ and $T_1', T_2', \ldots, T_s'$ such that
If $S_1 \subset \cup_{i=1}^{r} T_r$ and $S_2 \subset S_1$, then $S_2 \subset \cup_{i=1}^{r} T_r$.

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FOR FINAL TERM (FALL 2021,22)

Course Code: MTH631

EDIT BY:

SYED HUSSNAIN ALI NAQVI

MSC MATHEMATICS



Question No : 1 of 34

The graph of an odd function is symmetrical about ----------

Answer (Please select your correct option)

- x-axis
- y-axis
- origin
- None of these

even function y axis
odd function origion

Start Time 9:41 AM

87:00
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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

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If $h = f(x,y), x = f(r,\theta)$ and $y = f(r,\theta)$, then using the chain rule, $\frac{\partial h}{\partial \theta}$ = ...

Answer (Please select your correct option)

- $\frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta} + \frac{\partial f}{\partial r}\frac{\partial r}{\partial \theta}$
- $\frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta}$
- $\frac{\partial f}{\partial x} + \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} + \frac{\partial y}{\partial \theta}$
- None of these

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74:00
Time Left

13

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 14 of 34

If $h = f(x,y), x = r\cos\theta$ and $y = r\sin\theta$, where then using the chain rule, $\frac{\partial h}{\partial r}$ = ...

Answer (Please select your correct option)

- $\cos\theta\frac{\partial f}{\partial \theta} + \sin\theta\frac{\partial f}{\partial r}$
- $\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}$
- $\cos\theta\frac{\partial f}{\partial x} - \sin\theta\frac{\partial f}{\partial y}$
- None of these

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74:00 Time Left

14

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

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Question No : 3 of 50

_____ is the only region in R.

Answer (Please select your correct option)

- Set of finite points of R
- Interval
- Any finite sequence of R
- None of these

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81:00 Time Left

3

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

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MTH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$ ✓ p 87
- $-\frac{1}{2}$
- 0
- 1

Start Time 9:41 AM

80:00 Time Left

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 2 of 34

If H is an open covering of a compact subset S then, S can be covered by ----------

Answer (Please select your correct option)

- infinitely many sets from H
- finitely many sets from H ✓ p 79
- atleast one set from H
- None of these

Start Time 9:41 AM

83:00 Time Left

2

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

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MTH631 Real Analysis II

Question No : 1 of 26

The singleton set is always —

page 81

Answer (Please select your correct option)

- disconnected
- polygonally disconnected
- connected
- None of these

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MTH631 Real Analysis II

Question No : 2 of 26

Any polygonally connected set either it is open or not, is always _____

page 82

Answer (Please select your correct option)

- connected
- disconnected
- a region
- None of these

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MTH631 Real Analysis II

Question No : 5 of 26 **Marks: 1 (Budgeted Ti**

Suppose $U_0 \in T$ is a limit point of T. g is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then ——— is continuous at U_0.

page 102

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Answer (Please select yo[illegible]orrect option)

- $f \circ g$
- $\frac{f}{g}$
- $\frac{g}{f}$
- None of these

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MTH631 Real Analysis II

Question No : 7 of 26 **Marks:** 1 (Bud

A function f is _____ on a subset S of R^n if S is contained in an open set on which $f_1, f_2, \ldots, f_n$ are continuous

page 116

Answer (Please select your correct option)

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

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MTH631 Real Analysis II

Question No : 10 of 26

A function of the form $L(X) = m_1 x_1 + m_2 x_2 + \ldots + m_n x_n$ for some constants m_i's is called _____

page 112

Answer (Please select your correct option)

- linear function
- quadratic function
- constant function
- None of these

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MTH631 Real Analysis II

Question No : 20 of 34

If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|dx < \infty$, then $\int_a^b f(x)dx$ is _____

Answer (Please select your correct option)

- divergent
- conditionally convergent
- convergent ✓ b.pg 138
- None of these

Start Time: 9:41 AM

62:00 Time Left

20

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 21 of 34

Marks: 1 **(Budgeted Time** Min)

Suppose f is a continuous and $F(x) = \int_a^x f(t)dt$ is bounded on $[a,b)$. Let g' be an absolutely integrable on $[a,b)$ and suppose $\lim_{x \to b^-} g(x) = 0$

Then----------

Answer (Please select your correct option)

- $\int_a^b f(x)g(x)dx$ converges ✓ b.pg 139
- $\int_a^b f(x)g(x)dx$ diverges
- $\int_a^b f(x)g(x)dx$ conditionally converges
- None of these

Start Time: 9:41 AM

37:00 Time Left

21

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

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MTH631 Real Analysis II

Question No : 18 of 34 Marks: 1

A function f is said to be absolutely integrable on $[a,b)$ if f is locally integrable on $[a,b)$ and ----------

Answer (Please select your correct option)

- $\int_a^b |f(x)|dx < \infty$ ✓ b.pg 137
- $\int_a^b |f(x)|dx = \infty$
- $\int_a^b |f(x)|dx \geq \infty$
- None of these

Start Time: 9:41 AM

65:00 Time Left

18

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

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MTH631 Real Analysis II

Question No : 24 of 34

The lower integral $\underline{\int_R} f(X)d(X)$ of f over R is the supremum of all the _____.

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

Start Time: 9:41 AM

37:00 Time Left

24

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 19 of 34

An absolutely convergent integral is always ----------

Answer (Please select your correct option)

- divergent
- convergent
- conditionally convergent
- None of these

138

Start Time: 9:41 AM

62:00 Time Left

19

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 23 of 31

The upper integral $\overline{\int_R} f(X)d(X)$ of f over R is the infimum of all the----------

Answer (Please select your correct option)

- lower sums
- upper sums
- average sums
- None of these

Start Time: 9:41 AM

37:00 Time Left

23

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

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MTH631 Real Analysis II

Question No : 15 of 34

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called _____

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

Start Time: 9:41 AM

71:00 Time Left

15

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Question No :

If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^n$, then———

Answer (Please select your correct option)

- f is not integrable on R ✓
- f is integrable on R
- f is not differentiable on R
- None of these

145

Start Time: 9:41 AM

37:00 Time Left

22

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

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Question No : 12 of 34

A function of the form $L(X) = f_1(X_0)x_1 + f_2(X_0)x_2 + \ldots + f_n(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

Page#105

104

Start Time 9:41 AM

74:00 Time Left

12

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

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MTH631 Real Analysis II

Question No : 25 of 34

If f is integrable on a rectangle R, then ———

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$
- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X) = \int_R f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X) > \int_R f(X)d(X)$
- None of these

Start Time 9:41 AM

37:00 Time Left

25

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the .....

Answer (Please select your correct option)

- critical point
- point of inflection
- point of relative extrema
- None of these

Start Time: 9:41 AM

74:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23

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MTH631 Real Analysis II

Question No : 10 of 34

The rate of change of $f(x,y)$ in the direction of a unit vector is called the ---------- of $f(x,y)$.

Answer (Please select your correct option)

- differential
- directional derivative
- mixed derivative
- None of these

Start Time: 9:41 AM

74:00 Time Left

10

Question Summary : (Attempted Question)

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Question No : 11 of 34 Marks:

A function of several variables may have first order partial derivatives at a point X_0 but ---------- continuous at X_0.

Answer (Please select your correct option)

- fail to be
- must be
- always not
- None of these

102

Start Time: 9:41 AM
74:00
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11

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

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MTH631 Real Analysis II

Question No : 28 of 34

A ---------- surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

- parabolic
- elliptical
- differentiable
- None of these

Start Time: 9:41 AM

37:00 Time Left

28

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 27 of 34

Any bounded set E with only finitely many ---------- has zero content.

Answer (Please select your correct option)

- limit points
- neighbourhood points
- interior points
- None of these

Start Time: 9:41 AM

37:00 Time Left

27

Question Summary : (Attempted Question)

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Question No : 30 of 34

A set of points of the form $X = X_0 + tU, t_1 \leq t \leq t_2$ is called .....

Answer (Please select your correct option)

- the neighbourhood.
- line segment.
- the closure.
- None of these

75

Start Time: 9:41 AM

38:00 Time Left

30

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 5 of 34

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that _____

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$
- $f(X_0) = g(X_0) = 0$
- None of these

92

Start Time 9:41 AM

78:00 Time Left

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 29 of 34

If f is integrable on disjoint sets S_1 and S_2 then f is integrable on

Answer (Please select your correct option)

- $S_1 \cup S_2$
- S_1 / S_2
- S_2 / S_1
- None of these

Start Time: 9:41 AM

38:00 Time Left

29

Question Summary : (Attempted Question)

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MTH631 Real Analysis II

Question No : 28 of 34

A ---------- surface in $\mathbb{R}^n$ has zero content.

Answer (Please select your correct option)

- parabolic
- elliptical
- differentiable ✓
- None of these

153

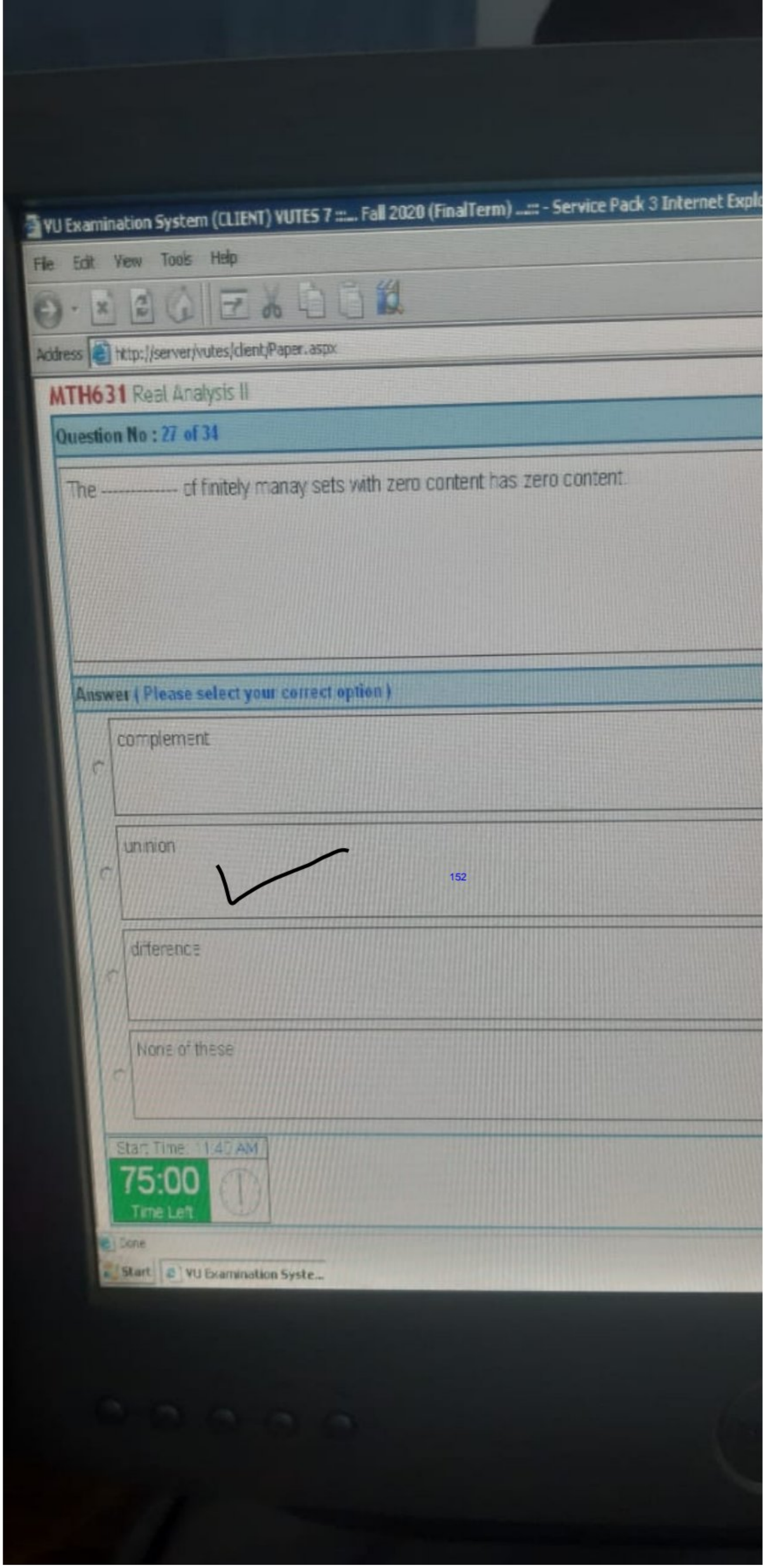
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MTH631 Real Analysis II

Question No : 14 of 31

If value of a becomes zero in the Taylor series $\sum_{n=0}^{\infty} \frac{f^n(a)}{n!}(x-a)^n$, then the series is called ------

Answer (Please select your correct option)

- Maclaurin series
- Fourier series
- Power series
- None of these

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77:00 Time Left

14

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MTH631 Real Analysis II

Question No : 30 of 34

A set is closed if and only if it contains all of its ——

Answer (Please select your correct option)

- Neighborhoods
- Limit points
- Exteriors
- None of these

78

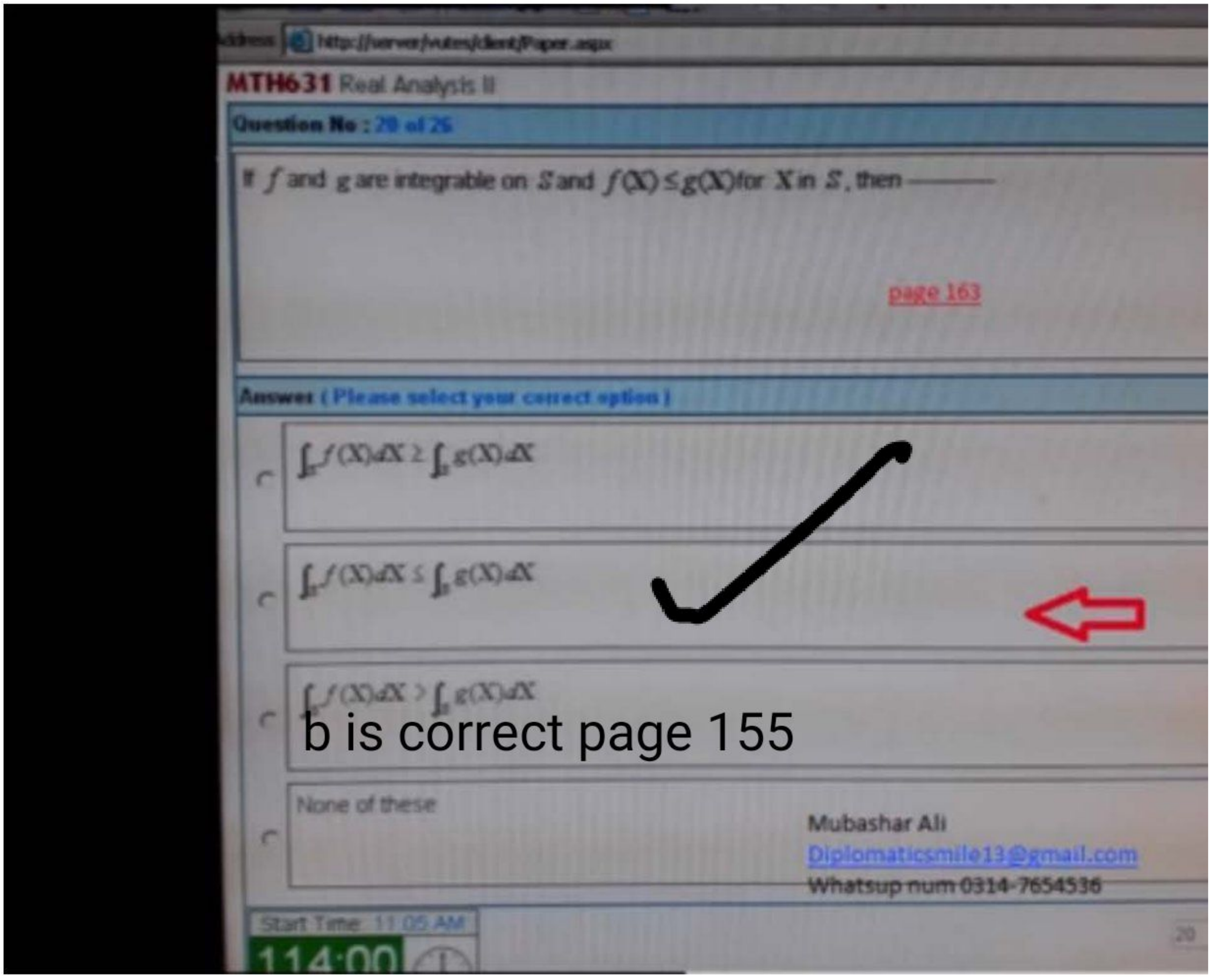
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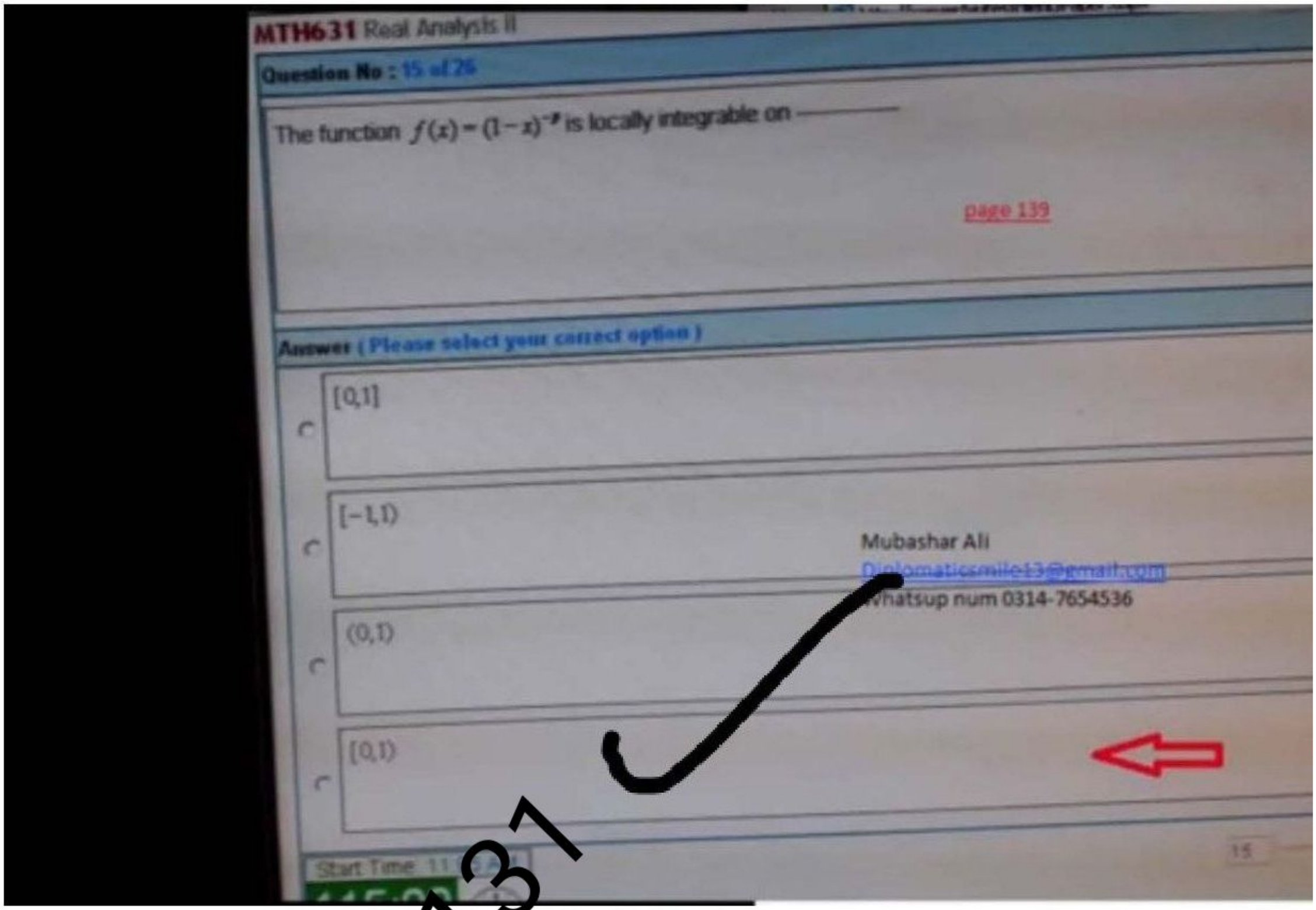
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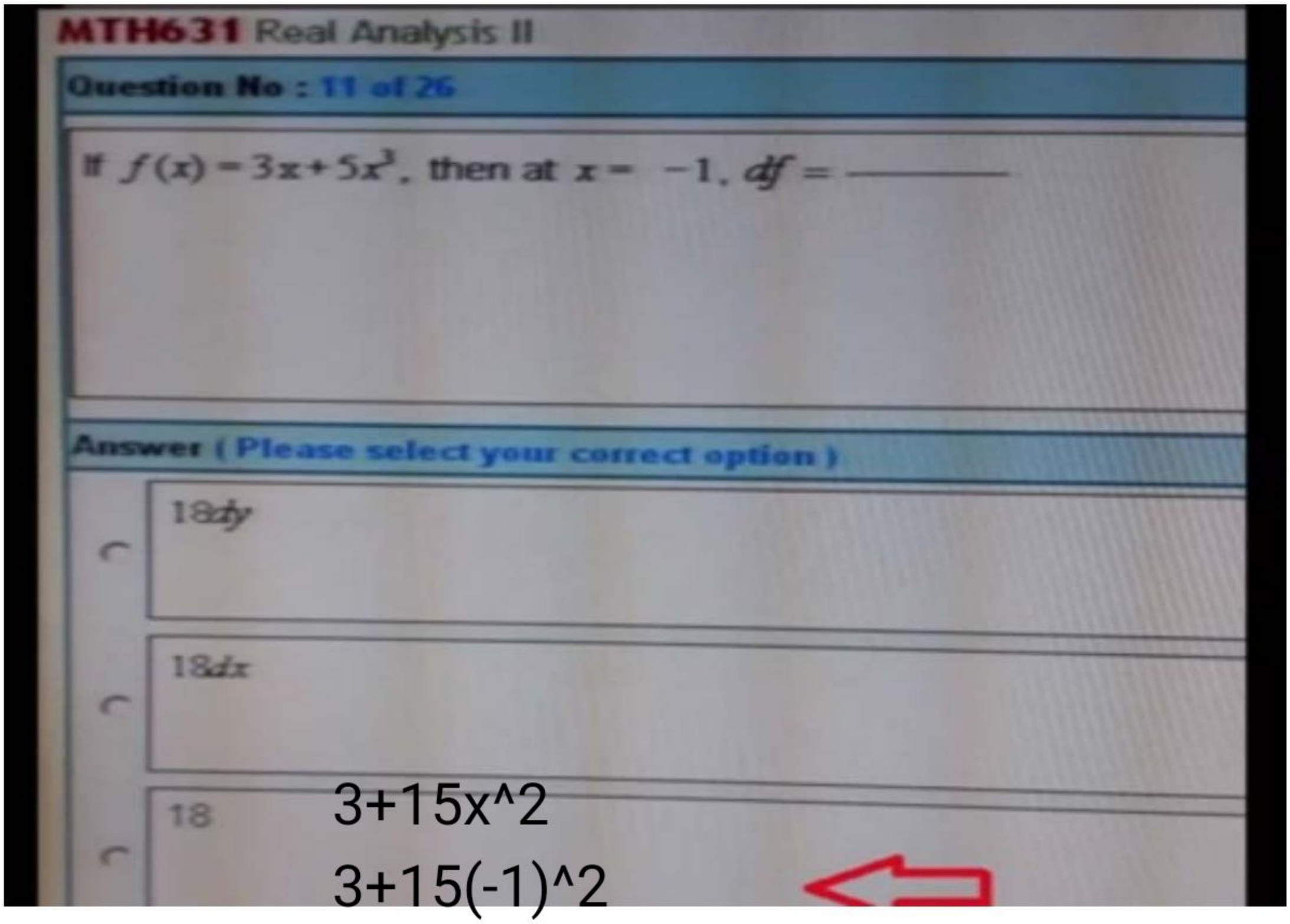


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page 131

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3+15x^2

3+15(-1)^2

3+15=18

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MTH631 Real Analysis II

Question No : 7 of 26

A $f_1, f_2, \cdots f_n$ exist on a neighborhood of X_0 and are continuous at X_0, then f is ——— at X_0.

page 115

Answer (Please select your correct option)

- uniformly converget
- piecewise convergent
- differentiable
- None of these

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Start Time: 11:05 AM

page 107

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MTH631 Real Analysis II

Question No : 6 of 26

A set A ⊆ R of real numbers is ———————— if there exists a real number M ∈ R, such that $x \leq M$ for every $x \in A$.

Answer (Please select your correct option)

- bounded below
- uniformly continuous
- bounded above
- None of these

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MTH631 Real Analysis II

Question No : 5 of 26

In a vector valued function $G=(g_1, g_2, \ldots, g_n)$. Then $g_1, g_2, \ldots, g_n$ are called the ———— functions of G.

page 114

Answer (Please select your correct option)

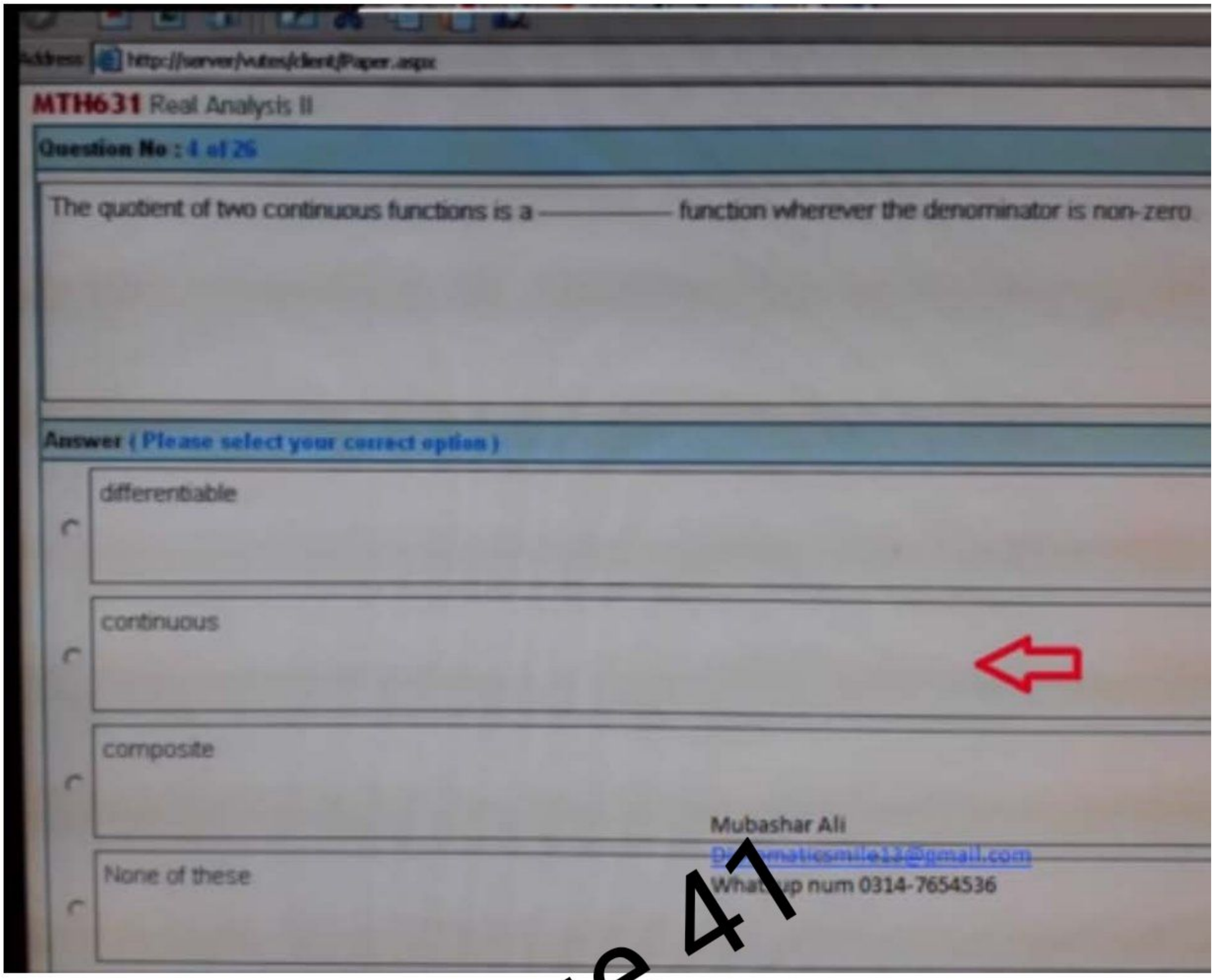
- bijective
- surjective
- component
- None of these

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page 114

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page 41

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MTH631 Real Analysis II

Question No : 3 of 26

The set $S = \{(x,y), x^2 + y^2 \leq 1 \text{ or } x^2 + y^2 \geq 4\}$ is not a region in R^2, since it is _____

page 90

Answer (Please select your correct option)

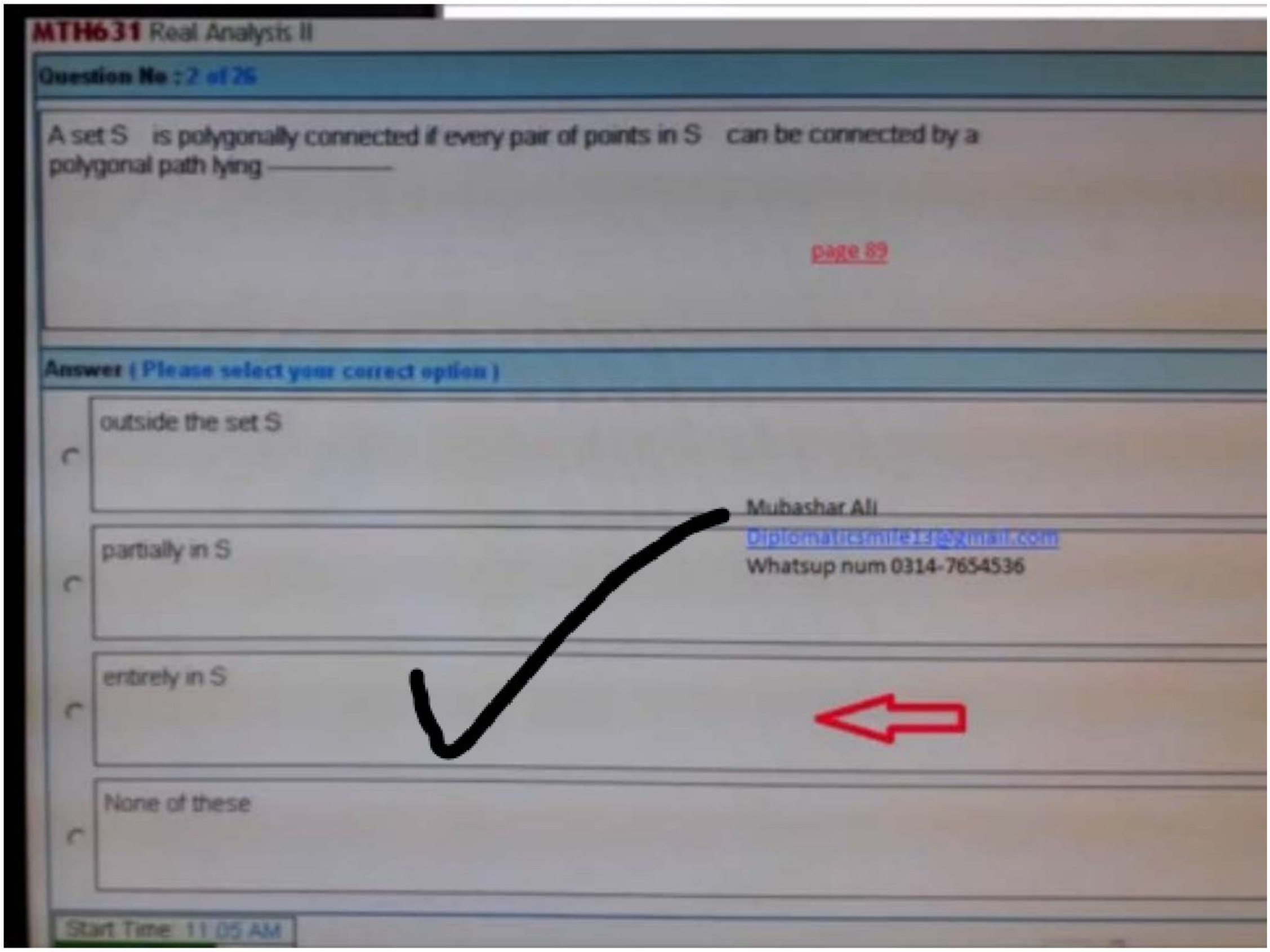
- not connected
- connected
- polygonally connected
- None of these

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page 82

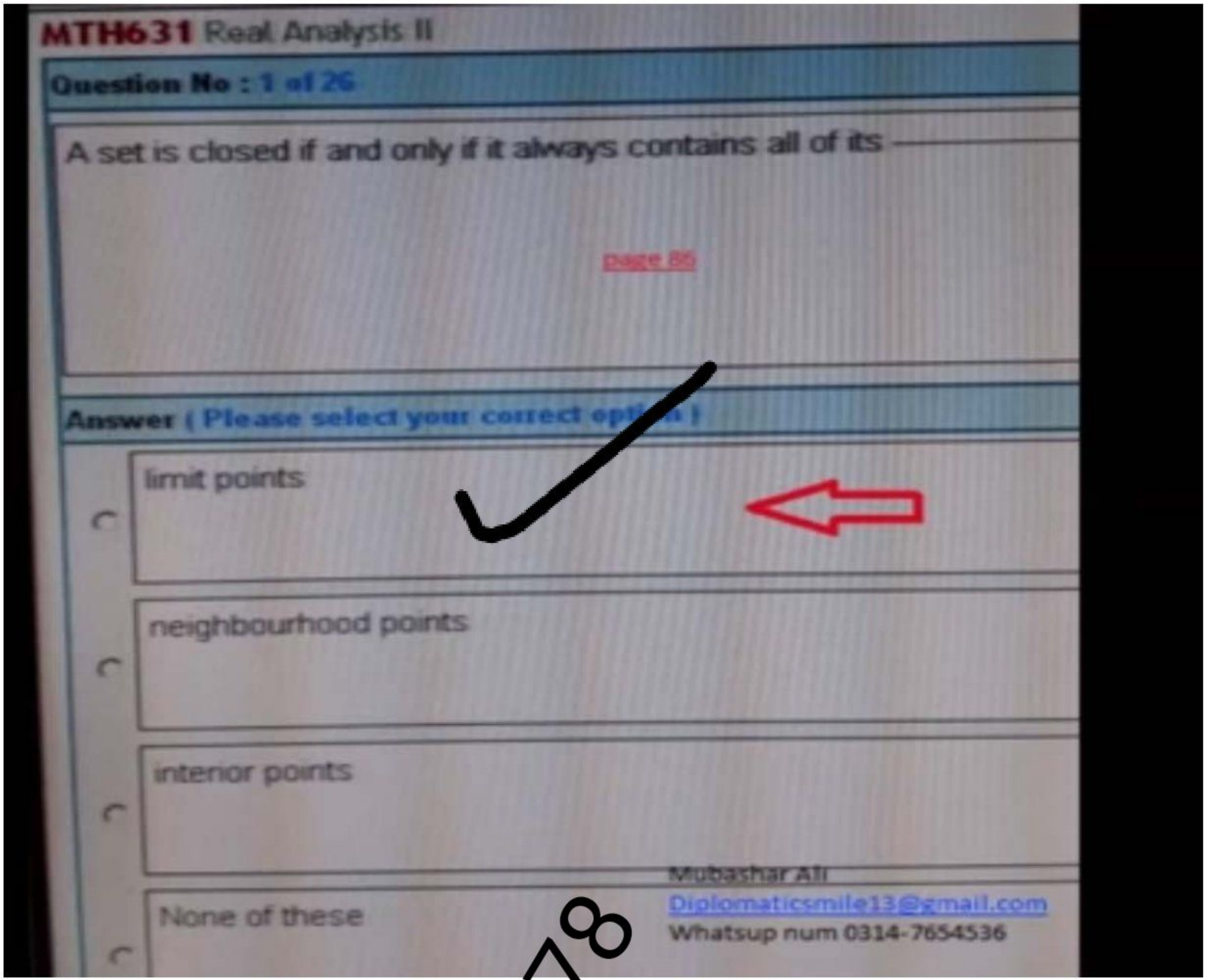
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page 87

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MTH631 Real Analysis II
Question No : 1 of 26
A set is closed if and only if it always contains all of its ——————
page 86
Answer (Please select your correct option)
limit points
neighbourhood points
interior points
None of these
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78

page

Question No : 20 of 26

A ———— surface in $\mathbb{R}^n$ has zero content.

PAGE 161

Answer (Please select your correct option)

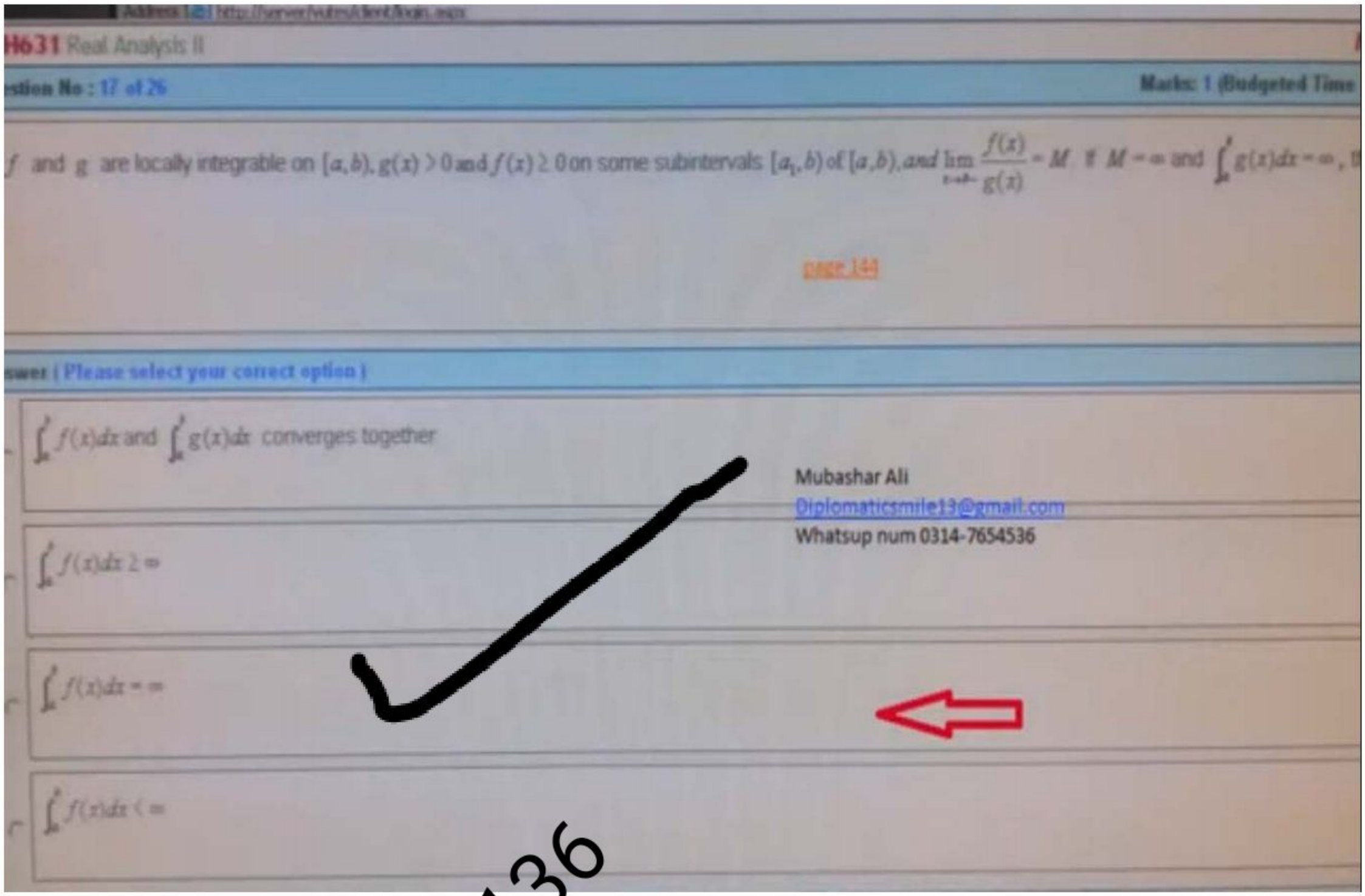
- parabolic
- elliptical
- differentiable
- None of these

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153

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page 136

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Question No : 11 of 26

If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$ If $p(X) \geq 0$ for all X, then p is called _____

page 134

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite
- negative definite

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page 126

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MTH631 Real Analysis II

Question No : 7 of 26 **Marks:** 1 (I

A function f is ———— on a subset S of R^n if S is contained in an open set on which $f_1, f_2, \ldots, f_n$ are continuous

page 116

Answer (Please select your correct option)

- continuously differentiable
- uniformly convergent
- piecewise convergent
- None of these

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If f is bounded on a rectangle R, then ----------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) \leq \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) \geq \underline{\int_R} f(X)d(X)$
- None of these

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A bounded function f is integrable on a rectangle R if and only if ----------

Answer (Please select your correct option)

- $\overline{\int_R} f(X)d(X) = \underline{\int_R} f(X)d(X)$ ✓
- $\overline{\int_R} f(X)d(X) > \underline{\int_R} f(X)d(X)$
- $\overline{\int_R} f(X)d(X) < \underline{\int_R} f(X)d(X)$
- None of these

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A set of points of the form $X = X_0 + tU, t_1 \leq t \leq t_2$ is called .....

Answer (Please select your correct option)

- the neighbourhood.
- line segment. ✓
- the closure.
- None of these

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38:00 Time Left

30

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22

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MTH631 Real Analysis II

Question No : 4 of 34

For the function $f(x,y)=\frac{xy}{x^2+y^2}$, the limit of $f(x,y)$ as $(x,y)\to(0,0)$ along the line $y=x$ is ----

Answer (Please select your correct option)

- $\frac{1}{2}$
- $-\frac{1}{2}$
- 0
- 1

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80:00 Time Left

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MTH631 Real Analysis II

Question No : 9 of 34

A point on the graph of a function at which the tangent line is parallel to x-axis is called the .....

Answer (Please select your correct option)

- critical point ✓
- point of inflection
- point of relative extrema
- None of these

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TH631 Real Analysis II

Question No : 5 of 34

If f and g are continuous functions at the point X_0 in R^n, then $\frac{f}{g}$ is continuous at X_0 provided that ——

Answer (Please select your correct option)

- $g(X_0) = 0$
- $g(X_0) \neq 0$ ✓
- $f(X_0) = g(X_0) = 0$
- None of these

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78:00 Time Left

Question Summary : (Attempted Question)

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If $p(X)$ is a homogeneous polynomial of degree r in $X - X_0$. If $p(X) \leq 0$ for all $X \neq X_0$. Then p is called ———

Answer (Please select your correct option)

- positive semidefinite
- positive definite
- negative semidefinite ✓
- negative definite

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A function of the form $L(X) = f_{x_1}(X_0)x_1 + f_{x_2}(X_0)x_2 + \ldots + f_{x_n}(X_0)x_n$ is called ———

Answer (Please select your correct option)

- quadractic function
- differential of f at X_0
- constant function
- None of these

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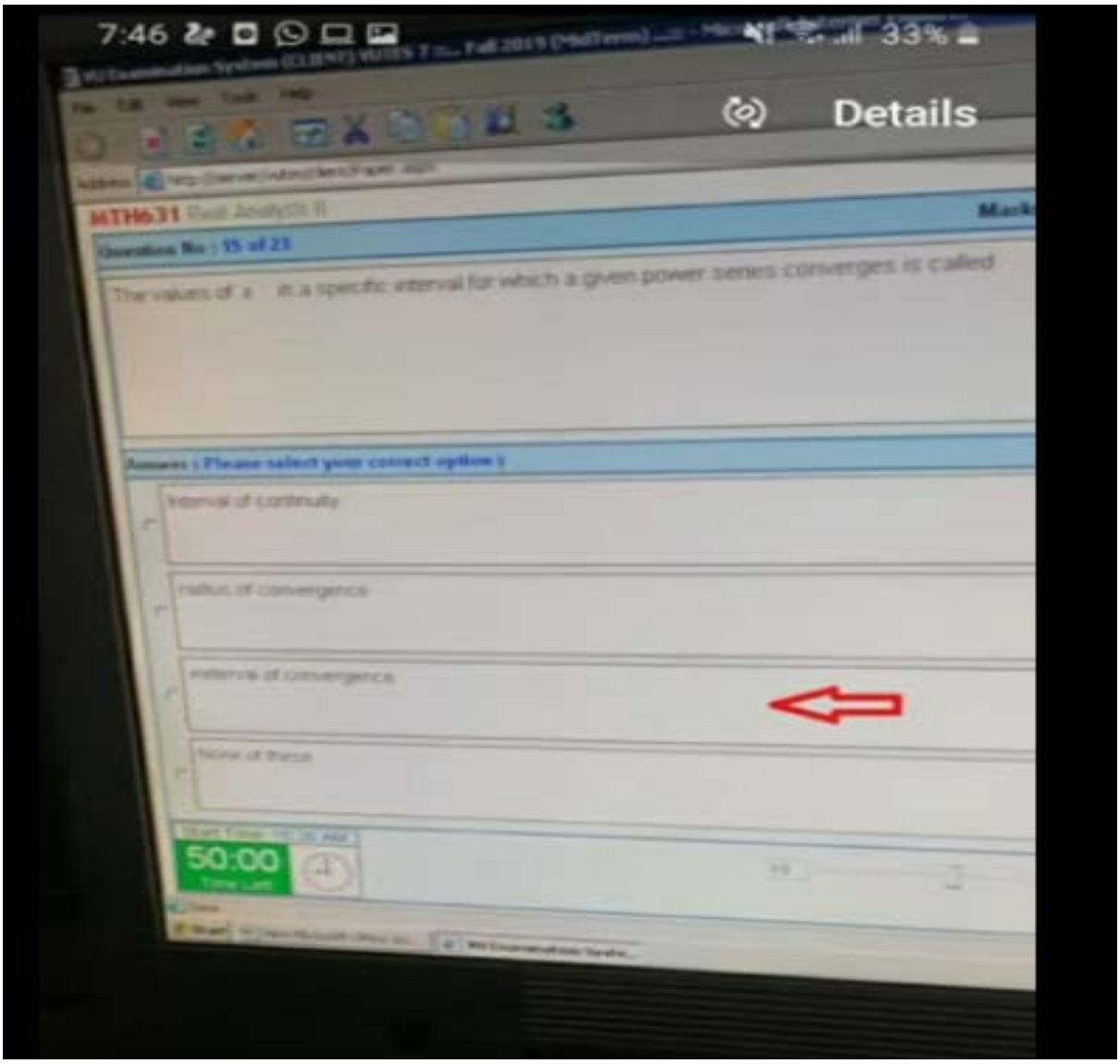
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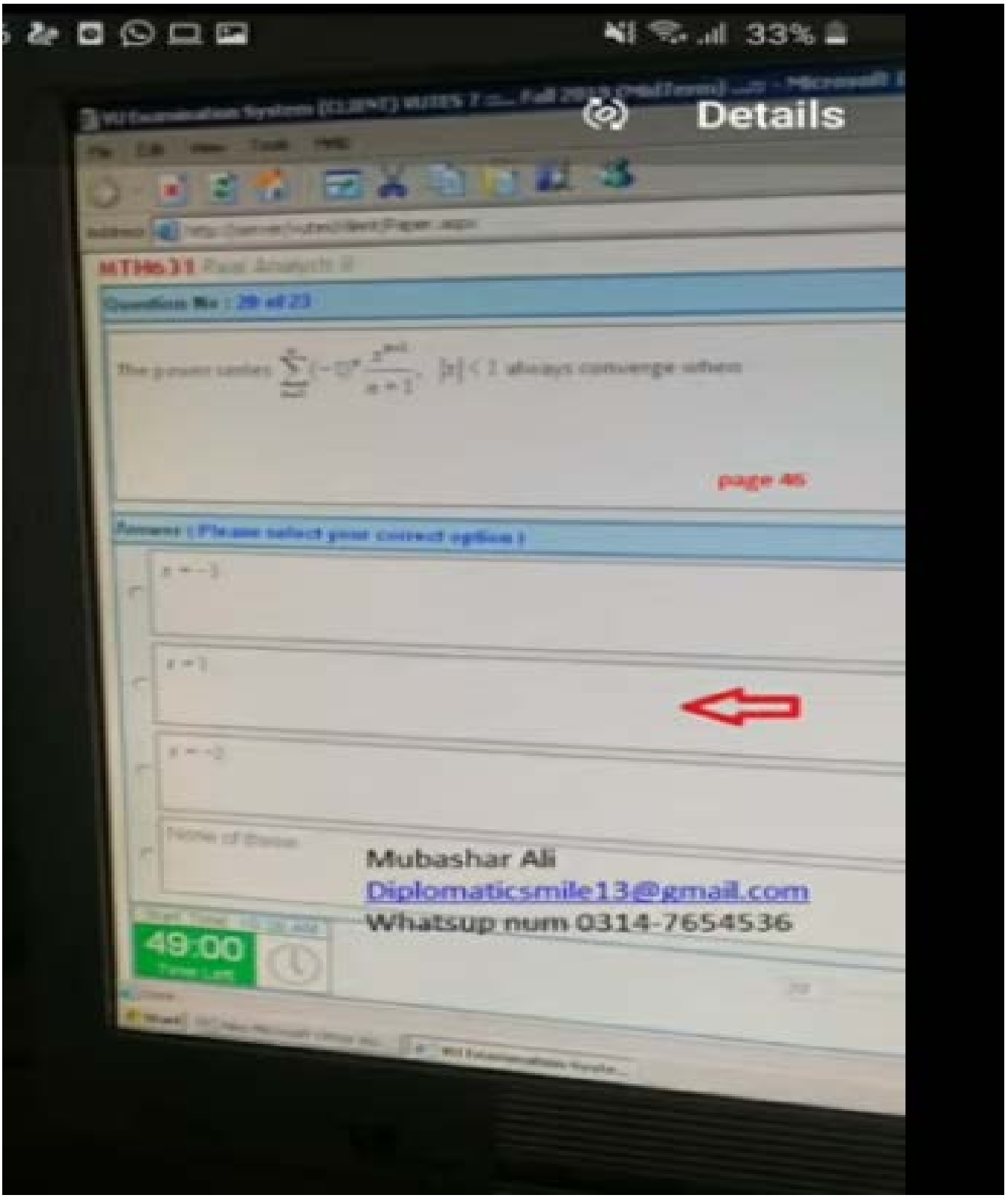
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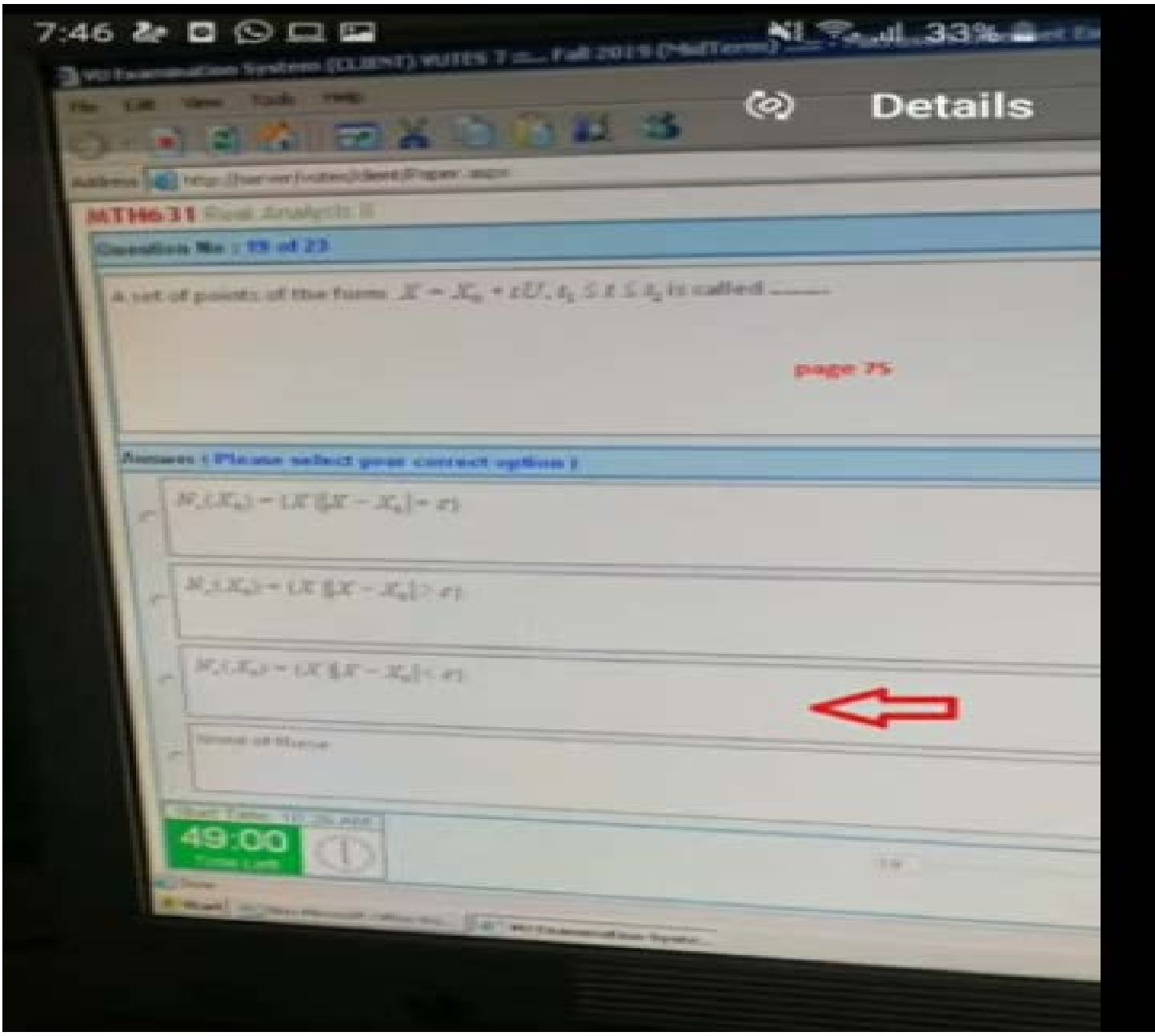
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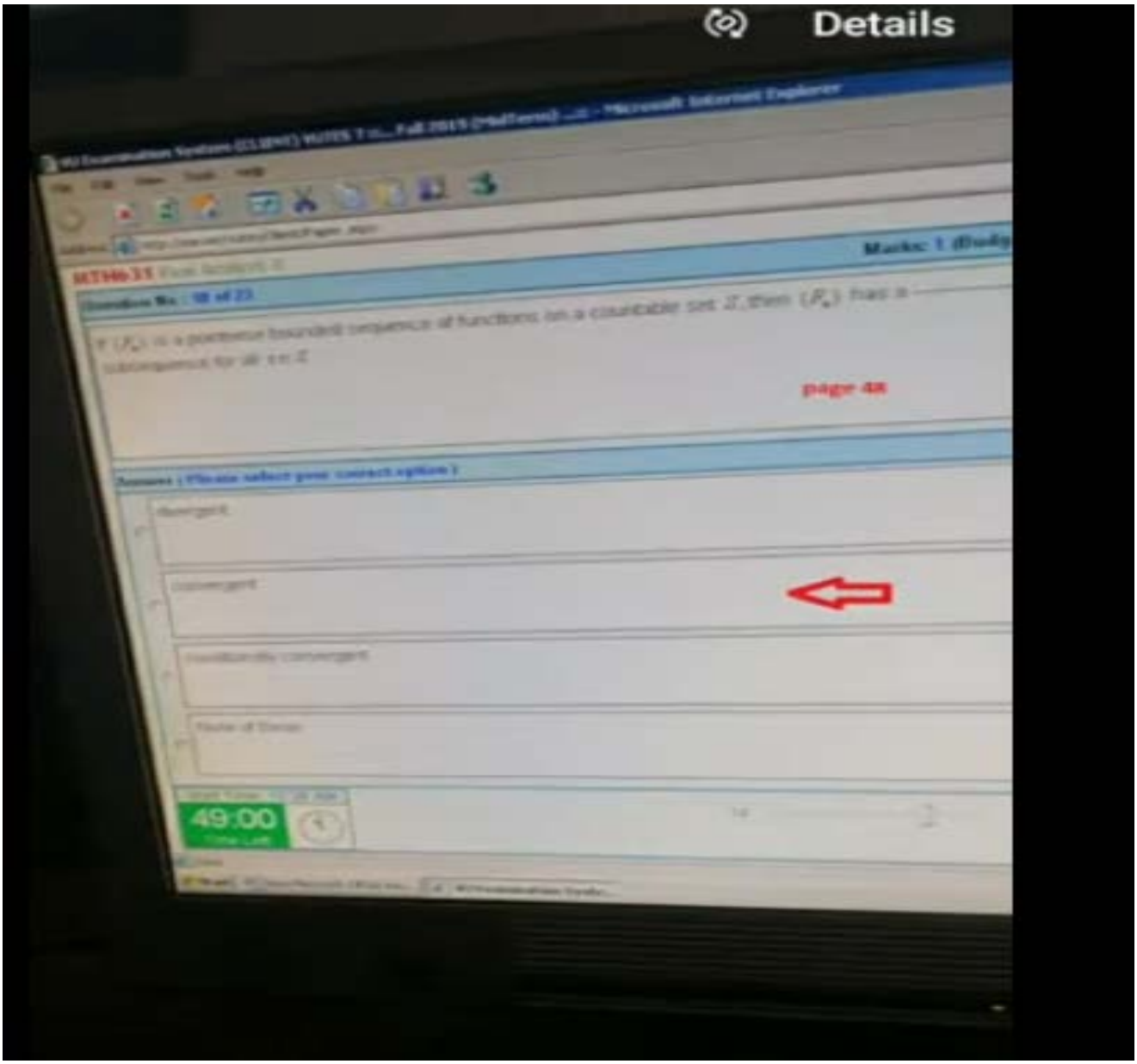
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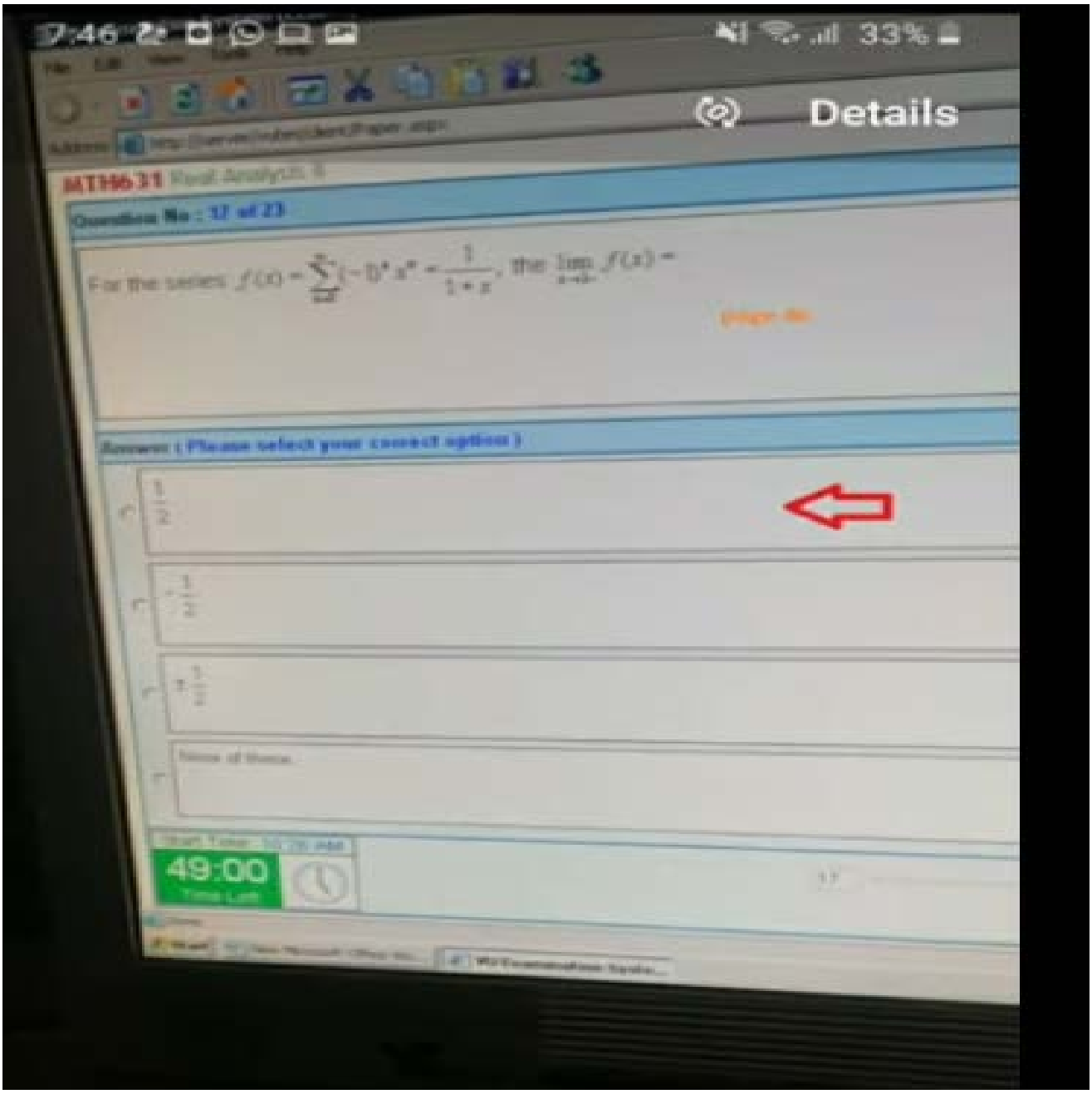
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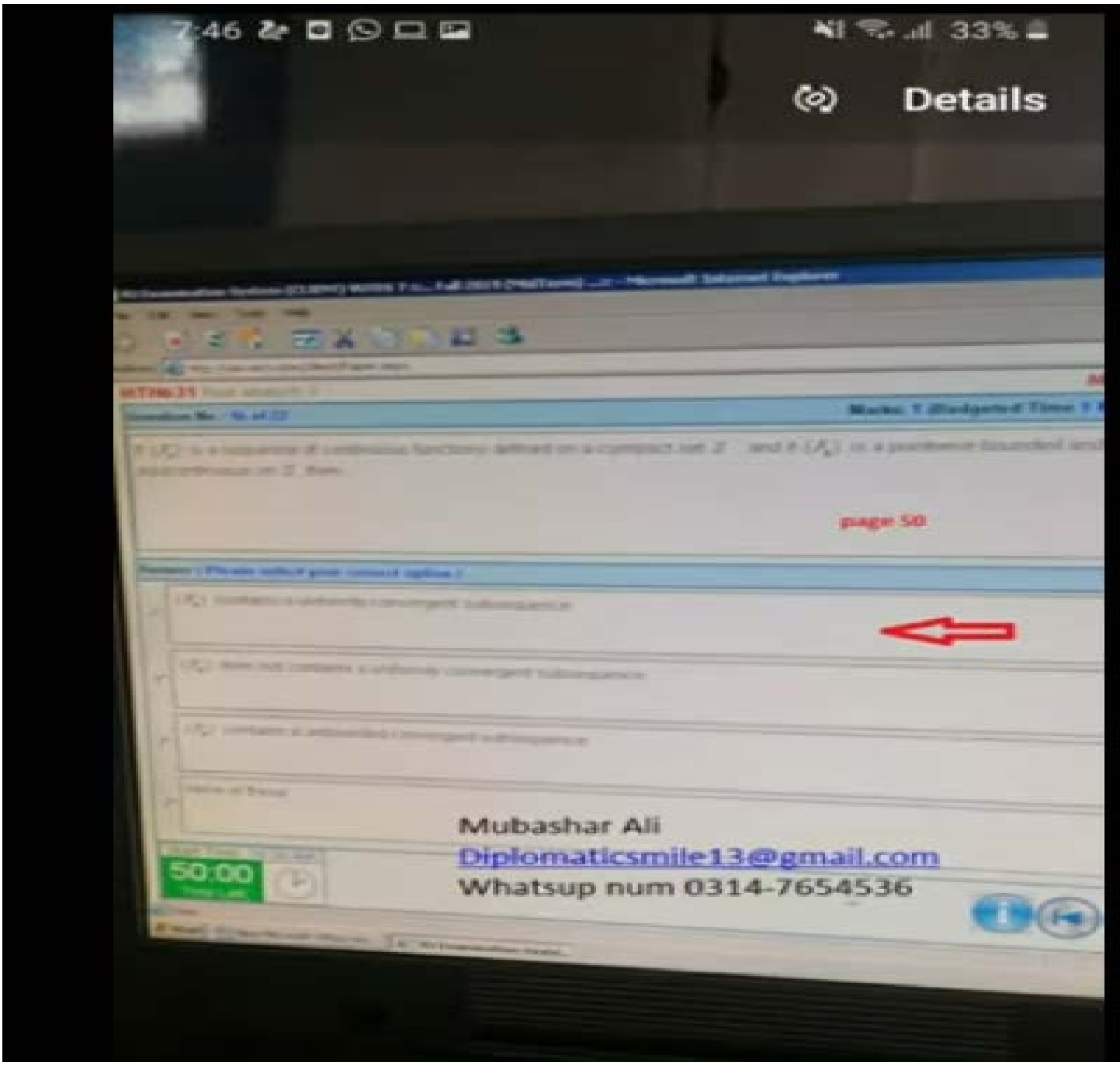

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Question No : 15 of 23
The values of x in a specific interval for which a given power series converges is called
Answers : Please select your correct option :
interval of continuity
radius of convergence
interval of convergence
None of these
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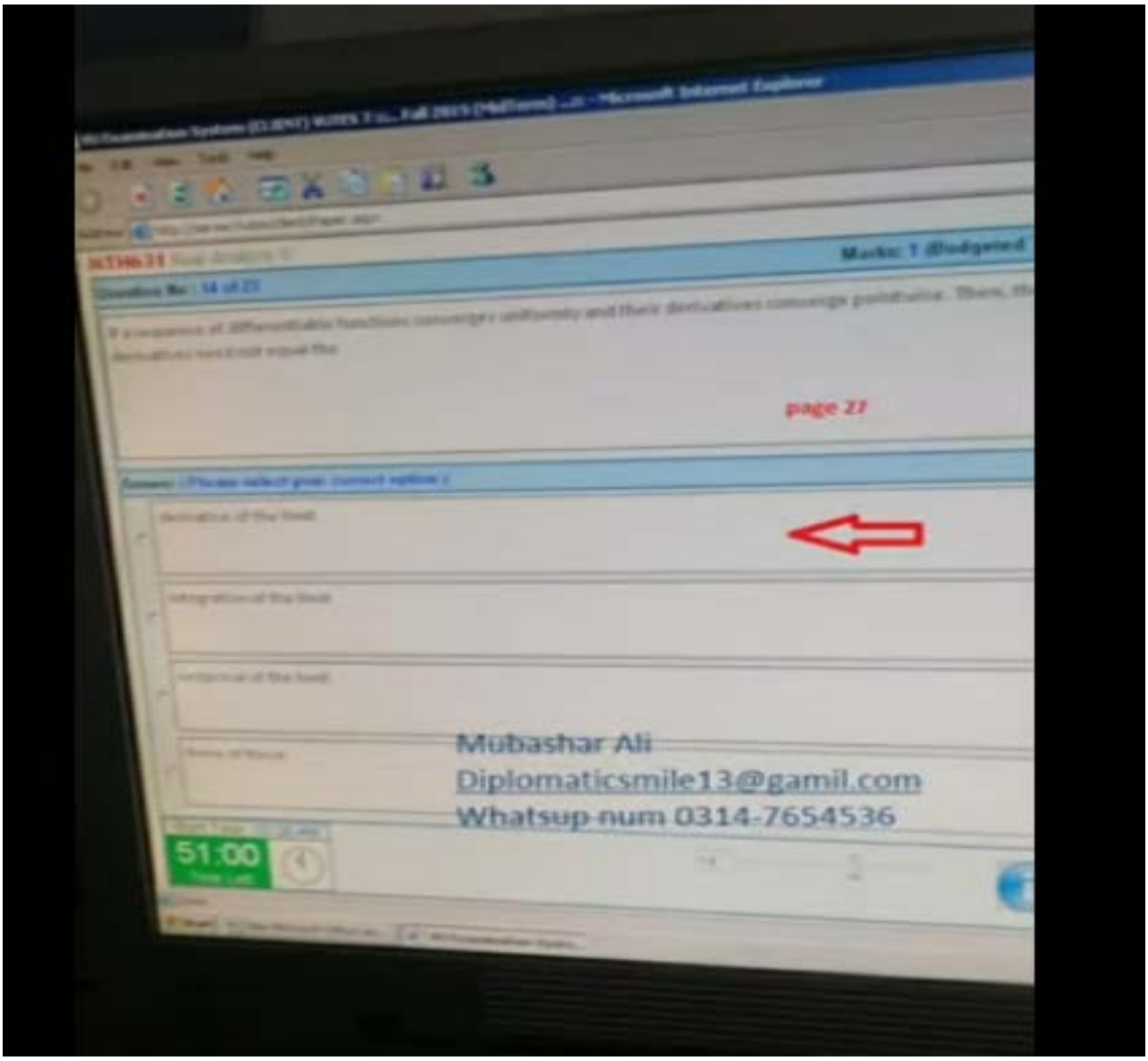

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page 46
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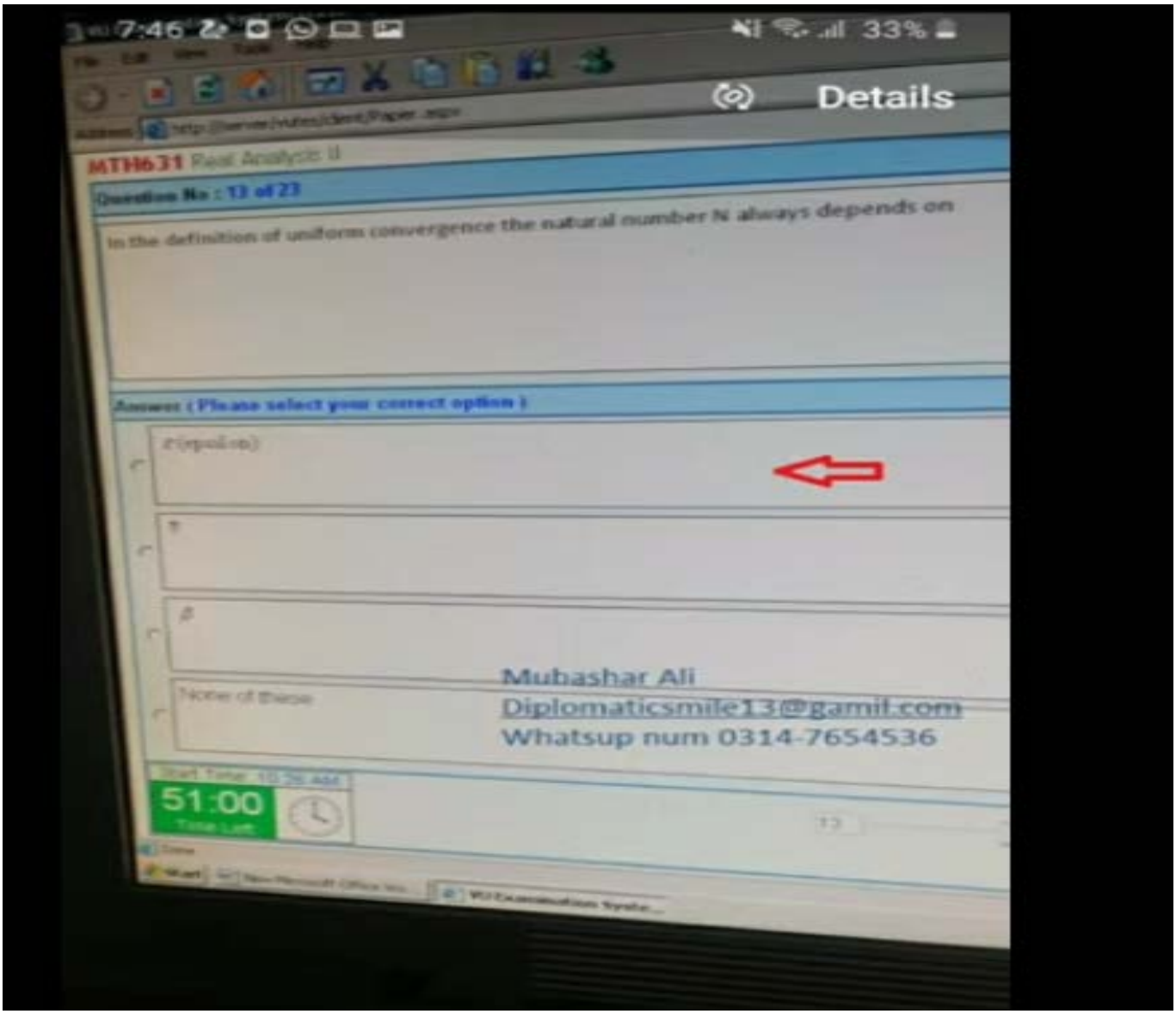

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Details
MTH631
page 75
49:00

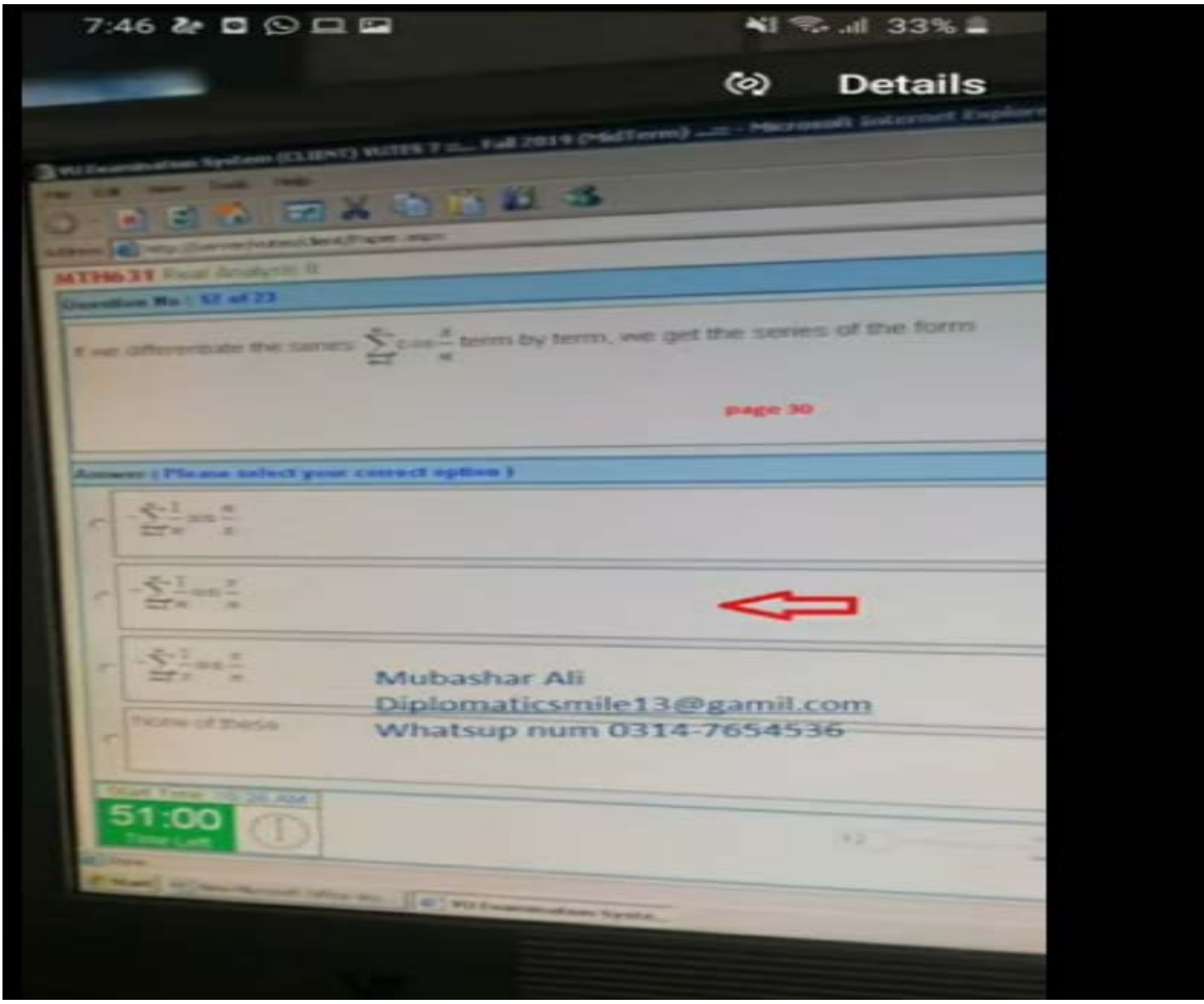

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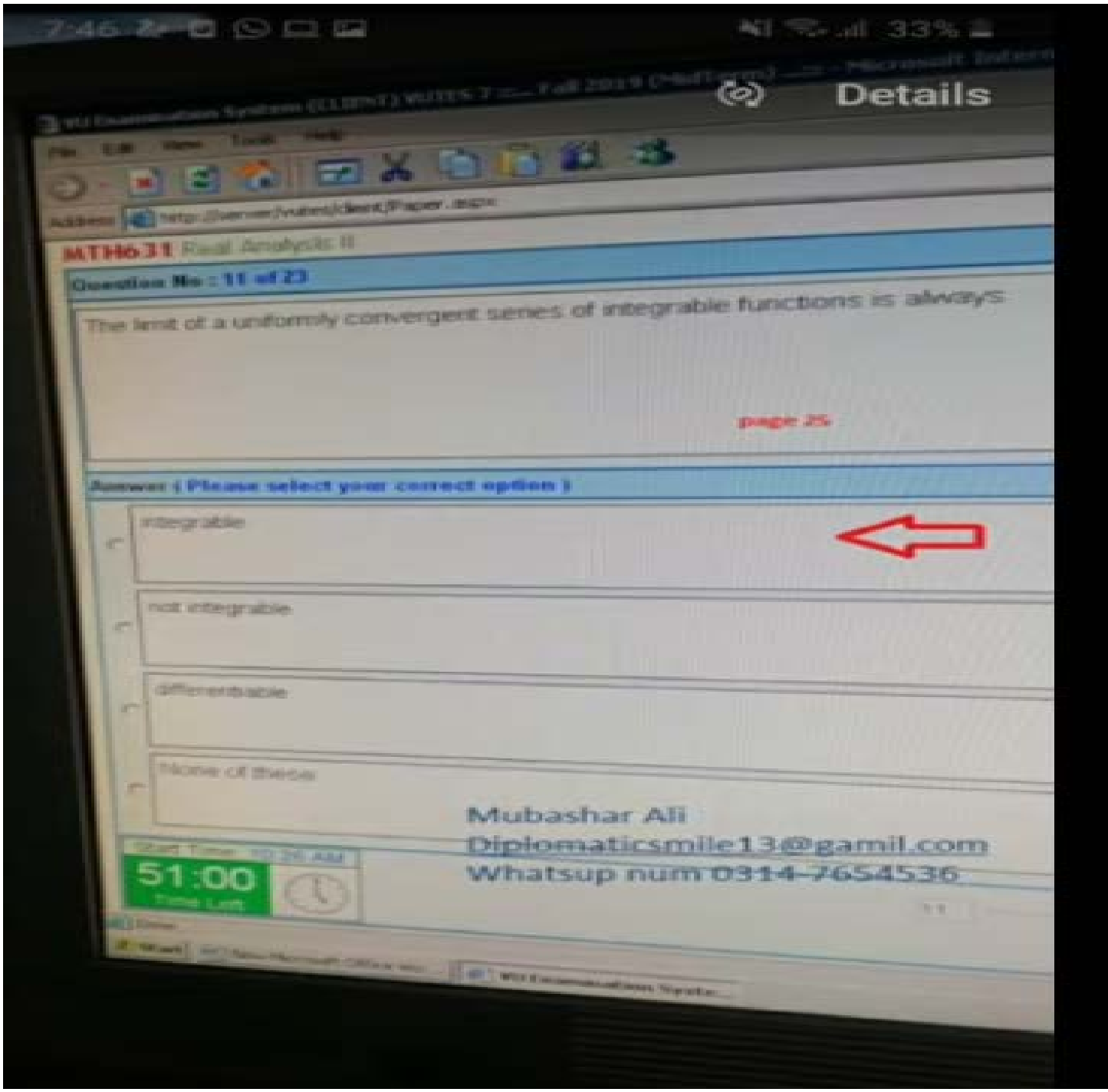

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Details
MTH6631
Question No : 12 of 23
For the series $f(x) = \sum_{n=0}^{\infty} (-1)^n x^n = \frac{1}{1+x}$, the $\lim_{x \to 1} f(x) =$
Answers : (Please select your correct option)
$\frac{1}{2}$
$-\frac{1}{2}$
$\pm\frac{1}{2}$
None of these
49:00
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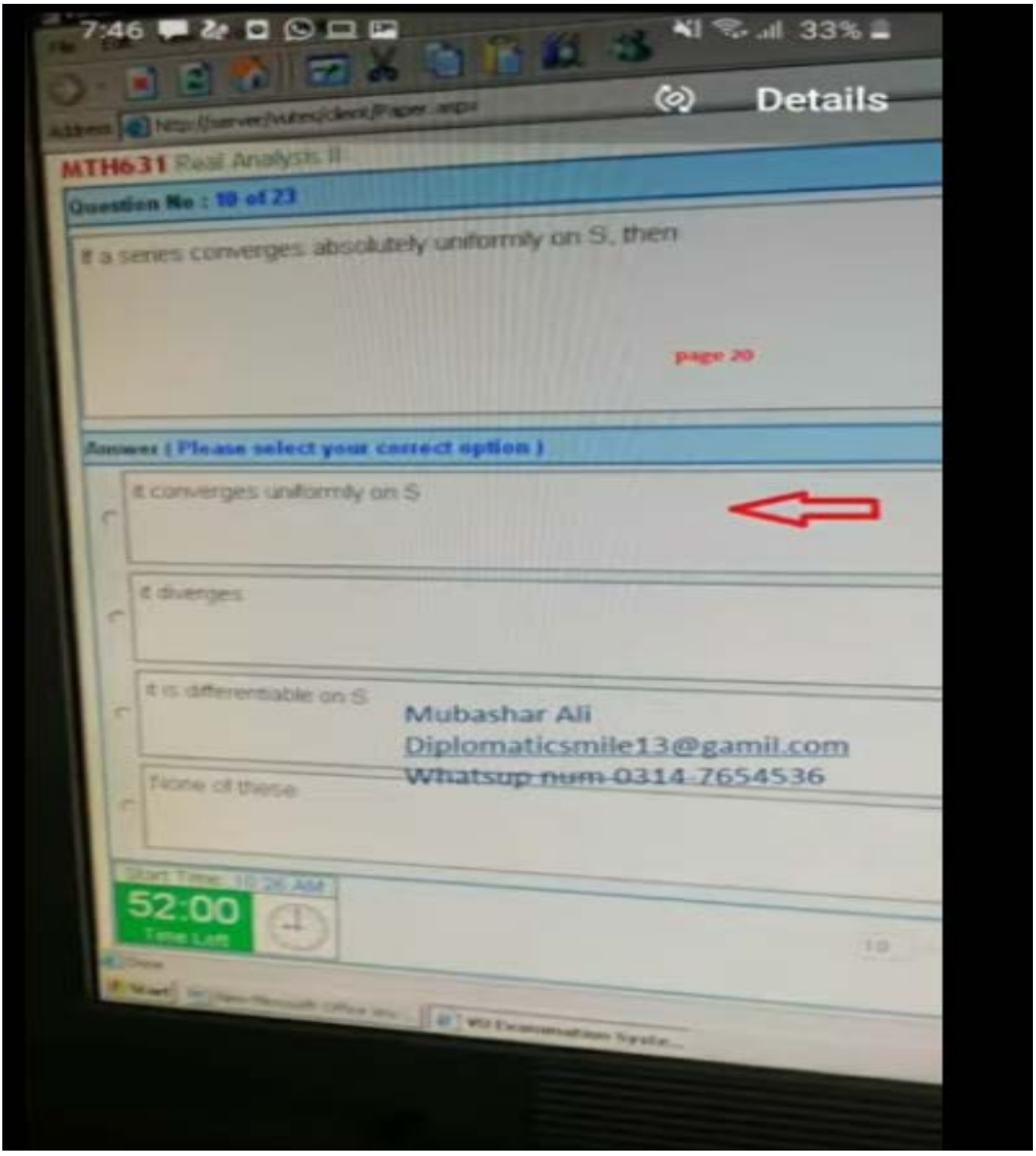

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Details
page 50
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50:00

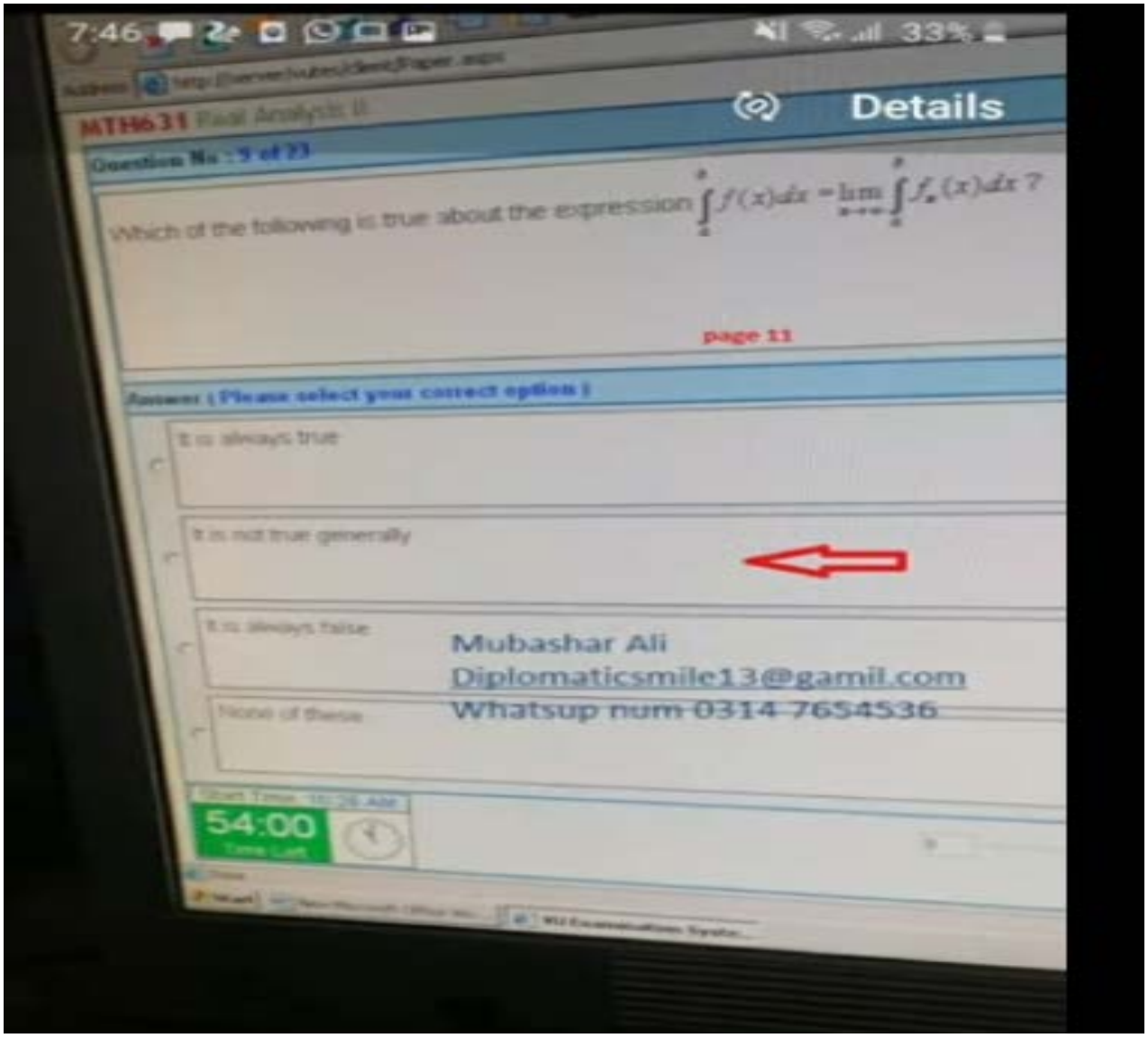

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Diplomaticsmile13@gamil.com
Whatsup num 0314-7654536
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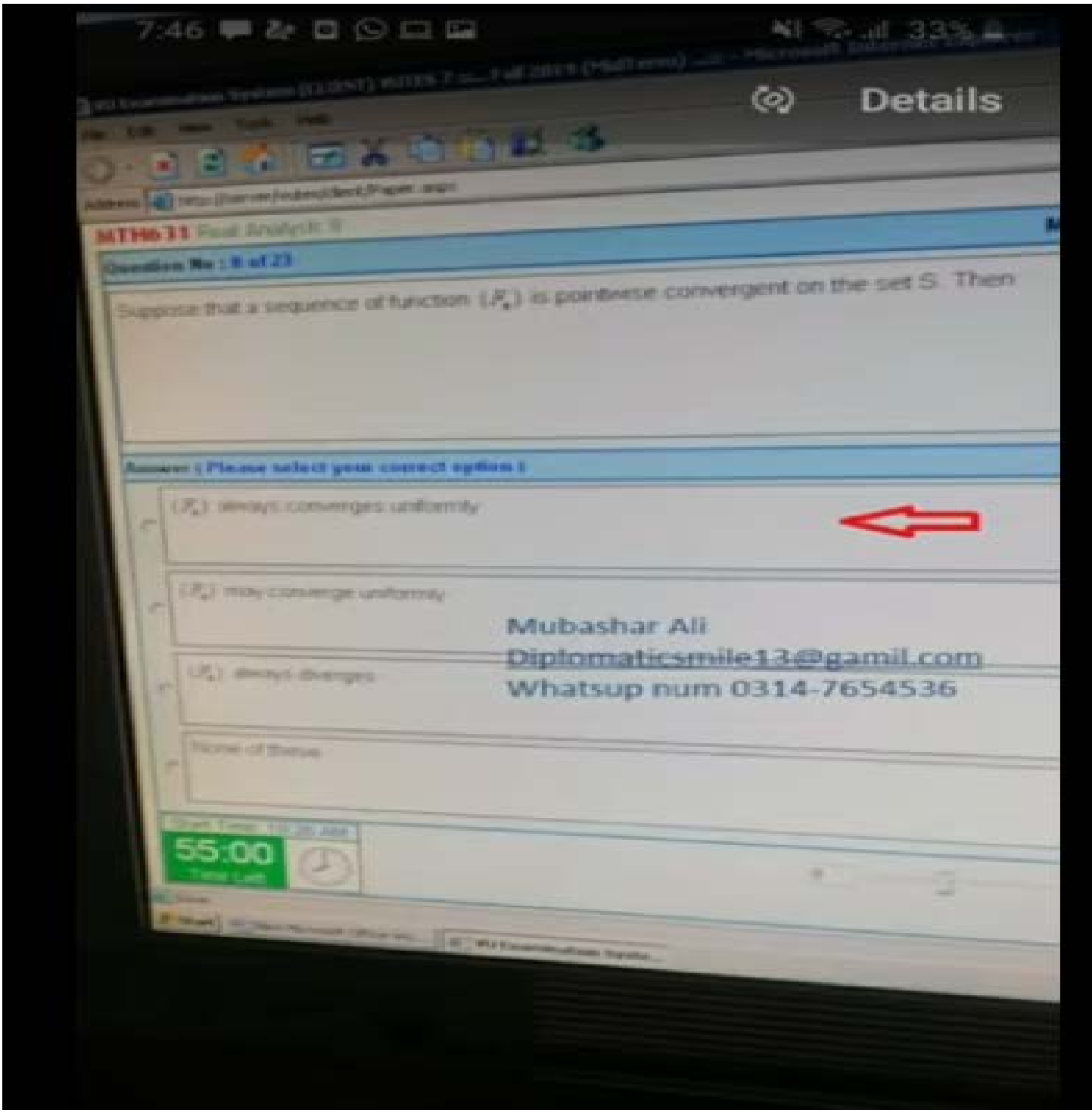

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Details
MTH631 Real Analysis II
Question No : 13 of 23
In the definition of uniform convergence the natural number N always depends on
Answer (Please select your correct option)
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Diplomaticsmile13@gamil.com
Whatsup num 0314-7654536
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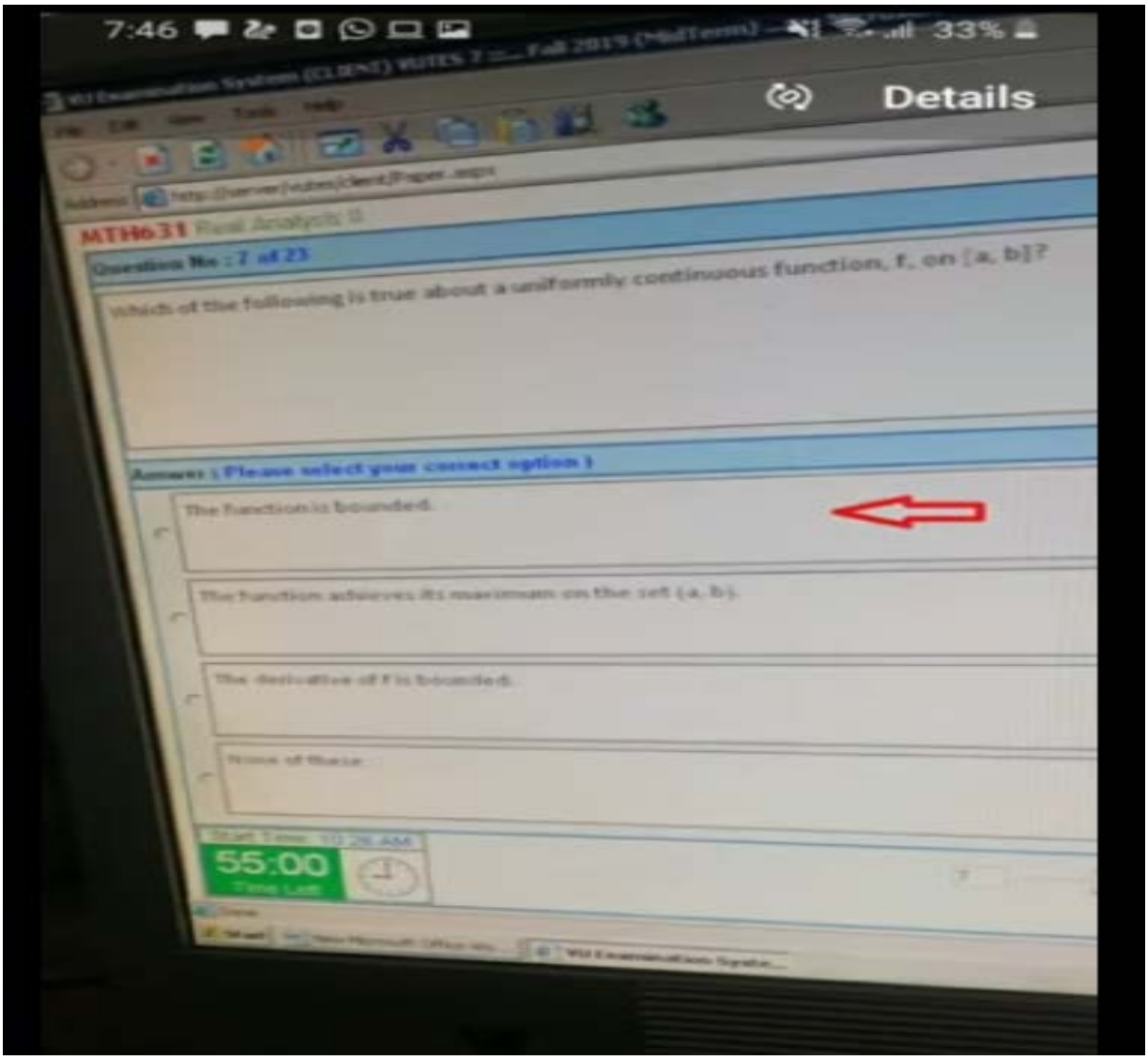

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Details
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Diplomaticsmile13@gamil.com
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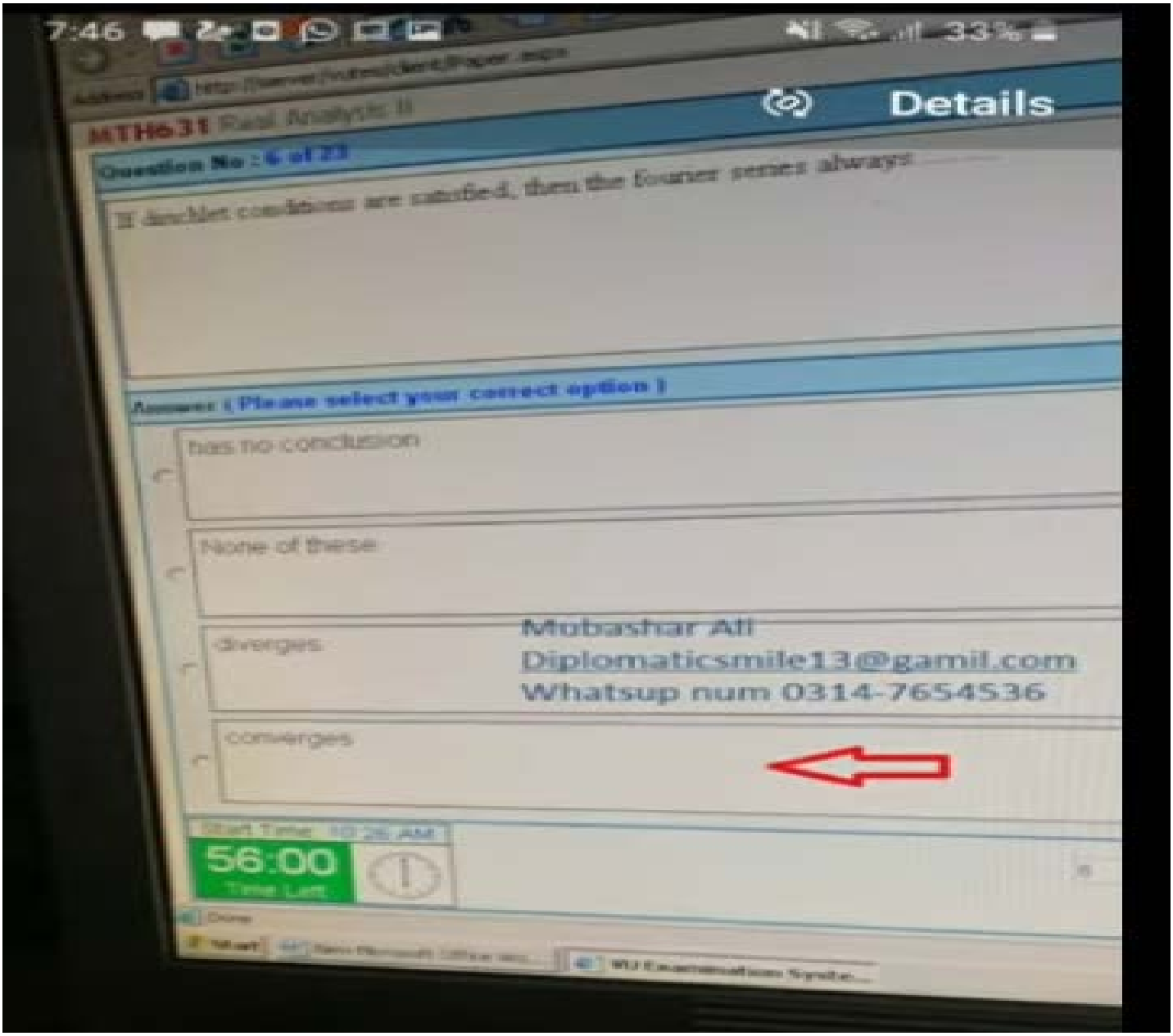

Details
MTH631
Question No : 11 of 23
The limit of a uniformly convergent series of integrable functions is always
page 25
Answer (Please select your correct option)
integrable
not integrable
differentiable
None of these
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Diplomaticsmile13@gamil.com
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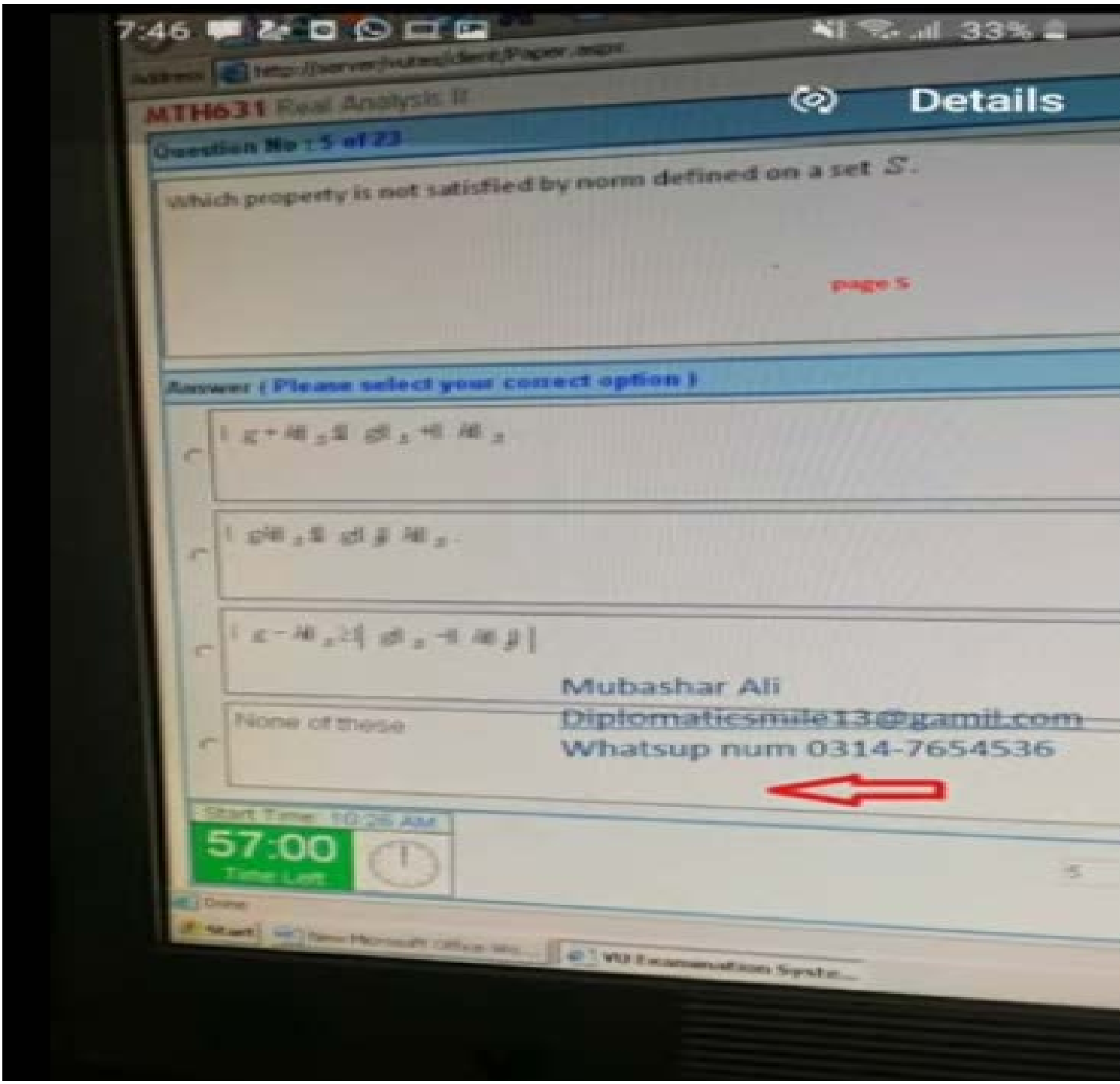

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Details
MTH631 Real Analysis II
Question No : 10 of 23
If a series converges absolutely uniformly on S, then
page 20
Answers (Please select your correct option)
it converges uniformly on S
it diverges
it is differentiable on S
None of these
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Diplomaticsmile13@gamil.com
Whatsup num 0314 7654536
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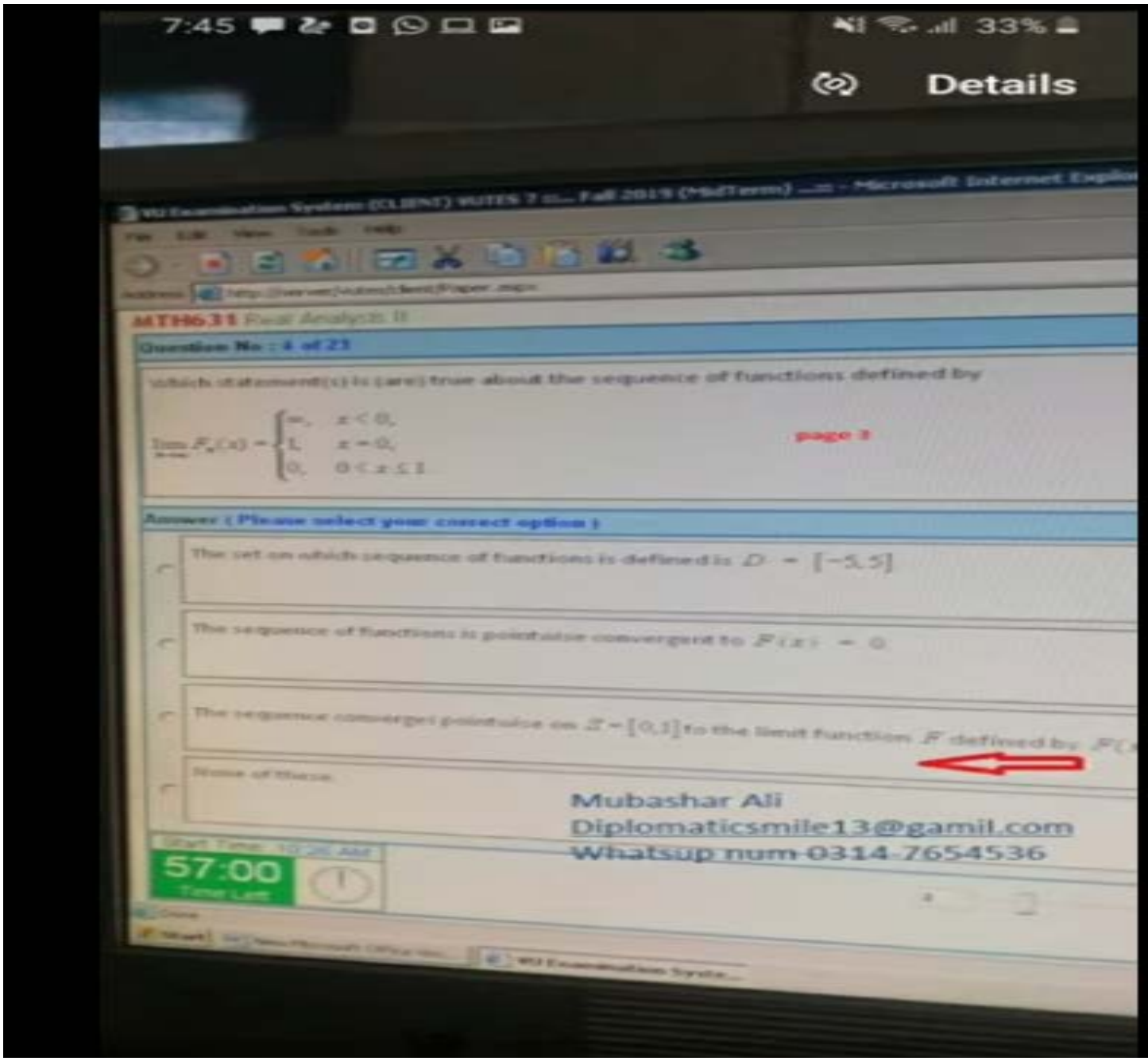

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MTH631 Real Analysis II
Details
Question No : 9 of 23
Which of the following is true about the expression $\int_a^b f(x)dx = \lim_{n\to\infty} \int_a^b f_n(x)dx$?
page 11
Answer: (Please select your correct option)
It is always true
It is not true generally
It is always false
None of these
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Diplomaticsmile13@gamil.com
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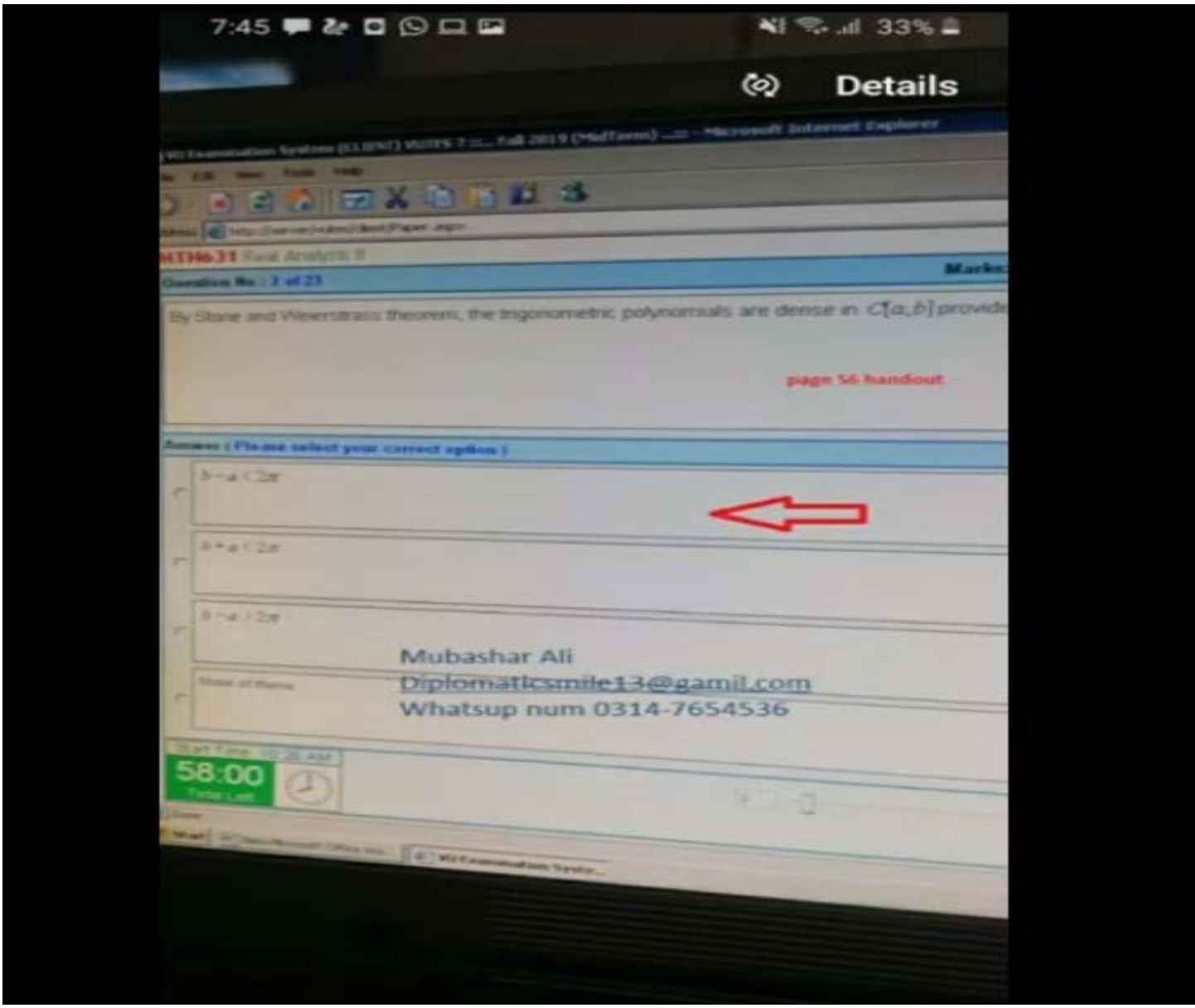

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Details
MTH631 Real Analysis II
Question No : 8 of 23
Suppose that a sequence of function (f_n) is pointwise convergent on the set S. Then
Answer: (Please select your correct option)
(f_n) always converges uniformly
(f_n) may converge uniformly
(f_n) always diverges
None of these
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Diplomaticsmile13@gamil.com
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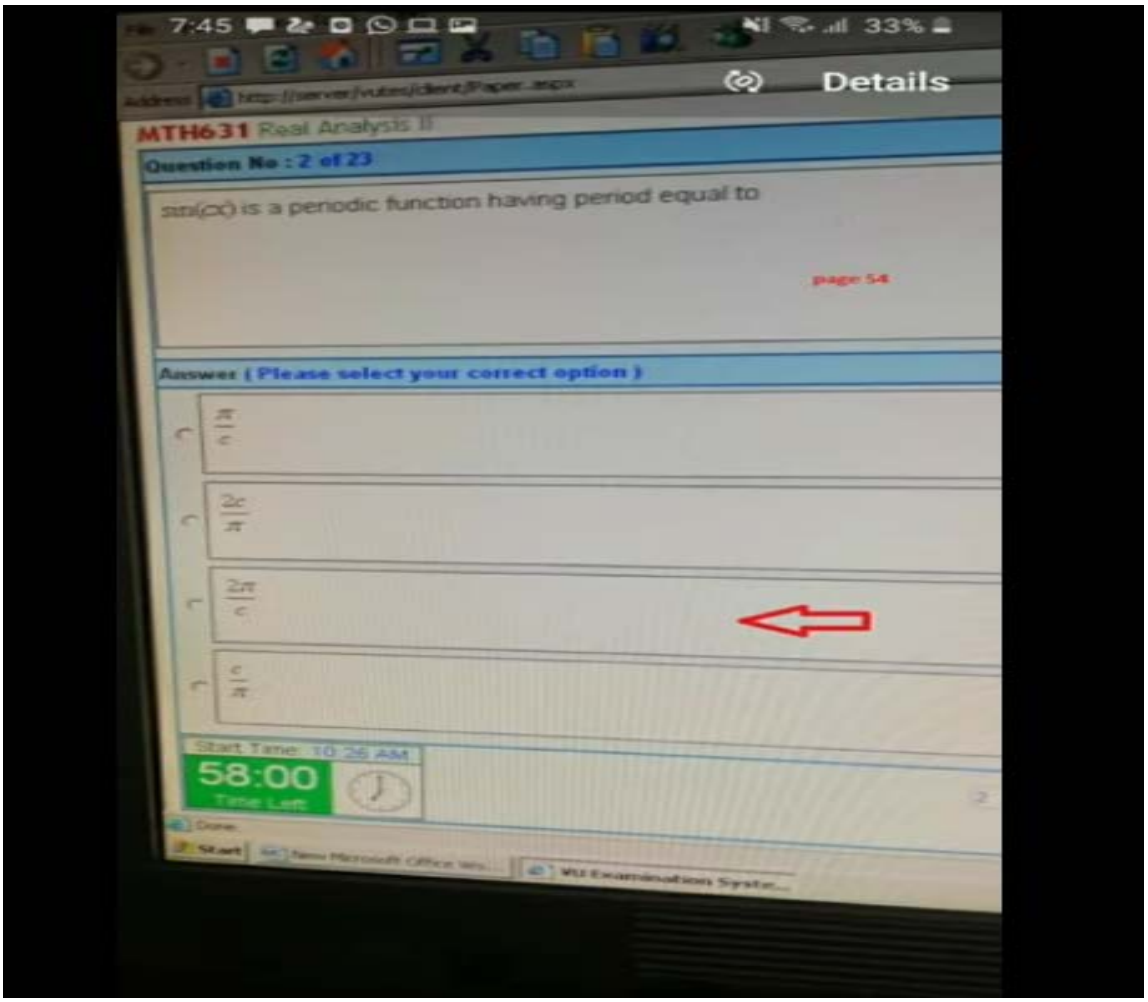

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Details
MTH6311 Real Analysis II
Question No : 7 of 23
Which of the following is true about a uniformly continuous function, f, on [a, b]?
Answer : (Please select your correct option)
The function is bounded.
The function achieves its maximum on the set [a, b].
The derivative of f is bounded.
None of these
55:00
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Details
MTH631
Question No :
has no conclusion
None of these
diverges
converges
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56:00
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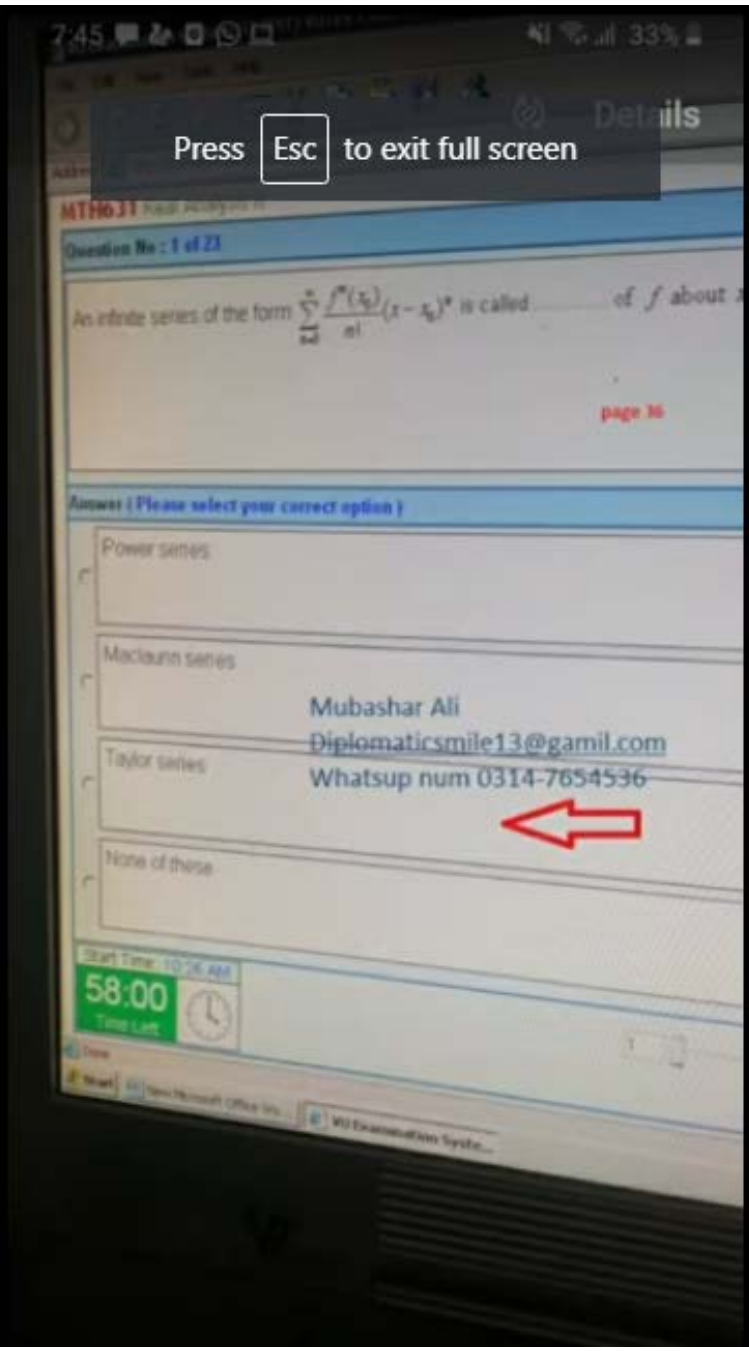

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MTH631 Real Analysis II
Details
Question No : 5 of 23
Which property is not satisfied by norm defined on a set S.
page 5
Answers: (Please select your correct option)
None of these
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Diplomaticsmile13@gamil.com
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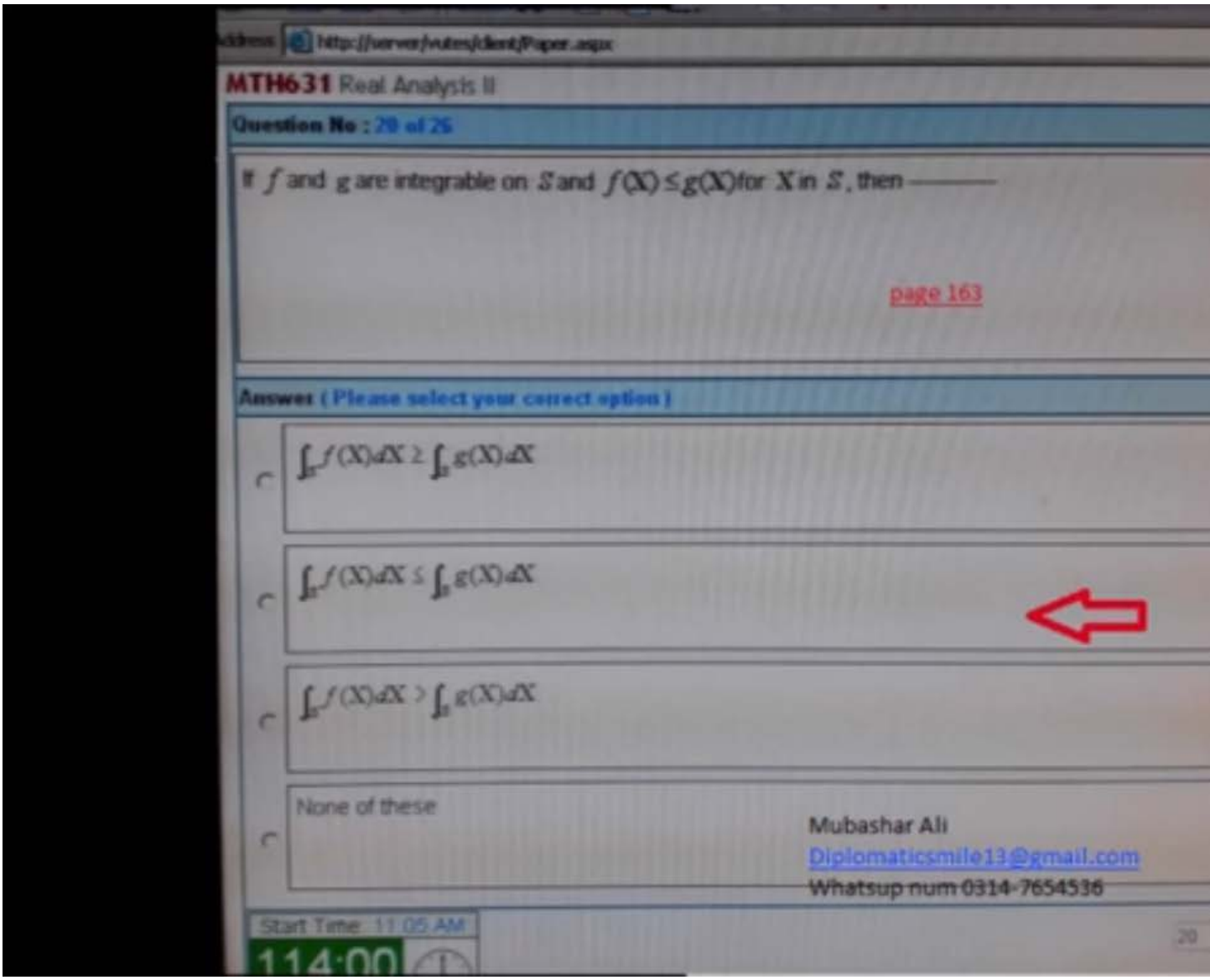

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Details
MTH631 Real Analysis II
Question No : 4 of 21
page 3
Answer : (Please select your correct option)
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Details
MTH631 Real Analysis II
Question No : 3 of 23
By Stone and Weierstrass theorem, the trigonometric polynomials are dense in C[a,b] provide
page 56 handout
Answer (Please select your correct option)
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Details
MTH631 Real Analysis II
Question No : 2 of 23
sin(cx) is a periodic function having period equal to
page 54
Answer (Please select your correct option)
$\frac{\pi}{c}$
$\frac{2c}{\pi}$
$\frac{2\pi}{c}$
$\frac{c}{\pi}$
Start Time: 10:26 AM
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2019


Press Esc to exit full screen
Question No : 1 of 23
An infinite series of the form $\sum \frac{f^n(x_0)}{n!}(x-x_0)^n$ is called ... of f about
page 36
Answer (Please select your correct option)
Power series
Maclaurin series
Taylor series
None of these
Mubashar Ali
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58:00


MTH631 Real Analysis II
Question No : 20 of 25
If f and g are integrable on S and $f(X) \leq g(X)$ for X in S, then ———
page 163
Answer (Please select your correct option)
$\int_S f(X)dX \geq \int_S g(X)dX$
$\int_S f(X)dX \leq \int_S g(X)dX$
$\int_S f(X)dX > \int_S g(X)dX$
None of these
Mubashar Ali
Diplomaticsmile13@gmail.com
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Start Time: 11:05 AM
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